

MAT351 Lecture Notes

EMILY

'25 Fall Semester

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§1 Day 1: Course Administrative Details (Sep. 3, 2025)

[I'll put any of my own remarks in Square Brackets or in footnotes. I love making connections to other ideas, and it feels like killing my child to exclude them just because the prof didn't mention it explicitly.]

Put "MAT351" in subject if you're emailing about the course. The syllabus tab will have the list of topics covered each week. Attendance is strongly encouraged.¹

- Problems are in *Partial Differential Equations: An Introduction* by Walter Strauss.
- Evans PDEs will cover the topics in more detail because it is a graduate textbook.
- PDEs in Action is more applied.

Midterm 2 will be $\geq 80\%$ problems from the Problem Sets. Final exam is gonna be 2 hours rather than 3 because 3 hours is too long.

Topics:

- Intro
- Examples
- General Concepts
- Classify (There are many different kinds). By the end of the semester, when you see a PDE you should know what kind it is. Also, if you see some behavior in the wild, you can reach into your toolbox and think about which kind of PDE properly models the behavior.

We will go over all the main classes of PDEs

- First order PDEs
- Wave equation (Hyperbolic PDEs)
- Heat equation/Diffusion (Parabolic PDEs)
- Laplace/Harmonic Functions (Elliptic PDEs)
- Boundary/Initial Value Problems (Fourier Series)
- First Order Nonlinear PDEs (Shocks)
- Eigenvalues (Hydrogen Atom)
- Schrödinger and Fourier Transform (If there's time)
- Distributions and Weak Solutions² (If there's time)
- ...

PDE is a vast subject. It is one of the most connected fields of math, with connections to Analysis, Number Theory, Physics, Differential Geometry, Algebra, Numerical Analysis, Experiments, and so on. The first part of the class will be more computational, finding explicit solutions to our first four PDEs mentioned above, and the second part will be more theoretical.

¹I'm looking at you, Isaac.

²This is my favourite topic in PDEs.

The rest of the lecture went over some examples of PDEs.

Example 1.1. A simple possible PDE would be as follows: Let $u = u(x, y)$ be a function of two variables. Then our PDE might be

$$u_x + u_y = 0.$$

Remark 1.2. $u_x = \frac{\partial}{\partial x} u$.

From here, a single solution would be $x - y$. We have a solution $u(x, y) = f(x - y)$ for any function f .

Remark 1.3. For ODEs, first order has one degree of freedom, second order has two degrees of freedom, etc. For PDEs, this first order PDE has infinitely many degrees of freedom.

We will also solve $u_x + c(x, y)u_y = 0$ next lecture (by method of characteristics).

Example 1.4 (Burgers' Equation). We will return to first order PDEs at some point to look at nonlinear cases. One such example is

$$u_x + uu_y = 0.$$

This is a simple example used for fluid dynamics and part of why fluid dynamics is hard.

Example 1.5 (Wave Equation). With the Wave Equation, there is one special coordinate, which is time. Take $u = u(t, x)$ with $t \in I \subseteq \mathbb{R}$, $x \in U \subseteq \mathbb{R}^m$, where I is an interval and U is an open set. Now, consider the $m = 1$ case. Then the wave equation is

$$u_{tt} - u_{xx} = 0.$$

For a general m , it's

$$u_{tt} - \Delta_x u = 0$$

Remark 1.6. If $f : \mathbb{R}^m \rightarrow \mathbb{R}$, then Δf is the Laplacian of f , $\frac{\partial^2}{\partial x_1^2} f + \cdots + \frac{\partial^2}{\partial x_m^2} f$. When f has two different variables ($f(t, x)$), then $\Delta_x f$ is the Laplacian with respect to x only. [An efficient way to write it is as the divergence of the gradient of a function, which will come up a bit later.]

We'll also look at

$$u_{tt} - \Delta u = f(x, t),$$

which is the non-homogeneous case.

Example 1.7 (Heat Equation). The heat equation is

$$u_t - \Delta u = 0.$$

The difference between this and the Wave Equation is that this is the first derivative, whereas the Wave Equation is the second derivative. Makes a huge difference.

Example 1.8 (Laplace's Equation). The Laplacian (Laplace's equation) is

$$-\Delta u = 0.$$

[Functions satisfying Laplace's Equation are sometimes called harmonic.] In this sense, Laplace's equation is the steady-state solution to the heat equation and the wave equation. Laplace's equation is very important for other fields as well. [Dirichlet's problem in the circle is just a special case of an initial value problem for Laplace's equation. We will almost certainly see this in 354. In this case, Complex Analysis turns out to be a very efficient way to solve this IVP.]

Remark 1.9. Why do we have the minus in front? One justification could be that Laplace's equation is inspired by physics. For example, in Electrostatics, if V is the electrical potential, then the electric field is given by $E = -\nabla V$. But from here, the electric charge is $\rho = \nabla \cdot E$ (up to a change in units), so it follows that $\rho = -\Delta V$.³ It follows that in regions of space with no charge, the electrical potential should be a harmonic function.

Another more mathematical justification is that $-\Delta$ is a non-negative operator. The professor kinda trailed off while talking about this and never finished the proof, but here's what I imagine he was trying to say:

$$\begin{aligned} \langle -\Delta g, g \rangle &= \int_{\mathbb{R}^m} -\Delta g \, g \\ &= \int_{\mathbb{R}^m} \operatorname{div}(-\nabla g(x)g(x)) + \int_{\mathbb{R}^m} |\nabla g|^2 dx \end{aligned}$$

from here, assuming that $\nabla g \cdot g \rightarrow 0$ sufficiently quickly, it would follow (by the divergence theorem) that

$$\int_{\mathbb{R}^m} \operatorname{div}(-\nabla g(x)g(x)) = 0,$$

and therefore that this inner product would be non-negative. That being said, he was not very precise here.

Next time we start with the formal definition of a PDE and write out Navier-Stokes.

³I *told* you it would come up later.

§2 Day 2: Definitions, Examples, Solutions, Oh My! (Sep. 8, 2025)

Definition 2.1 (Partial Differential Equation). A Partial Differential Equation is an equation for an unknown function u of more variables involving its partial derivatives, in general written as

$$F(D^k u, D^{k-1} u, \dots, Du, u, x) = 0. \quad (*)$$

The convention is that $D^k u$ is the k th partial derivatives of u .⁴

We'll generally look for $u : x \in U \subseteq \mathbb{R}^m \rightarrow X$, where X is typically $\mathbb{R}, \mathbb{C}, \mathbb{R}^m$. We also want $u \in C^k(U)$ (although, there are more general setups).⁵ For the PDE $(*)$, we say the order is k (order will mainly be 1 or 2 in this class.) If $(*)$ can be written as

$$\mathcal{L}[u] = 0,$$

where $\mathcal{L}[u + v] = \mathcal{L}[u] + \mathcal{L}[v]$, then we call it linear. We also can include $\mathcal{L}[cu] = c\mathcal{L}[u]$, but this is usually redundant. Otherwise it's nonlinear. If $(*)$ has the form $\mathcal{L}[u] = f$ it's called (linear) non-homogeneous.

Example 2.2. The general second order pde linear: $u : (x, y) \in U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$

$$Q_{11}(x, y)u_{xx} + Q_{12}(x, y)u_{xy} + Q_{22}(x, y)u_{yy} + b_1(x, y)u_x + b_2(x, y)u_y + c(x, y)u = 0.$$

We say it's constant coefficient if Q_{ij}, b_k, c are constant.

Some examples from last class were constant coefficient

Example 2.3 (Wave Equation).

$$u_{tt} - u_{xx} = 0$$

Example 2.4 (Heat Equation).

$$u_t - u_{xx} = 0$$

Example 2.5 (Laplace's Equation).

$$u_{xx} + u_{yy} = 0$$

Example 2.6.

$$u_x + Q(x, y)u_y = 0$$

Some more examples of linear PDEs:

Example 2.7 (Schrödinger Equation).

$$u : (t, x) \in I \times \mathbb{R}^m \rightarrow \mathbb{C}$$

where $I \subseteq \mathbb{R}$

$$iu_t - \Delta_x u = 0$$

We'll also look at $iu_t + (-\Delta + V(x))u = 0$.

⁴I imagine something like D^k is a k -tensor analogous to—in the case of $k = 2$ —the Hessian matrix. Such formalism is pointless, however.

⁵Much of 457/8 is dedicated to things that make PDEs more general

On the other hand, here are some nonlinear PDEs:

Example 2.8 (Burgers' Type).

$$u_t + Q(u)u_x = 0$$

Example 2.9.

$$u_t + u_x = u^2.$$

Since the right is in terms of u and nonlinear, it's a nonlinear PDE.

These are the only two types of nonlinear PDEs we'll solve. In some sense, the first one is "more" nonlinear because we're multiplying the partial derivative by something nonlinear, whereas the second is only multiplying u by something.

Example 2.10 (Nonlinear Wave Equation).

$$u_{tt} - u_{xx} = u^p,$$

This might be a good model for general relativity, as general relativity is all about the nonlinear wave equation.

Example 2.11 (Maxwell's Equations). Maxwell's Equations are an example of a linear system of PDEs.⁶

$$\begin{aligned}\frac{1}{c}\vec{E}_t &= \nabla \times \vec{B} \\ \frac{1}{c}\vec{B}_t &= -\nabla \times \vec{E}.\end{aligned}$$

Here, we take

$$\vec{E}, \vec{B} : (t, x) \in \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}^3,$$

and c is the speed of light. We also have some constraints that

$$\begin{aligned}\nabla_x \cdot \vec{E} &= 0 \\ \nabla_x \cdot \vec{B} &= 0,\end{aligned}$$

and these are added for physical reasons and also to make the ODE solvable—in real life we indeed observe that the Electric field and the Magnetic field are conservative, so adding these requirements doesn't cause problems. If we don't have these, then there's too many degrees of freedom to solve. [Cf. Helmholtz Theorem]

Example 2.12 (Navier-Stokes' Equations). These are a nonlinear system of PDEs.

$$\vec{u}_t + \vec{u} \cdot \nabla_x \vec{u} = -\nabla p + \nu \Delta_x \vec{u}.$$

Here, we have that $\vec{u} : (t, x) \in [0, \infty) \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is the velocity field, $\nu > 0$ is a constant which represents viscosity, we have that $p : (t, x) \rightarrow \mathbb{R}$ is pressure. We say also that

$$(\vec{u} \cdot \nabla \vec{u})_k = \sum_{j=1}^3 \vec{u}_j \partial_j \vec{u}_k$$

and then we have

$$\nabla \cdot \vec{u} = 0$$

because otherwise there would be too many unknowns to solve it. [Another way to think about this term is that it represents conservation of mass, so this boundary condition makes the fluid incompressible—so water, rather than air].

⁶“Emily gave a FANTASTIC example here. Kudos to her.” – Isaac

§3 Day 3: I'm kind of getting sick of only doing examples (Sep. 10, 2025)

Definition 3.1. A PDE is fully nonlinear if it is nonlinear in the highest order derivative. This is, if the PDE is of the form

$$F(D^k u, D^{k-1} u, \dots, u, x) = 0,$$

where F is nonlinear in the $D^k u$ component.

Example 3.2.

$$u_t + uu_x = 0 \tag{1}$$

$$u_{tt} + u_x u_{xx} = 0 \tag{2}$$

$$u_t + u_x = u^2 \tag{3}$$

$$u_t^2 + u_x = 0 \tag{4}$$

$$u_t u_x + u = 0 \tag{5}$$

We see that (1) is not fully nonlinear, likewise with (2), and (3). On the other hand, (4) and (5) are fully nonlinear, because they are nonlinear in Du , and their order is 1. Finally, m

Example 3.3 (Mean Curvature). This is from Differential Geometry.

$$\nabla \cdot \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right) = 0.$$

It is fully nonlinear as well.

We will now solve some equations “by hand”.

Example 3.4. Let $u = u(x, y)$, $(x, y) \in \mathbb{R}^2$, with

$$u_{xx} = 0.$$

We can just integrate twice to get the general solution here.

$$\begin{aligned} u_{xx} &= 0 \\ u_x &= f(y) \\ u &= xf(y) + g(y). \end{aligned}$$

This is clearly the general solution as we just integrated twice. Now, the question is how much information do we need to provide for there to be only one solution to the IVP? For this, if we specify

$$u(0, y) = H(y) \quad \text{and} \quad u_x(0, y) = L(y),$$

then this would give that

$$u(x, y) = L(y)x + H(y),$$

which is sufficient.

Example 3.5 (267 Exposure Therapy). “I just want to test your ODE knowledge.”

$$u_{xx} + u = 0.$$

Solutions are

$$u = a(y) \sin x + b(y) \cos x,$$

He then started talking about like factoring polynomials and exponentials to find the solution to the ODE but I fainted from the experience, so I’m not sure what he did.

Remark 3.6. When drawing, functions of t and x , make t the vertical axis, and x the horizontal axis. I don’t know why. Also, he writes $u(t, x)$ but the book writes $u(x, t)$. I personally prefer $u(t, x)$, but I might end up using both accidentally.

Example 3.7. Let $u = u(t, x)$ with

$$u_{tt} - u_{xx} = 0. \tag{w}$$

Then we can write this as

$$(\partial_t - \partial_x)(\partial_t + \partial_x)u = 0.$$

If we know $(\partial_t + \partial_x)u = 0$, then u is a solution of (w), but this can be solved by $F(t - x)$, for all function F . Similarly, if $(\partial_t - \partial_x)u = 0$, then u solves (w), which would yield solutions $G(t + x)$ of (w), for all functions G .

It turns out that

$$u(t, x) = F(t - x) + G(t + x)$$

is actually the general solution—we will prove this later.

We saw that solving a PDE gives infinitely many degrees of freedom. So how do we pose a PDE so that we can identify a specific solution? Answer: Give additional info in the form of (A) initial conditions, (B) boundary conditions. This motivates the following definition:

Definition 3.8 (Well-Posed Problem). We say that a Well-Posed Problem is a PDE along with Boundary Conditions and Initial Conditions such that the solutions have

1. Existence of Solutions
2. Uniqueness of Solutions
3. Continuous dependence⁷ on this additional data.

Example 3.9 (Poisson’s Equation). Let u be the unknown, and let our equation be

$$-\Delta u = f.$$

Then there’s two ways to make it well-posed. If we say that $u = u(x)$, and $x \in D \subseteq \mathbb{R}^m$, and $g : \partial D \rightarrow \mathbb{R}$, then we can either do

1. Let $u = g$, $x \in \partial D$. These are the Dirichlet Boundary Conditions [a.k.a. The Dirichlet Circle Problem]

⁷I felt a visceral sense of dread typing this.

2. Let

$$\frac{\partial u}{\partial \hat{n}} = \nabla u \cdot \hat{n},$$

where $\hat{n}$ is the unit outward normal of ∂D . Then the Neumann boundary conditions are

$$\frac{\partial u}{\partial \hat{n}} = g,$$

where $x \in \partial D$

3. The Robin boundary conditions are

$$\frac{\partial u}{\partial \hat{n}} + Qu = g$$

where $x \in \partial D$.

For the Neumann boundary conditions we need some care to prove this is well-posed, which will appear in the homework. Giving both the Dirichlet and Neumann boundary data is not a good idea. One might remark that this seems strange—after all, it's a second-order PDE, so it might look like (as with a second-order ODE) we'd have to specify stuff about both the function and the derivatives. But we will see this is not the case.

Example 3.10 (Dirichlet's Boundary Conditions). Consider Poisson's equation along with the Dirichlet boundary conditions:

$$-\Delta u = f \quad (x \in D)$$

$$u = g \quad (x \in \partial D)$$

Are solutions of this problem unique? Assume there exist u_1, u_2 solutions, then look at $v = u_1 - u_2$. Then this yields

$$-\Delta v = 0 \quad (x \in D)$$

$$v = 0 \quad (x \in \partial D)$$

The idea is to integrate our PDE against v . First, notice the identity that

$$\begin{aligned} \int_D \Delta v \cdot v \, dx &= \int_D \sum_{i=1}^m v \partial_{x_i}^2 v \, dx \\ &= \int_D \sum_{i=1}^m \partial_{x_i} (v \partial_{x_i} v) \, dx - \int_D \sum_{i=1}^m \partial_{x_i} v \cdot \partial_{x_i} v \, dx. \\ &= \int_D \nabla \cdot (v \nabla v) - \int_D \nabla v \cdot \nabla v \, dx. \end{aligned}$$

Using this, we can derive

$$\begin{aligned} 0 &= \int_D v \Delta v \, dx \\ &= \int_D \nabla \cdot (\nabla v) v \, dx \\ &= \int_D \nabla \cdot (\nabla v v) \, dx - \int_D \nabla v \cdot \nabla v \, dx \\ &= \int_{\partial D} \Delta v \cdot v \cdot \hat{n} \, dx - \int_D |\nabla v|^2 \, dx \end{aligned} \tag{1}$$

Since $v \equiv 0$ on the boundary, $\Delta v \cdot v \cdot \hat{n} \equiv 0$ on the boundary, so it follows that $\int_D |\nabla v|^2 dx = 0$. It follows that $\nabla v \equiv 0$ on D , thus v is constant. But we know $v \equiv 0$ on ∂D so $v \equiv 0$ on the whole domain. This proves uniqueness.

Example 3.11 (Vibrating String). Let $D = [0, \infty) \times [0, L]$. A vibrating string can be modelled by the laws

$$\begin{aligned} u_{tt} - c^2 u_{xx} &= 0 \\ u(t, 0) &= w_1(t) \\ u(t, L) &= w_2(t) \end{aligned}$$

where $u : D \rightarrow \mathbb{R}$ [a longer derivation of these laws can be found in Stein and Shakarchi Book 1], where $w_1, w_2 : [0, \infty) \rightarrow \mathbb{R}$ are how the end-points of the string move. It turns out this is not enough to get a unique solution. We also need

$$u(0, x) = f(x), \quad u_t(0, x) = g(x).$$

He did not show this, but he showed show in the case of Laplace's equation, these boundary conditions don't yield a well-posed problem.

Example 3.12 (Laplace's Equation). Suppose we have $u : \mathbb{H}^n \rightarrow \mathbb{R}$. Then impose the initial conditions

$$\begin{aligned} u_{xx} + u_{yy} &= 0 \\ u(x, 0) &= 0 \\ u_y(x, 0) &= \frac{1}{m} \sin(mx). \end{aligned}$$

Notice that this is analogous to providing both the Dirichlet and Neumann boundary conditions for Poisson's equations above (but it's not, because it's not bounded). This is a sequence of PDEs, which we'll call $u^{(m)}$ the solution.

As $m \rightarrow \infty$ the boundary conditions go to 0, so they roughly go to the solution $u \equiv 0$. If we solve for m large, then our solution should be close to 0 to get continuous dependence. But is this the case?

We can solve this by ODEs like we did earlier (rewriting as $(\partial_t - i\partial_x)(\partial_t + i\partial_x)u = 0$ then solving each root individually—he left actually doing this as an exercise). This gives solutions of

$$u^{(m)} = \frac{e^{-\sqrt{m}}}{m^2} \sin(mx) \sinh(my).$$

The question is whether this goes to 0 as $m \rightarrow \infty$. But notice that $\sinh$ is basically exponential, so this is going to go to infinity everywhere. Thus, in this case, continuous dependence on boundary conditions doesn't hold, so this isn't a well-posed problem.

§4 Day 4: Solving some linear PDEs (Sep. 15, 2025)

Well-Posed Problems have three parts

- Existence, which we discussed in the last class, and non-existence, which we saw in the homework
- Uniqueness, which we saw examples and a proof of last class
- Continuous dependence on boundary data, which we saw a counterexample of last week for the Laplace problem.

We will mostly study well-posed problems almost all the time. We'll have to prove that we have well-posedness for a given problem (but we might have to leave this for another class if it's too hard). [Also, one can ask what's called a regularity question—this won't come up in this class, but it's about when we can prove a weak solution is continuous or differentiable or smooth or etc. This can be defined precisely using methods of 436/457/458.]

First order PDEs (we are doing $\mathbb{R}^2$ for now) can be classified in a couple different ways.

1. Linear homogeneous, with constant coefficients, which is of the form

$$au_x + bu_y + cu = 0,$$

2. Linear homogeneous with non-constant coefficients, which is of the form

$$a(x, y)u_x + b(x, y)u_y + c(x, y)u = 0$$

3. Linear non-homogeneous, which is of the form

$$a(x, y)u_x + b(x, y)u_y = f(x, y)$$

4. Semilinear, of the form

$$au_x + bu_y = f(u),$$

where f is nonlinear.

5. Quasilinear, which we will study rigorously later. An example is

$$u_x + uu_y = 0.$$

[This is where the regularity problem I discussed above can pop up.]

We will now solve these.

1. Previously, we mentioned that the solutions to $au_x + bu_y = 0$ are of the form

$$u(x, y) = F(bx - ay).$$

However, we will do this more rigorously now. We will add additional data (boundary/initial conditions) later—working backwards in some sense. There are three methods we can employ to solve this problem.

- a) The geometric approach. This PDE says $\nabla u \cdot (a, b) = 0$. In other words, $\nabla u \perp (a, b)$. It follows that starting at any point, our function must be constant in the direction of (a, b) . As such, to make the problem well posed, we just need to provide initial conditions along any (smooth) curve which is never tangent to the level sets. These level sets are called coordinate curves. One can deduce that $u(x, y) = F(bx - ay)$ is the general solution from this reasoning, but he did not do this explicitly.

Example 4.1.

$$\begin{cases} au_x + bu_y = 0 \\ u(x, 0) = g(x) \end{cases}$$

We know $u(x, y) = F(bx - ay)$, so

$$g(x) = u(x, 0) = F(bx),$$

thus we have $F(x) = g\left(\frac{x}{b}\right)$, so

$$u(x, y) = g\left(x - \frac{a}{b}y\right).$$

- b) Change coordinates—this method is very important in any PDE, as you might be able to reduce the dimensionality of your problem if you are clever. We will do this next time.
- c) The general approach is the method of characteristics, which we will use later for semilinear, quasilinear, etc. We will also do this next time.

§5 Day 5: More on First-Order Equations (Sep. 17, 2025)

Problem set 2 due next Thursday, as the TAs are going to have office hours on Wednesday.

- (1) We solved $au_x + bu_y = 0$ for $u = u(x, y)$, a, b constants by the geometric method last time.
- (2) Let's use change of variables. Set $u = u(x, y)$ and set $v(x', y') = u(x, y)$, where x', y' represent a change of coordinates from x, y . Differentiating $u(x, y)$ yields

$$\begin{aligned} u_x &= v_{x'} \frac{dx'}{dx} + v_{y'} \frac{dy'}{dx} \\ u_y &= v_{x'} \frac{dx'}{dy} + v_{y'} \frac{dy'}{dy}. \end{aligned}$$

From here, set

$$\begin{cases} x' = ax + by \\ y' = bx - ay \end{cases}$$

Then we have that

$$\begin{cases} u_x = av_{x'} + bv_{y'} \\ u_y = bv_{x'} - av_{y'} \end{cases}$$

so plugging into our PDE yields

$$a(av_{x'} + bv_{y'}) + b(bv_{x'} - av_{y'}) = 0,$$

and then simplifying,

$$a^2 v_{x'} + b^2 v_{x'} = 0.$$

But if $a^2 + b^2 = 0$, then our PDE is quite boring, so assuming it is positive, this implies $v_{x'} = 0$. From here, we get that v is constant with respect to x' , so

$$v = f(y') = f(bx - ay) = u(x, y),$$

which solves the PDE.

- (3) The book *mentions* this method, but not really systematically, only briefly.⁸ It's a more general method. Let's look at curves in the plane $\gamma(s) = (x(s), y(s))$ in parametric form.⁹ Look at u along this curve. We define

$$z(s) = u(x(s), y(s)),$$

then differentiate along this curve

$$\dot{z}(s) = \dot{x}(s)u_x(x(s), y(s)) + \dot{y}(s)u_y(x(s), y(s)).$$

Using the original ODE, we can get $\dot{z} \equiv 0$ if $\dot{x} \equiv a$ and $\dot{y} \equiv b$. If we can merely deduce what all the curves satisfying the system

$$\begin{cases} \dot{x} \equiv a \\ \dot{y} \equiv b \end{cases},$$

⁸See Evans 3.2 if you want the systematic development. Fair warning, Evans is not for the faint of heart.

⁹It gets a bit hard to follow, because he switches between x and y being functions, and being independent variables at times.

then we will have curves along which $\dot{z} \equiv 0$, so u is constant along these curves. We call these curves the characteristic curves of our PDE, and these equations are the characteristic ODEs. The solutions to these ODEs are

$$(x(s), y(s)) = (as + x(0), bs + y(0)),$$

which implies that the solution can be determined by values along all curves of this form. This is equivalent to the general solution provided by the other two methods.

Now, let's suppose that we have data on the x -axis. Suppose $y(0) = 0$. Then we have that

$$u(as + x(0), bs) = u(x(0), 0), \quad s \in \mathbb{R},$$

so it follows that $u(x, y) = u(x - \frac{a}{b}y, 0)$. This gives us a unique solution for any [sufficiently regular] data on the x -axis.

Example 5.1. Suppose we are given

$$\begin{cases} u_x + yu_y = 0 \\ u(0, y) = y^2 \end{cases}$$

Of the three methods, try to do the first two as an exercise. But we will use the method of characteristics once again. Set $z(s) = u(x(s), y(s))$. Then once again we have $\dot{z} = u_x \dot{x} + u_y \dot{y}$, so set

$$\begin{cases} \dot{x}(s) = 1 \\ \dot{y}(s) = y(s) \end{cases}$$

to get $\dot{z}(s) = 0$. These yield curves of the form

$$x(s) = s + x_0, \quad y(s) = y_0 e^s.$$

Rearranging, we can get that

$$y(x) = y_0 e^{x-x_0},$$

which makes things slightly nicer. From here, to solve the ODE, notice that $u(x, Ce^x) = u(0, C)$ and so $u(x, y) = f(ye^{-x})$. To solve f given our constraints, notice that

$$f(ye^0) = u(0, y) = y^2,$$

so our solutions are

$$u(x, y) = (ye^{-x})^2 = y^2 e^{-2x}.$$

This is the outline of the general method. Find some way to make the function nice along curves z [they need not be constant, just solvable—if we had $\dot{z}(s) = s^2$ we could solve something like this], then determine what data is required to specify the characteristic curves (in the above, y_0 uniquely determined the curve), then find the curve that intersects (x, y) in terms of this data, and this will yield the value of u at this point.

Example 5.2.

$$u_x + 2xy^2 u_y = 0,$$

then we have that

$$\dot{x} = 1, \dot{y} = 2xy^2,$$

so we can get that

$$\frac{dy}{dx} = 2xy^2.$$

From here, we need to use our ODE knowledge to solve this (it's separable).

$$\begin{aligned}\int \frac{1}{y^2} dy &= \int 2x dx \\ \frac{-1}{y} &= x^2 + C \\ y &= (-x^2 + C)^{-1},\end{aligned}$$

and thus these are our characteristic curves, so u is constant along them. He then graphed these functions as C varied. You can do this on desmos if you want. Notably, its asymptotes are at $\pm\sqrt{C}$ and its y -intercept is $\frac{1}{C}$. It follows that $u(0, \frac{1}{C}) = f(C) = f_0(\frac{1}{y} + x^2)$

Remark 5.3. Let $u = u(t, x)$, $t \in \mathbb{R}$, $x \in \mathbb{R}^m$. In general, we can solve first order equations like

$$u_t + a(t, x)u_x + b(t, x)u = f(t, x). \quad (*)$$

By the same method as before, we want to reduce this into a system of ODEs. Set $z(s) = u(t(s), x(s))$. Then we have that

$$\begin{aligned}z'(s) &= t'(s)u_t(t(s), x(s)) + x'(s)u_x(t(s), x(s)) \\ &= f(t(s), x(s)) - b(t(s), x(s))u(t(s), x(s)) \quad (\text{plugging into } (*)) \\ &= f(t(s), x(s)) - b(t(s), x(s))z(s).\end{aligned}$$

Thus for $(*)$, we say that the system of characteristics is

$$\begin{cases} \dot{t} = 1 \\ \dot{x} = a(t, x) \\ \dot{z} = f(t, x) - b(t, x)z \end{cases}$$

We set $\dot{t} = 1$ because the parameterization of the path doesn't matter. This is a nonlinear system of ODEs, but it works.

Example 5.4 (Semilinear). Find characteristics for

$$\begin{cases} u_t + 2tu_x = u^2 \\ u(0, x) = g(x) \end{cases}$$

To do this, we have that the system of ODEs is

$$\begin{cases} \dot{t} = 1 \\ \dot{x} = 2t \\ \dot{z} = z^2 \end{cases},$$

so if we let our initial data be $t(0) = 0$, $x(0) = x_0$, $z(0) = g(x_0)$, then solving yields

$$\begin{cases} t = s \\ x = t^2 + x_0 \\ z = \frac{g(x_0)}{1 - sg(x_0)} \end{cases}.$$

Rearranging, we have that

$$\begin{cases} s = t \\ x_0 = x - t^2 \\ u(t(s), x(s)) = \frac{g(x_0)}{1 - sg(x_0)} \end{cases},$$

so therefore

$$u(t, x) = \frac{g(x - t^2)}{1 - tg(x - t^2)}.$$

This solution could blow up depending on the values of g , so just like nonlinear ODEs ($x' = 1 + x^2$), we have less assurances of nice properties when we don't have linearity.

§6 Day 6: Method of Characteristics (Sep. 22, 2025)

Theorem 6.1 (Non-Homogeneous 1st Order). Non-homogeneous 1st-order PDEs are of the form

$$u_t + b \nabla u = f(t, x),$$

where $b \in \mathbb{R}^m$ (constant), $u : (t, x) \in \mathbb{R} \times \mathbb{R}^m \rightarrow \mathbb{R}$, and $f : (t, x) \in \mathbb{R} \times \mathbb{R}^m \rightarrow \mathbb{R}$. Given the homogeneous case

$$v_t + b \nabla v = 0,$$

we have that the general solution is $F(x - bt)$.

We'll solve the transport equation with forcing. First, we will apply the method of characteristics. Let $(t(s), x(s), z(s))$ be the characteristic curves. Then we have that our system of ODEs is

$$\begin{cases} \dot{t}(s) = 1 & t(0) = 0 \\ \dot{x}(s) = b & x(0) = x_0 \\ \dot{z}(s) = f(t, x) & z(0) = g(x_0) \end{cases}$$

where the right side gives some additional data with which we can hopefully have a unique solution. It follows that $t(s) = s$ and $x(s) = x_0 + bs$. From here,

$$\dot{z}(s) = \frac{d}{ds} u(s, x_0 + bs) = f(x_0 + bs),$$

so it follows that

$$z(t) = \int_0^t f(s, x_0 + bs) ds = \mu(t, x_0 + bt) - \mu(0, x_0).$$

Rearranging gives

$$\mu(t, x_0 + bt) = \mu(0, x_0) + \int_0^t f(s, x_0 + bs) ds$$

thus we have that ($x = x_0 + bt$)

$$\mu(t, x) = g(x - bt) + \int_0^t f(s, x - b(t - s)) ds.$$

Remark 6.2. Solution of non-homogeneous is the superposition of solution of homogeneous problem and the solution of the non-homogeneous problem without any data. Namely, the $g(x - bt)$ is the homogeneous solution, and the integral is the solution of the non-homogeneous without any data.

Remark 6.3. This is an example of Duhamel's principle.

$$u(t, x) = u_0 + \int_0^t v,$$

where u_0 solves the homogeneous problem, and v solves

$$\begin{cases} v_t + b \cdot \nabla v = 0 \\ v(t) = v(s) = f(s, x) \end{cases}.$$

Example 6.4. We will tackle

$$\begin{cases} u_t + uu_x = 0 \\ u(0, x) = x^2 \end{cases}$$

If we have constant coefficients, (constant c instead of u) then c would be the speed of transport. As such, this is a model for shocks/explosions. Compressible fluids and the like.

We will try the method of characteristics. Our system with data is

$$\begin{cases} \dot{t} = 1 & t(0) = 0 \\ \dot{x} = z & x(0) = x_0 \\ \dot{z} = 0 & z(0) = x_0^2 \end{cases}$$

so it has turned this Quasilinear first order PDE into a system of ODEs. This yields the solution

$$\begin{cases} t = s \\ x = sx_0^2 + x_0 \\ z = x_0^2 \end{cases}$$

From here we have to invert the relation. The above tells us that

$$u(s, sx_0^2 + x_0) = z(s) = x_0^2,$$

so we just need to find x_0 as a function of t and x . On Wednesday he is going to ask us for the answer, so be prepared (jk I know most of you will not be there).

§7 Day 7: Lots of Calculations (Sep. 22, 2025)

Remark 7.1 (Method of Characteristics). For the PDE

$$P(t, x, u)u_t + Q(t, x, y)u_x = f(t, x, u)$$

the characteristic curves $(t(s), x(s))$ are given as solutions of

$$\begin{cases} \dot{t} = P(t, x, z) \\ \dot{x} = Q(t, x, z) \\ \dot{z} = f(t, x, z). \end{cases}$$

These are not necessarily level curves.

We will do second-order PDEs now. We start with the 2D case, for linear 2nd order. The general case is

$$\begin{aligned} a_{11}(x, y)u_{xx} + 2a_{12}(x, y)u_{xy} + a_{22}(x, y)u_{yy} \\ + a_1(x, y)u_x + a_2(x, y)u_y + c(x, y)u = 0. \end{aligned} \quad (\text{lower order terms})$$

The PDE can have very different behavior depending on the coefficients a_{11} , a_{12} , and a_{22} . So first we will start with the constant coefficient case.

Let our a all be real values, and likewise u . Then our constant coefficient case is given by

$$a_{11}u_{xx} + 2a_{12}u_{xy} + a_{22}u_{yy} + \text{lower order} = 0. \quad (*)$$

We will classify it based on the discriminant of the polynomial $a_{11}\lambda^2 + 2a_{12}\lambda + a_{22}$.

Theorem 7.2. By a linear change of independent variables, the equation $(*)$ can be written in one of the 3 forms below:

1. If $a_{12}^2 - a_{11}a_{22} < 0$, then it's

$$u_{xx} + u_{yy} + \text{lower order} = 0.$$

2. If $a_{12}^2 - a_{11}a_{22} = 0$ then it's,

$$u_{xx} + \text{lower order} = 0.$$

3. If $a_{12}^2 - a_{11}a_{22} > 0$ then it's,

$$u_{xx} - u_{yy} + \text{lower order} = 0.$$

Zander pointed out that this is naturally motivated by algebraic properties of quadratic forms. He then said the word “sheaf” and I ran away to hide.

Proof. We will automatically disregard any lower order terms. We can rescale $a_{11} = 1$ without loss of generality (unless $a_{11} = a_{22} = 0$). Then we have

$$u_{xx} + 2a_{12}u_{xy} + a_{22}u_{yy} = 0.$$

We can therefore rewrite this as

$$(\partial_x + a_{12}\partial_y)^2 u + (a_{22} - a_{12}^2)u_{yy} = 0.$$

From here, we're going to use a change of variables

$$\begin{cases} x = \xi \\ y = a_{12}\xi + \eta \end{cases}$$

which yields

$$\begin{cases} \partial_\xi = \partial_x + a_{12}\partial_y \\ \partial_\eta = \partial_y \end{cases},$$

so we get that

$$\partial_\xi^2 u + b^2 \partial_\eta u = 0,$$

where $b^2 = (a_{22} - a_{12}^2)$. This gives us 1 and 2. For 3, set $b^2 = a_{12}^2 - a_{22} > 0$. PDE is

$$(\partial_x + a_{12}\partial_y)^2 u - b^2 u_{yy} = 0.$$

□

In HW 3 we will see problems on classification of PDEs and the relation to the characteristic polynomial $a_{11}\lambda^2 + 2a_{12}\lambda + a_{22}$.

For the general case of dimensions higher than 2, we look at the linear PDE

$$\sum_{i,j=1}^m a_{ij}(x) \partial_{x_i x_j}^2 u + b(x) \nabla u + c(x) u = 0,$$

where $x \in \mathbb{R}^m$. From here we look at the Eigenvalues and all that stuff.

Theorem 7.3. We can classify according to the signature of the matrix $A = (a_{i,j})_{i,j=1}^m$. Notice that it is real symmetric, so¹⁰ it's diagonalizable.

1. If all the eigenvalues have the same sign (nonzero), then the PDE is elliptic.
2. If all the eigenvalues have the same sign except one is zero, it's parabolic.
3. If all the eigenvalues have one sign, except one with the opposite sign, it's hyperbolic.

We don't classify all the possible cases, but these are the main things we see anyways. If we restrict it to 2 dimensions.

We'll start with the wave equation. Surely this won't be so hard. This is just the equation

$$u_{tt} - c^2 u_{xx} = 0.$$

We will

1. Find the general solution rigorously ($f(x+ct) + g(x-ct)$),
2. Find the solution to the IVP (D'Alembert's Equation)
3. Look at physical consequences (e.g. finite speed)
4. Look at a derivation of equation

Proposition 7.4. The general solution to the 1d wave equation is $u(x,t) = f(x+ct) + g(x-ct)$.

¹⁰by spectral theorem :-)

Proof. We have shown that this is a collection of possible solutions, but we haven't shown it's the general solution. To do this, there's a bunch of possible ways to do it, but we'll do it rigorously. There's two possible ways to go about this,

1. Using the method of change of variables, set

$$\begin{cases} \phi = x - ct \\ \psi = x + ct. \end{cases}$$

which yields

$$\begin{cases} \partial_\phi = \frac{1}{2}\partial_x - \frac{1}{2c}\partial_t \\ \partial_\psi = \frac{1}{2}\partial_x + \frac{1}{2c}\partial_t. \end{cases}$$

"Why did I choose this? Good question. Anyways, [...]" - Fabio. We can rewrite this as

$$\begin{cases} 2c\partial_\psi = c\partial_x - \partial_t \\ 2c\partial_\phi = c\partial_x + \partial_t. \end{cases}$$

From here, it follows that our PDE is just

$$\partial_\phi \partial_\psi v = 0,$$

so it turns out that

$$\partial_\psi v = F(\psi),$$

and thus

$$v = f(\psi) + g(\phi),$$

which is exactly what we want.

2. For the method of characteristics, we are going to use the following trick:

$$\begin{aligned} u_{tt} - c^2 u_{xx} &= 0 \\ \partial_t^2 u - c^2 \partial_x^2 u &= 0 \\ \frac{\partial_t^2}{\partial_x^2} u - c^2 u &= 0 && \text{(Dividing by } \partial_x^2) \\ \lambda^2 u - c^2 u &= 0 && (\lambda = \frac{\partial_t}{\partial_x}) \\ \lambda^2 = c^2 &\implies \lambda = \pm c. && \text{(Dividing by } u. \text{ Wait, huh?)} \end{aligned}$$

This *does* work formally, although not for the (tongue-in-cheek) reasons I wrote. Fabio didn't formalize it himself, but my best guess is that the third line is using the chain rule, then the fifth line just uses the fact that if λ^2 and c^2 agree on every function, then they must be the same operator.

□

Theorem 7.5 (D'Alembert's Formula). Consider the IVP for the wave equation.

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 \\ u(0, x) = \phi(x) \\ u_t(0, x) = \psi(x). \end{cases}$$

Then

$$u(t, x) = \frac{1}{2} (\phi(x + ct) + \phi(x - ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(s) ds.$$

[Fabio didn't write the formula until the end of the proof. I only wrote it earlier than he did for completeness.]

Proof. We know $u(t, x) = F(x + ct) + G(x - ct)$. From here, we will impose the initial conditions.

$$\begin{cases} u(0, x) = F(x) + G(x) = \phi(x) \\ u_t(0, x) = cF'(x) - cG'(x) = \psi(x), \end{cases}$$

so we have the equation

$$\begin{cases} F' + G' = \phi \\ F' - G' = \frac{1}{c}\psi \end{cases}$$

thus solving this yields

$$2F' = \phi' + \frac{1}{c}\psi \quad 2G' = \phi' - \frac{1}{c}\psi.$$

From here, we can determine that

$$\begin{cases} F = \frac{1}{2}\phi + \frac{1}{2c} \int_0^x \psi(s)ds + A \\ G = \frac{1}{2}\phi - \frac{1}{2c} \int_0^x \psi(s)ds + B. \end{cases}$$

Since we need $F + G = \phi$, it follows that $A + B = 0$. We can therefore do

$$u(t, x) = \frac{1}{2}\phi(x + ct) + \frac{1}{2c} \int_0^{x+ct} \psi(s)ds + \frac{1}{2}\phi(x - ct) - \frac{1}{2c} \int_0^{x-ct} \psi(s)ds,$$

which yields a final formula of

$$u(t, x) = \frac{1}{2} (\phi(x + ct) + \phi(x - ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(s)ds.$$

Notice that this proof simultaneously shows uniqueness, because at each step of the proof, our hand was forced. There was no step along the way at which we could've deviated and gotten a different final solution, so it must be the unique solution. We can show uniqueness independently as well, of course, this is just a quick way to verify it. $\square$

We can deduce many properties of the solution of the wave equation from this formula. We can also deduce some abstractly.

Example 7.6. Consider the system

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 \\ u(0, x) = e^{-x^2} \\ u_t(0, x) = 0. \end{cases}$$

Then this is just solved by

$$u(t, x) = \frac{e^{-(x+ct)^2} + e^{-(x-ct)^2}}{2}.$$

Example 7.7. Consider the system

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 \\ u(0, x) = \chi_{[-1,1]}(x) \\ u_t(0, x) = 0 \end{cases}$$

By D'Alembert, this is just

$$u(t, x) = \frac{1}{2}\chi_{[-1,1]}(x + ct) + \frac{1}{2}\chi_{[-1,1]}(x - ct).$$

He then graphed out how this would look as time evolved.

This models the propagation of signals, or spacetime. If we look at the supports of a function subject to the wave equation over time, it will resemble this graph, which is why we can use it to model the above. A plot of it reveals that after a finite time, the signal will leave where it originated ($\frac{1}{c}$ seconds). We see that the edges of where the signal have propagated to form a sort of “light cone,” which represents all the points in spacetime that the signal could have reached after the initial transmission. This is because there’s a finite speed of propagation. Given the initial conditions, nothing can move faster than c , so we can call c the speed of light, or something like that. It’s interesting stuff, but probably not required for the class lol.

§8 Day 8: Wave Equation (Sep. 29, 2025)

The wave equation is time-reversible.

Theorem 8.1 (D'Alembert's Equation). The unique solution to the system

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 \\ u(x, 0) = \phi(x) \\ u_t(x, 0) = \psi(x) \end{cases}$$

is given by

$$u(x, t) = \frac{1}{2} [\phi(x + ct) + \phi(x - ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(s) ds.$$

Remark 8.2. We assumed some regularity for the data ϕ, ψ (for example, $\phi \in C^1$). Nevertheless, the formula makes sense for rougher ϕ and ψ .

Remark 8.3 (Trust me on this). 1. When deriving identities or inequalities, we are allowed to think our functions are very nice. This is because reasonable functions can approximate most functions we care about, and then we can take the limit. Usually this works. [The Stone-Weierstrass approximation theorem tells us that actually almost any collection of functions you can think of that's closed under pointwise addition and multiplication will be dense in this sense.]

2. If $\phi \in C^2$ and $\psi \in C^1$, then $u \in C^2$, which makes it a perfectly valid solution of our function. We call this a classical solution. But, there are ways to say that rougher functions solve the PDE. Not in the classical sense, in some other sense. Most things in the world aren't necessarily smooth. So for the Burgers' equation and such we will see solutions in some other sense.

Remark 8.4 (Finite Speed). If the data ϕ, ψ vanishes in $|x| > R$ (for some $R \in \mathbb{R}$), then $u(x, t)$ vanishes on the set

$$S = \{(t, x) : |x| > R + |ct|\}.$$

We can imagine this as drawing cones over every point in our initial data whose slope is $\frac{1}{c}$ along the boundary.

Let's choose a point (x_0, t_0) . Which points in the plane "care" about the value of u at this point? We call this the domain of influence of (x_0, t_0) . We also say this is the light cone of this region. It consists precisely of a (two-sided) cone centered around our point, with slope $\frac{1}{c}$. The one in the positive t direction is the forward light cone, and the one in the negative t direction is the backwards light cone.

We call the points on our initial data within the light cone of our point the domain of dependence, because outside of this region, the data would not have enough time to reach our point, and so the value of our point depends exclusively on the value of our function at this point.

Claim 8.5. The finite speed can be seen from the PDE alone without the explicit formula for the solution using "energy".¹¹

Definition 8.6. The energy density is given by

$$\sigma(x, t) = \frac{1}{2} (u_t^2 + c^2 |\nabla u|^2(x, t),$$

¹¹Isaac losing his mind rn

so we have that the total energy of u at a time t is just integrating σ over $\mathbb{R}^d$. This gives

$$E(t) = \int_{\mathbb{R}^d} \frac{1}{2} (u_t^2 + c^2 |\nabla u|^2) dx,$$

where we have that our wave is in $\mathbb{R} \times \mathbb{R}^d$. We say that u_t^2 is the kinetic energy, and $c^2 |\nabla u|^2$ is the potential energy.

Claim 8.7. $\dot{E}(t) \equiv 0$.

Proof.

$$\frac{d}{dt} E(t) = \int_{\mathbb{R}^d} \frac{d}{dt} \frac{1}{2} (u_t^2 + c^2 |\nabla u|^2) dx$$

Notice that the interchange of the limit and derivative imposes some requirements. Both because we need some sort of smoothness of the integrand, but also because of problems with our integral diverging. To solve this, we can assume that the data of our PDE has compact support—call it P .

$$\begin{aligned} &= \int_{\mathbb{R}^d} u_t u_{tt} + c^2 \nabla u \nabla u_t dx \\ &= \int_{\mathbb{R}^d} u_t u_{tt} - c^2 \Delta u u_t dx && \text{(Gauss)} \\ &= \int_{\mathbb{R}^d} u_t (u_{tt} - c^2 \Delta u) dx \\ &= 0. && \text{(Plug in Wave Equation)} \end{aligned}$$

Recall that the multi-variable wave equation is

$$u_{tt} - c^2 \Delta u = 0,$$

which is where this last line derives. For the application of Gauss to get this last part, we need some domain for which our functions are identically 0 over the boundary. Fortunately, since we assumed our support is compact¹² we can find a region for which this works. He remarked that we will need this type of argument a lot, especially for Elliptic PDEs, so we should make sure we are very comfortable with it. $\square$

¹²Fabio called it finite support, which I figure just means bounded. Of course this means its closure is compact, so it still works the same.

§9 Day 9: More on Wave Equation, Heat Equation (Sep. 29, 2025)

Let's look at the wave equation in $d + 1$ dimensions. This is

$$\begin{cases} u_{tt} - c^2 \Delta_x u = 0 & t \in \mathbb{R}, x \in \mathbb{R}^d \\ u(x, 0) = \phi(x) \\ u_t(x, 0) = \psi(x) \end{cases}$$

We defined the energy density $\sigma(x, t) = \frac{1}{2}|u_t|^2 + \frac{1}{2}c^2|\nabla u|^2$, and the total energy at the time t as

$$E(t) = \int_{\mathbb{R}^d} \sigma(x, t) dx.$$

We proved for sufficiently nice solutions that $\dot{E} \equiv 0$.

Theorem 9.1 (Finite Speed of Propagation). If ϕ, ψ vanish for $|x| > R$ then $u(x, t) = 0$ for $|x| > R + ct$ where $t > 0$. If $d = 1$ this is just D'Alembert.

Proof. The past cone (the domain of dependence of (x_0, t_0)) is

$$|x - x_0| = c(t_0 - t). \quad (0 \leq t \leq t_0)$$

Taking the gradient of the function

$$F(x, t) = |x - x_0| - c(t_0 - t)$$

yields the unit outward normal to our cone C . It is precisely

$$\left(x - x_0, -c \frac{t - t_0}{|t - t_0|} \right) \frac{1}{\sqrt{1 + c^2}}.$$

Define $R = x - x_0$ and write

$$\hat{m} = \left(\frac{R}{|R|}, c \right) \frac{1}{\sqrt{1 + c^2}}$$

on C . Then we have that

$$\partial_t \sigma(x, t) - \nabla \cdot p(x, t) = 0,$$

where

$$\sigma(x, t) = \frac{1}{2}|u_t|^2 + \frac{1}{2}c^2|\nabla u|^2, \quad p(x, t) = c^2 u_t \nabla u.$$

This can be found just by plugging these in to the above and cancelling out. Call D the solid region of the cone between 0 and t . For the rest of this proof refer to [this desmos graph](#) for the sets. Then

$$\int_D \partial_t \sigma - \nabla \cdot p dV = 0,$$

but this is just the divergence (in both x and t) of the vector field $(-p, \sigma)$, so applying Gauss, this is

$$\int_{\partial D} (-p, \sigma) \cdot \hat{m} da,$$

where $\partial D = T \cup B \cup K$, where T is the top of our slice of the cone, B is the bottom slice, and K is the boundary connecting them (again, refer to our graph).

$$\begin{aligned}
 \int_T (-p, \sigma) \cdot (0, \dots, 0, 1) dx &= \int_T \sigma dx \\
 &= \frac{1}{2} \int_T u_t^2 + c^2 |\nabla u(t)|^2 dx \\
 \int_B (-p, \sigma) \cdot (0, \dots, -1) dx &= \frac{-1}{2} \int_B \psi^2 + c^2 |\nabla \phi|^2 dx \\
 \int_K (-p, \sigma) \cdot \hat{m} &= \int_K -p \hat{m}_x + \sigma \hat{m}_t \\
 &= \frac{1}{\sqrt{1+c^2}} \int_K -c^2 u_t \nabla u \cdot \frac{R}{|R|} + c \left(\frac{1}{2} u_t^2 + \frac{c^2}{2} |\nabla u|^2 \right) da \\
 \int_T \frac{1}{2} u_t^2 + \frac{c^2}{2} |\nabla u|^2 &\leq \int_B \frac{1}{2} \psi^2 + \frac{1}{2} c^2 |\nabla \phi|^2 - \int_K \dots
 \end{aligned} \tag{*}$$

We will show that our integral over K is positive to show that we can drop this term in the inequality. To do this, set $u_r = \nabla u \cdot \frac{R}{|R|}$, to represent the radial derivative of u . Then

$$\begin{aligned}
 \frac{1}{c} I &= -c u_t u_r + \frac{1}{2} u_t^2 + \frac{c^2}{2} |\nabla u|^2 \\
 &= \frac{1}{2} (u_t - c u_r)^2 + \frac{c^2}{2} (|\nabla u|^2 - u_r^2) \\
 &= \frac{1}{2} (u_t - c u_r)^2 + \frac{c^2}{2} \left| \nabla u - u_r \cdot \frac{R}{|R|} \right|^2 \\
 &\geq 0.
 \end{aligned}$$

It follows that

$$\int_T \frac{1}{2} u_t^2 + \frac{c^2}{2} |\nabla u|^2 \leq \int_B \frac{1}{2} \psi^2 + \frac{c^2}{2} |\nabla \phi|^2,$$

so if $\phi, \psi = 0$ on B , then $\int_T u_t^2 + |\nabla u|^2 dx = 0$. This implies $u \equiv 0$ on T so $u \equiv 0$ on the whole cone. Also, $u(x_0, t_0)$ is zero if data is zero on the base of the past cone of (x_0, t_0) (i.e. the domain of dependence). $\square$

What we showed also implied finite speed property for $d+1$ dimensional wave equation. “No more cones for a while. We may have ellipses and spheres, but no cones. Well, maybe I’ll say one thing...” Think about how the energy inequality implies uniqueness of our solution. Also, think about how $\dot{E}(t) = 0$ implies uniqueness.

Now let’s start on the Heat/Diffusion equation in 1d. Recall this is

$$u_t - k u_{xx} = 0. \tag{k > 0}$$

We may assign initial data $u(x, 0) = \phi(x)$. We may also look at boundary conditions. Here, only take $t \geq 0$. For $t < 0$ the problem is generally ill-posed. Notice that if you change t to $-t$ this is not the same equation, because we are requiring $k > 0$.

We will solve it explicitly later. First, let’s look at some properties that can be obtained only from the equation.

Theorem 9.2 (Weak Maximum Principle). Consider the rectangle $0 \leq x \leq \ell$ and $0 \leq t \leq T$. Suppose we have a solution here. Namely, we have that $u_t = k u_{xx}$ in $(x, t) \in [0, \ell] \times [0, T] = R$. Then the max value for u in R is obtained on either $x = 0$, $x = \ell$ or $t = 0$.

Remark 9.3. The strong maximum principle says the max can only be on the boundaries mentioned (unless the function is constant). The weak maximum principle is provable easily, the strong maximum principle is very hard.

We will save the proof for next time. It's a basic trick in analysis. It's different in flavour from what we've done so far—no integration by parts or integrals at all.

We also have the min principle. Just apply the max principle to $-u$. We will also have this same idea for the elliptic equation.¹³

¹³There was something else at the end but I really had to pee so I wasn't paying attention at the very end. It was something about uniqueness in the heat equation and boundary conditions, but I figure he'll go over it again if it's important.

§10 Day 10: Heat/Diffusion Equation (Oct. 6th, 2025)

Theorem 10.1. Let u solve $u_t = ku_{xx}$, for $k > 0$ in the rectangle $R = [0, \ell] \times [0, T]$ with initial/boundary conditions

$$\begin{cases} u(x, 0) = \phi(x) & 0 \leq x \leq \ell \\ u(0, t) = g(t) & 0 \leq t \leq T \\ u(\ell, t) = h(t) & 0 \leq t \leq T. \end{cases}$$

Notice here that the $u(0, t)$ and $u(\ell, t)$ terms represent some sort of “forcing,” which represents either pumping in or removing heat or something of that sort.

Then we have that u attains its max somewhere along $B = \partial R \setminus \{(x, T) : 0 < x < \ell\}$.

Proof. If u has a max at a point p in the interior of R , then $u_t|_p = 0$ and $u_{xx}|_p \leq 0$. This is almost violating the PDE. The idea is to create a big of room.

Let $v(x, t) = u(x, t) + \varepsilon x^2$. Then v solves

$$v_t - kv_{xx} = u_t - ku_{xx} - 2\varepsilon k < 0,$$

if $\varepsilon > 0$. But then $v = \mu + \varepsilon x^2$, $\varepsilon > 0$ arbitrary cannot have an interior max Q , as otherwise, $v_t|_Q = 0$, $v_{xx}|_Q \leq 0$, which is a contradiction. Thus, v attains its max on the boundary.

Assume that v attains its max along the top. Specifically, suppose it attains a max at (x_0, T) . It follows that

$$\lim_{\delta \rightarrow 0} \frac{v(x_0, T) - v(x_0, T - \delta)}{\delta} \geq 0.$$

Then we still get a contradiction to $v_t - v_{xx} = 0$.

To show u attains its max along the boundary, let $M = \max_R u$. We have therefore that $\max_R v = \max_B v$. This implies

$$\max_R u + \varepsilon x^2 = \max_B u + \varepsilon x^2$$

and thus

$$\max_R u \leq \max_B u + \varepsilon \ell^2,$$

for ε arbitrary. This implies that $\max_R u \leq \max_B u$. □

Remark 10.2. We also have a minimum principle, as discussed last time. The PDE is linear, so just negate our solution and apply the above.

From here, uniqueness follows and also continuous dependence on data.

Say that $g = h = 0$. Then u_j solves with $u_j(x, 0) = \phi_j(x)$, $j = 1, 2$. Because $u_1 - u_2$ and $u_2 - u_1$ are solutions, we get

$$\max_R |u_1 - u_2| \leq \max_{0 \leq x \leq \ell} |\phi_1(x) - \phi_2(x)|.$$

This gives stability/continuous dependence in the sup norm.

Proposition 10.3. We can find the uniqueness/continuous dependence using the energy method. Define

$$E(t) = \frac{1}{2} \int_0^\ell u^2(x, t) dx.$$

Then we have

$$\begin{aligned}\dot{E}(t) &= \int_0^\ell u(x, t) u_t(x, t) dx \\ &= \int_0^\ell u k u_{xx} dx \\ &= -k \int_0^\ell (u_x)^2(x, t) dx \leq 0.\end{aligned}$$

Then this gives uniqueness and continuous dependence. Uniqueness because given u_1, u_2 solving, $v = u_1 - u_2$ satisfies

$$\frac{1}{2} \int_0^\ell v^2(x, t) dx \leq \frac{1}{2} \int_0^\ell v^2(x, 0) dx = 0,$$

so thus $u_1 = u_2$. Continuous dependence because given ϕ_1, ϕ_2 that solve, $v = u_1 - u_2$. Then we have by the same argument above that solutions depend continuously wrt the L^2 norm. Here, we have that the L^2 norm of ϕ is

$$\|\phi\|_{L^2} = \left(\int_0^\ell \phi^2(x) dx \right)^{\frac{1}{2}}.$$

Remark 10.4. Continuous dependence in ODEs is nice and simple, because the initial conditions are just points, so just take the standard Euclidean norm. Furthermore, since most norms on $\mathbb{R}^n$ are equivalent, we don't have to worry too much about which norm the dependence is continuous with respect to. For PDEs, it's not quite the same. The L^∞ norm (sup norm) and the L^2 norm are indeed not equivalent, so when we talk about continuous dependence, we have to specify which norm it depends continuously upon. That is, even if solutions are continuously dependent on the data (which is a function) with respect to the L^2 norm, it might not depend continuously on the data with respect to the L^∞ norm.

§11 Day 11: Heat/Diffusion Again (Oct. 8, 2025)

We will consider the following equation where $x \in \mathbb{R}$ and $t \geq 0$. We have a fixed parameter $k > 0$.

$$\begin{cases} u_t - ku_{xx} = 0 \\ u(x, 0) = \phi(x) \end{cases}$$

The goal is to find an explicit formula for the solution for the IVP above. We will eventually state the full theorem.

The idea is to use invariance properties and symmetries of our equation to reduce it to ODEs. Here are some nice invariances of our equation. These are only invariances of the PDE and not the initial data.

1. If $u(x, t)$ is a solution, then this implies that $u(x - y, t)$ is a solution, for all $y \in \mathbb{R}$.
2. If u is a solution, then $\partial_x u$, $\partial_t u$, $\partial_{xx} u$ are also solutions.
3. Linear combinations of solutions are solutions (superposition principle).
4. If $S = S(x, t)$ is a solution, then

$$\int_{\mathbb{R}} S(x - y, t) \psi(y) dy$$

is also a solution, for any nice enough ψ .

Remark 11.1. The general solutions of many linear PDEs have solutions in the form

$$\int_{\mathbb{R}} K(x - y, t) \phi(y) dy,$$

for some K called the fundamental solution. This is called the convolution of $K(\cdot, t)$ and ϕ .

5. If u is a solution, then $u(\sqrt{a}x, at)$ is a solution for all $a > 0$.

Remark 11.2 (By me, not Fabio). We are going to solve our PDE with initial conditions $u(x, 0) = \chi_{\{x \geq 0\}}$. As usual, I use χ for the indicator function.¹⁴ This is the distribution function for the Dirac delta function. For those unaware, the Dirac delta function is a generalized function δ such that

$$\int_a^b \delta(x) dx = \begin{cases} 1 & 0 \in [a, b] \\ 0 & \text{otherwise.} \end{cases}$$

Obviously this function doesn't actually exist because it would have to be infinite height at 0, but using measure theory this can be made perfectly formal. In some sense as well, we have that $\partial_x \phi = \delta$. Not in the classical sense of course, because ϕ is discontinuous, but we *would* have that an analog of the fundamental theorem of calculus holds:

$$\int_a^b \delta(x) dx = \phi(b) - \phi(a).$$

This is idea behind the Radon-Nikodym derivative, for those curious.

But why do we care about all this fancy stuff? Suppose that we have some function $\phi : \mathbb{R} \rightarrow \mathbb{R}$. Then it would follow that

$$\phi(x) = \int_{\mathbb{R}} \delta(x - y) \phi(y) dy.$$

¹⁴If you object, please DM me how to write `\mathbb{1}` for future.

So if we can set $K(x, t)$ to have $K(x, 0) = \delta(x)$ (whatever this means) and K solves our ODE (whatever *this* means), then by symmetry 4, we should have that the function

$$u(x, t) = \int_{\mathbb{R}} K(x - y, t) \phi(y) dy$$

solves our ODE, and also

$$u(x, 0) = \int_{\mathbb{R}} K(x - y, 0) \phi(y) dy = \int_{\mathbb{R}} \delta(x - y) \phi(y) dy = \phi(x).$$

In other words, u would satisfy the initial conditions $u(x, 0) = \phi(x)$.

So we will start by trying to solve the PDE for the (mostly normal) function $u(x, 0) = \chi_{\{x \geq 0\}}(x)$, then applying symmetry 2, we should get a solution for

$$v(x, 0) = u_x(x, 0) = \delta(x).$$

From here, we should have a formula to solve our PDE with any initial conditions. If we have done this, then we can retroactively verify it works, instead of having to formalize these middle steps. Alright, back to what Fabio said.

We will find the fundamental solution of $u_t - u_{xx} = 0$. Let's start by solving with initial conditions

$$u(x, 0) = \chi_{\{x \geq 0\}}.$$

By symmetry 5, we look for $Q(x, t) = g(p)$ with $p = \frac{x}{\sqrt{4t}}$. Calculating the PDE for Q yields

$$\partial_t Q = g'(p) \frac{x}{\sqrt{4t}} \left(\frac{-1}{2t} \right) = \frac{-1}{2t} p g'(p).$$

Also,

$$\begin{cases} \partial_x Q = g'(p) \frac{1}{\sqrt{4t}} \\ \partial_{xx} Q = g''(p) \frac{1}{4t} \end{cases}$$

Which means Q solves if

$$\begin{aligned} -\frac{1}{2t} p g'(p) - \frac{1}{4t} g''(p) &= 0 \\ v' + 2pv &= 0 & (v = g') \\ v(p) &= e^{-p^2} C_1. \end{aligned}$$

So it follows that for some $C_1, C_2 \in \mathbb{R}$,

$$g(p) = C_1 \int_{-\infty}^p e^{-q^2} dq + C_2,$$

and therefore we can solve the PDE with

$$Q(x, t) = C_1 \int_{-\infty}^{\frac{x}{\sqrt{4t}}} e^{-p^2} dp + C_2.$$

Now, we want to find C_1, C_2 for the given data ϕ . Formally, take $t \searrow 0$, because we want $\lim_{t \searrow 0} Q(x, t) = \phi(x)$. Then

$$\begin{aligned} \lim_{t \searrow 0} Q(x, t) &= C_1 \frac{\sqrt{\pi}}{2} + C_2 & (x > 0) \\ \lim_{t \searrow 0} Q(x, t) &= -\frac{C_1 \sqrt{\pi}}{2} + C_2 \left[\frac{\sqrt{\pi}}{2} \int_0^{\infty} e^{-p^2} dp \right] & (x < 0) \\ \begin{cases} C_1 \frac{\sqrt{\pi}}{2} + C_2 &= 1 \\ -C_1 \frac{\sqrt{\pi}}{2} + C_2 &= 0 \end{cases} \end{aligned}$$

and thus $C_2 = \frac{1}{2}$ and $C_1 = \frac{1}{\sqrt{\pi}}$. It follows that

$$Q(x, t) = \frac{1}{\sqrt{\pi}} \int_0^{\frac{x}{\sqrt{4t}}} e^{-p^2} dp + \frac{1}{2}$$

is a solution. Therefore $\partial_x Q = S(x, t)$ is a solution, and

$$S(x, t) = \frac{1}{\sqrt{4\pi t}} e^{-\frac{x^2}{4t}}.$$

This is just the Gaussian. From here, for any data ϕ ,

$$\frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} e^{-\frac{(x-y)^2}{4t}} \phi(y) dy$$

is a solution.

Claim 11.3. This is $u(x, t)$ solving

$$\begin{cases} u_t - u_{xx} = 0 \\ u(x, 0) = \phi(x) \end{cases}$$

for all given ϕ ($t \geq 0$).

Proof. The first part is satisfied (well, we will be more rigorous later) by our properties above. The second part we have also verified in some sense, but we will be more rigorous here. Our question is whether

$$u(x, t) = \frac{1}{\sqrt{4\pi t}} \int_{-\infty}^{\infty} e^{-\frac{(x-y)^2}{4t}} \phi(y) dy \rightarrow \phi(x) \quad (*)$$

as $t \searrow 0$. To show it does, notice that

$$\begin{aligned} &= \int_{-\infty}^{\infty} \partial_x Q(x-y) \phi(y) dy \\ &= \int_{-\infty}^{\infty} -\partial_y Q(x-y) \phi(y) dy \\ &= [-Q(x-y) \phi(y)]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} Q(x-y) \phi'(y) dy \\ &= \int_{-\infty}^{\infty} Q(x-y, t) \phi'(y) dy \end{aligned}$$

We assume that $\phi(\infty) = \phi(-\infty) = 0$ to make the boundary conditions cancel out. You should know (*) by heart.

Notice that (*) makes sense for all $t > 0$, when ϕ is not growing too fast at $y = \pm\infty$. This is because the first term (our Gaussian) becomes very very small, so it won't end up affecting the integral too much. Now, let's take

$$\begin{aligned} \lim_{t \searrow 0} u(x, t) &= \int_{-\infty}^{\infty} Q(x-y, t) \phi'(y) dy \\ &= \int_{-\infty}^x \phi'(y) dy = \phi(x) \end{aligned}$$

□

As such we say that $S(x, t)$ is the fundamental solution of the heat equation on $\mathbb{R}$. Actually, $S(x, t)$ is singular as $t \searrow 0$ if $x = 0$ [as you should expect by my remark]. But it's smooth in x and if we integrate over $\mathbb{R}$, then for all $t > 0$,

$$\int_{\mathbb{R}} \frac{1}{\sqrt{4\pi t}} e^{-\frac{x^2}{4t}} dx = 1.$$

and S solves $S_t - S_{xx} = 0$, when $t > 0$.

Remark 11.4. For the wave equation, singularities would be propagated, but for the heat equation, they are immediately smoothed out. Also, we have that the heat equation in some sense has infinite speed of propagation. To prove this, just look at the formula.

$$u(x, t) = \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} e^{-\frac{(x-y)^2}{4t}} \phi(y) dy.$$

If ϕ has compact support (also, suppose it's positive), then for any $x \in \mathbb{R}$, we can choose some $y \in \mathbb{R}$ such that $\phi(y)$ is nonzero, and although the integrand will be multiplied by $e^{-\frac{(x-y)^2}{4t}}$, this is strictly positive (albeit very small). Accordingly, every point on $\mathbb{R}$ will immediately be nonzero.

Theorem 11.5. Let ϕ be continuous and bounded. Then $(*)$ is a solution to

$$\begin{cases} u_t - u_{xx} = 0 \\ u(x, 0) = \phi \end{cases}$$

and moreover,

1. $u \in C^\infty(\mathbb{R} \times (0, \infty))$
2. $\lim_{t \searrow 0} u(x, t) = \phi(x)$ for all x .

Proof.

$$\begin{aligned} u(x, t) &= \int_{\mathbb{R}} \frac{1}{\sqrt{4\pi t}} e^{-\frac{z^2}{4t}} \phi(x - z) dz & (z = x - y) \\ &= \int_{\mathbb{R}} \frac{1}{\sqrt{4\pi t}} e^{-\frac{p^2}{4}} \phi(x - \sqrt{t} p) dp & (p = \frac{z}{\sqrt{t}}) \end{aligned}$$

We write convolutions as $f * g(x) = \int f(x - y)g(y)dy$. Now, let's argue that we can bring partial derivatives inside the integral.

$$\begin{aligned} u(x, t) &= \int_{-\infty}^{\infty} e^{-\frac{(x-y)^2}{4t}} \phi(y) dy \\ |u(x, t)| &\leq \int_{-\infty}^{\infty} \frac{e^{-\frac{(x-y)^2}{4t}}}{\sqrt{4\pi t}} |\phi(y)| dy \\ &\leq \sup_y |\phi(y)|, \end{aligned}$$

so we want to show that

$$\partial_x u(x, t) = \int_{-\infty}^{\infty} \partial_x S(x - y, t) \phi(y) dy.$$

To do this, think of this as a limit

$$\begin{aligned} \frac{\partial}{\partial x} \int_{-\infty}^{\infty} F(x, y) dy &\cong \lim_{m \rightarrow \infty} \int_{-\infty}^{\infty} F_m(x, y) dy \\ &= \int_{-\infty}^{\infty} \lim_{m \rightarrow \infty} F_m(x, y) dy \end{aligned}$$

provided $F_m(y)$ converges as $m \rightarrow \infty$ uniformly and this limit is absolutely integrable. $\square$

§12 Day 12: Heat/Diffusion Part 3 (Oct. 15, 2025)

Theorem 12.1. Let ϕ be bounded and continuous, $\phi = \phi(x)$, $x \in \mathbb{R}$. Then the formula $u(x, t) = \int_{\mathbb{R}} S(x - y, t)\phi(y)dy$, where

$$S(z, t) = \frac{1}{\sqrt{4\pi t}} e^{-\frac{z^2}{4t}},$$

is such that

1. $u \in C^\infty(\mathbb{R} \times (0, \infty))$,
2. u solves $u_t - u_{xx} = 0$.
3. $\lim_{t \searrow 0} u(x, t) = \phi(x)$ for all x .

Notably, even if ϕ is not smooth, u is.

Proof. We can write

$$u(x, t) = \int_{\mathbb{R}} S(z, t)\phi(x - z)dz.$$

This is just the convolution $S(\cdot, t) * \phi$. Now, if we set $z = \sqrt{t}p$, then this yields

$$u(x, t) = \int_{\mathbb{R}} e^{-\frac{p^2}{4}} \phi(x - \sqrt{t}p)dp.$$

Notice that $|u(x, t)| \leq \sup_y |\phi(y)|$.

1. To prove that $u \in C^\infty(\mathbb{R} \times (0, \infty))$, first we will state a nice general lemma.

Lemma 12.2.

$$\lim_{m \rightarrow \infty} \int_{\mathbb{R}} F_m(y)dy = \int_{\mathbb{R}} \lim_{m \rightarrow \infty} F_m(y)dy,$$

provided that $\lim F_m(y)$ exists for all $y \in \mathbb{R}$, this limit is absolutely integrable, and $F_m \Rightarrow F$.

From here, notice that if $F_m(x) = \frac{u(x+h_m, t) - u(x, t)}{h_m}$, then applying this to suitable sequences h_m yields that

$$\frac{\partial}{\partial x} u(x, t) = \int_{\mathbb{R}} \frac{\partial}{\partial x} S(x - y, t)\phi(y) dy$$

To do this, let us check our conditions. For absolute integrability,

$$\begin{aligned} \int_{\mathbb{R}} \frac{\partial}{\partial x} S(x - y, t)\phi(y) dy &= \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} -\frac{x - y}{2t} e^{-\frac{(x-y)^2}{4t}} \phi(y) dy \\ &= \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} \frac{1}{2\sqrt{t}} e^{-p^2} p \phi(x - p\sqrt{t}) dp. \quad \left(\frac{x-y}{\sqrt{t}} = p\right) \\ &\Rightarrow \left| \int_{\mathbb{R}} \frac{\partial}{\partial x} S(x - y, t)\phi(y) dy \right| \leq \frac{C}{t} \sup |\phi|. \end{aligned}$$

This proves that this middle part is absolutely integrable. Also, by Taylor's formula,

$$\begin{aligned} \left| \frac{S(z + h_m, t) - S(z, t)}{h_m} - \frac{\partial}{\partial z} S(z, t) \right| &= \left| \frac{\partial}{\partial z} S(z + \xi, t) - \frac{\partial}{\partial z} S(z, t) \right| \\ &\quad \text{(For some } 0 \leq |\xi| \leq |h_m|) \\ &= |\xi| \cdot |S''(z + \xi_0, t)| \quad (0 \leq |\xi_0| \leq |\xi|) \\ &\leq |h_m| \sup |S''(z, t)| \end{aligned}$$

and thus we have that our convergence is uniform. It follows that we can interchange the integral and limit. It follows that the first derivative of u exists. We can do similar proofs to show it is C^k for any k , so it is C^∞ .

2. In the proof of (i) we showed that

$$\frac{\partial^2}{\partial x^2} u(x, t) = \int_{\mathbb{R}} \frac{\partial^2}{\partial x^2} S(x - y, t) \phi(y) dy.$$

We also showed that

$$\frac{\partial}{\partial t} u(x, t) = \int_{\mathbb{R}} \frac{\partial}{\partial t} S(x - y, t) \phi(y) dy$$

thus, since S solves the heat equation for $t > 0$,

$$u_t - u_{xx} = 0.$$

3. We want to show that $\lim_{t \searrow 0} u(x, t) = \phi(x)$.

$$u(x, t) - \phi(x) = \int_{\mathbb{R}} \frac{1}{\sqrt{4\pi}} e^{-\frac{p^2}{4}} \phi(x - \sqrt{t}p) - \frac{1}{\sqrt{4\pi}} e^{-\frac{p^2}{2}} \phi(x) dp,$$

so we need to show that

$$\lim_{t \searrow 0} \int_{\mathbb{R}} e^{-\frac{p^2}{4}} [\phi(x - \sqrt{t}p) - \phi(x)] dp = 0.$$

To do this, we can just use the continuity of ϕ . It $\sqrt{t}p \ll 1$, this should be roughly 0 close to x , and $e^{-\frac{p^2}{4}}$ makes things small elsewhere. Specifically,

$$\int_{\mathbb{R}} e^{-\frac{p^2}{4}} [\phi(x - \sqrt{t}p) - \phi(x)] dp = I_1 + I_2,$$

where we have that

$$\begin{aligned} I_1 &= \int_{|p\sqrt{t}| < \delta} e^{-\frac{p^2}{4}} [\phi(x - \sqrt{t}p) - \phi(x)] dp \\ I_2 &= \int_{|p\sqrt{t}| \geq \delta} e^{-\frac{p^2}{4}} [\phi(x - \sqrt{t}p) - \phi(x)] dp. \end{aligned}$$

Fix $\varepsilon > 0$ arbitrary. We want to show that there exists $t_0 > 0$ such that for all $t \leq t_0$ we have that

$$|I_1| < \frac{\varepsilon}{2}, \quad |I_2| < \frac{\varepsilon}{2}.$$

To do this, let δ be such that

$$|\phi(x - y) - \phi(x)| < \frac{\varepsilon}{2}, \quad |y| < \delta.$$

a) To estimate I_1 , notice that

$$\begin{aligned} |I_1| &\leq \int_{|p\sqrt{t}| < \delta} e^{-\frac{p^2}{4}} |\phi(x - \sqrt{t}p) - \phi(x)| dp \\ &\leq \int_{\mathbb{R}} e^{-\frac{p^2}{4}} \frac{\varepsilon}{2\sqrt{4\pi}} dp \leq \frac{\varepsilon}{2}. \end{aligned}$$

b) To estimate I_2 ,

$$\begin{aligned} |I_2| &\leq \left(2 \sup_{y \in \mathbb{R}} \phi(y) \right) \int_{|p\sqrt{t}| \geq \delta} e^{-\frac{p^2}{4}} dp \\ &= \left(2 \sup_{y \in \mathbb{R}} \phi(y) \right) \int_{|p| < \frac{\delta}{\sqrt{t}}} e^{-\frac{p^2}{4}} dp \\ &= 4 \sup |\phi| \int_{\frac{\delta}{\sqrt{t}}}^{\infty} e^{-\frac{p^2}{4}} dp. \end{aligned}$$

Given ε and δ , pick t_0 small enough such that

$$\int_{\frac{\delta}{\sqrt{t_0}}}^{\infty} e^{-\frac{p^2}{4}} dp < \frac{\varepsilon}{8 \sup |\phi|},$$

so we are done. □

Remark 12.3. We always said that $u(x, t) = S(\cdot, t) * \phi$. This is a solution to the IVP

$$\begin{cases} u_t = u_{xx} & t > 0 & x \in \mathbb{R} \\ u = \phi & t = 0 & x \in \mathbb{R}. \end{cases}$$

We said this because it's not unique. In particular, there exists u solving

$$\begin{cases} u_t = u_{xx} & t > 0 \\ u(x, 0) = 0, \end{cases}$$

but with $u \not\equiv 0$. At the same time, under some conditions (see homework) we can get uniqueness.

Remark 12.4. $u_t = u_{xx}$ is ill-posed for $t < 0$. To see this, consider the data

$$\frac{1}{m} \sin(mx) = \phi_m(x).$$

Then the solution is

$$u_m(x, t) = \frac{1}{m} \sin(mx) e^{-m^2 t}.$$

If $t = -\varepsilon$, then $u_m(x, -\varepsilon) = \frac{1}{m} \sin(mx) e^{\varepsilon m^2}$, so there's no continuous dependence on data.

Example 12.5 (Diffusion with a source). This is the non-homogeneous case.

$$u_t - u_{xx} = f(x, t).$$

We'll say that $t > 0$, and that f has to be nice enough such that the following stuff still works. We want to find a solution using Duhamel's formula/operator method. To do this, we want to "think of the pde as an ode." To do this, recall how Duhamel works.

$$\begin{cases} u_t + Au = f \\ u(0) = \phi, \end{cases}$$

where A is a matrix (linear operator). Use integrating factors.

$$v = e^{At}u \implies v_t = e^{At}f.$$

This yields that

$$v(t) = \underbrace{v(0)}_{\phi} + \int_0^t e^{As} f(s) ds$$

$$u(t) = e^{-At} \phi + \int_0^t e^{-A(t-s)} f(s) ds.$$

In the ODE case, $e^{-At}\phi$ is the solution of the homogeneous problem. In the PDE case, this is given by

$$e^{-At}\phi \leftrightarrow S_t * \phi.$$

We can also write this as $e^{t\partial_x^2}\phi$ [I do hate this, sorry Fabio]. Thus, writing our Duhamel's formula for PDEs yields

$$u(x, t) = S_t * \phi + \int_0^t S_{t-s}(\cdot) * f(\cdot, s) ds.$$

We will work with this more next time.

§13 Day 13: Heat/Diffusion Part 4 (Oct. 15, 2025)

Midterm: everything we covered in class up to and including section 3.5.

I was a bit late (why have you forsaken me TTC) so I missed the start. Notes were harder to infer because I missed the critical initial explanation. I've done my best to fill it in.

He had rewrote the formula for solutions of IVP. I presume before I arrived he discussed that we would apply Duhamel's to the case of ODEs, and then show that it extends naturally to the case of PDEs. I believe the correspondence between the PDEs and ODEs comes from essentially just ignoring the x derivatives. So the heat equation $u_t - u_{xx} = f$ gives $u_t - u = f$. Likewise, the wave equation $u_{tt} - u_{xx} = f$ gives $u_{tt} - u = f$. It is kind of intuitive that this should work, but I did not catch Fabio's justification of it.

With that aside, we started with the heat equation. Use Duhamel's on

$$u'(t) = u(t) + f(t)$$

to get $u(t) = e^t \phi + \int_0^t e^{t-s} f(s) ds$.

Translated to the PDE, in the above, the $e^t \phi$ gives the solution of homogeneous problem at time t . We solved the homogeneous heat equation as

$$\frac{e^{-\frac{x^2}{4t}}}{\sqrt{4\pi t}} * \phi,$$

where $*$ is convolution, so we would hope that in the Duhamel's formula, this substitutes for $e^t \phi$. But notice that actually

$$\begin{cases} u_t = u_{xx} + f(x, t) \\ u(x, 0) = \phi(x) \end{cases}$$

is solved by

$$u(t) = S(t)\phi + \int_0^t S(t-s)ds,$$

where

$$S(t)g = \frac{e^{-\frac{x^2}{4t}}}{\sqrt{4\pi t}} * g,$$

so indeed we see that it is the case.

This is what the book called using the “operator method.”

Let's do the same thing for the wave equation. I strongly recommend you check out page 77 and 78 of Strauss. I couldn't follow what Fabio was doing.

$$\begin{cases} u_{tt} - u_{xx} = f(x, t) \\ u(x, 0) = g(x) \\ u_t(x, 0) = h(x) \end{cases}$$

The ODE analogue is

$$\begin{cases} \ddot{u} + Bu = f(t) \\ u(0) = g \\ \dot{u}(0) = h \end{cases}$$

(where g and h are constants—this is the notation he used). Solving this requires we have some information about B . So let's assume $B > 0$. Then, we can solve this as roughly looking like sin and cos. Specifically, the solution of homogeneous problem is

$$\cos(Bt)g + \frac{\sin(Bt)}{B}h.$$

For the non-homogeneous problem, it's

$$u(t) = \cos(Bt)g + \frac{\sin(Bt)}{B}h + \int_0^t \frac{\sin(B(t-s))}{B} f(s)ds.$$

Notice that this is Duhamels, where $S(t) = \frac{\sin(Bt)}{B}h$. Specifically

$$u(t) = S'(t)g + S(t)\psi + \int_0^t S(t-s)f(s)ds.$$

Comparing with D'Alembert's, the homogeneous solution $\frac{\sin(Bt)}{B}h$ looks unrecognizable.

We will see that¹⁵

$$\cos(Bt)g \cong \frac{1}{2}(g(x+t) + g(x-t))$$

and

$$\frac{\sin(Bt)}{B}h \cong \frac{1}{2} \int_{x-t}^{x+t} h(y)dy$$

Later we'll see why $\cos(i\partial_x t)g = \frac{1}{2}(g(x+t) + g(x-t))$. [He mentioned pset 5, but I forget what it was regarding.]

From here, Duhamel's gives

$$\frac{1}{2} \int_0^t \int_{x-(t-s)}^{x+(t-s)} f(y,s)dyds = \frac{1}{2} \iint_{(x,t)} f dS.$$

[We have to prove this formula of course, but he didn't do this in class.]

Odd even extension to solve heat/wave on half-line. This is assigned reading.

Fabio then asked if we want him to manually write out the solution for heat with source on the whole line. The class said yes. Here it is (assuming he wrote it right and I copied it right).

$$u(x,t) = \int_{-\infty}^{\infty} \frac{e^{-\frac{(x-y)^2}{4t}}}{\sqrt{4\pi t}} \phi(y)dy + \int_0^t \int_{-\infty}^{\infty} e^{-\frac{(x-y)^2}{4(t-s)}} \cdot \frac{1}{\sqrt{4\pi(t-s)}} f(y,s)dyds.$$

Now, we just have to check that this is correct. A posteriori we have to show it makes sense. Of course we got here from theory so we know it is correct, but we still have to prove it.

¹⁵He did not write $\cong$, he just wrote $=$. This pains me, so I did not.

§14 Day 14: Conserved Quantities and Fourier Series (Oct. 22, 2025)

We talked about conserved quantities/energy. For example, energy in the wave equation and momentum in the wave equation. But we found both of these based on physical intuition. How do we find conserved quantities?

One way is that invariances of the equation (technically the Lagrangian) and conserved quantities have a correspondence (Noether's Theorem).

Connection back to ODEs is that we use the principle of least action [this is from classical mechanics and the calculus of variations], and then if the action is invariant under symmetries, then applying these symmetries to our solution will preserve their satisfying the principle of least action, so they will also be solutions.

Let's use a different method from what we previously had to prove that energy is conserved by the wave equation.

Example 14.1. Recall that the wave equation is $u_{tt} - u_{xx} = 0$. From here, we're going to basically guess a factor that should make things invariant.

$$\int_{-\infty}^{\infty} (au + bu_x + cu_t)(u_{tt} - u_{xx})dx = 0$$

But we're not actually guessing at *random* what this factor is. We can cancel out some factors based on the symmetries of the equations.

$$\begin{aligned} & \int_{-\infty}^{\infty} u_t u_{tt} - u_t u_{xx} dx \\ &= \int_{-\infty}^{\infty} \partial_t \frac{u_t^2}{2} + \partial_x u_t u_x dx & \text{(IBP)} \\ &= \frac{d}{dt} \int_{-\infty}^{\infty} \frac{u_t^2}{2} + \frac{u_x^2}{2} dx \end{aligned}$$

Specifically, if $u(x, t) = u(x, t + h)$, then taking the derivative with respect to h yields

$$\left. \frac{\partial}{\partial h} \right|_{h=0} u(x, t + h) = u_t.$$

So using our Lagrangian, this motivates the choice of u_t .

From here, we will start something completely different. We haven't yet solved our wave and heat equations on a finite interval. Rather, we've only done it on the whole line. The goal is to solve it on some interval $x \in [0, \ell]$, and $t \in \mathbb{R}$ with the conditions

$$\begin{cases} u_{tt} - u_{xx} = 0 \\ u(x, 0) = \phi(x) \\ u_t(x, 0) = \psi(x) \\ \text{Boundary Conditions at } x = 0, \ell \end{cases}$$

and a similar initial value problem for $u_t - u_{xx}$ with boundary conditions on an interval. [This method is going to be similar in theory to how we solved the heat equation on the whole line, where we find one particular solution, then combine it with itself in ways such that they approximate our initial conditions well.]

To do this, we will look for a solution of the wave equation in separated form

$$u(x, t) = X(x)T(t),$$

and we will impose that it solves the equation (only the equation, not the boundary condition). But this uses quite a clever trick [in my opinion—Fabio said it's boring].

$$X(x)T''(t) = c^2 X''(x)T(t)$$

$$\frac{X''(x)}{X(x)} = \frac{T''(t)}{c^2 T(t)}$$

thus since the left side is a function of x and the right side is a function of t , they must both be constant. We will assume (and justify later) that the constant is negative. This gives

$$\frac{X''}{X} = \frac{T''}{c^2 T} = -\beta^2.$$

For later we will say $\beta^2 = \lambda$. But then if $X'' = -\beta^2 X$ and $T'' = -c^2 \beta^2 T$, then we have that

$$\begin{cases} X(x) = C \cos(\beta x) + D \sin(\beta x) \\ T(t) = A \cos(c\beta t) + B \sin(c\beta t). \end{cases}$$

Now, we'll impose the Dirichlet boundary conditions (later we will use Neumann, Mixed and Robin boundary conditions). That is, $u(0, t) = 0 = u(\ell, t)$. It follows that $X(0)T(t) = 0 = X(\ell)T(t)$ so $X(0) = 0 = X(\ell)$, so

$$\begin{cases} X(0) = C = 0 \\ X(\ell) = D \sin(\beta \ell) = 0 \end{cases}$$

But we don't want to set $D = 0$ or else our solution sucks [that's the technical term] so we instead use this to find $\beta = \frac{\pi n}{\ell}$.

Remark 14.2. We found that $\beta = \frac{\pi n}{\ell}$. But $-\beta^2$ is an eigenvalue for the problem

$$\begin{cases} X'' = (-\beta^2)X \\ X(0) = 0 = X(\ell). \end{cases}$$

and so what we've done here is find the eigenfunction $D_n \sin\left(\frac{\pi n}{\ell}x\right)$.

Therefore we found that

$$u_n(x, t) = \sin\left(\frac{\pi n}{\ell}x\right) \left(A_n \cos\left(c\frac{\pi n}{\ell}t\right) + B_n \sin\left(c\frac{\pi n}{\ell}t\right)\right)$$

In particular, we have that linear combinations of these solutions will be a solution. Further, even an infinite series will be a solution, assuming the convergence is nice and yields a nice enough function.

What is left is that we have to make our function satisfy the initial conditions ϕ and ψ . So given a solution of the form

$$\sum_{m \geq 1} \sin\left(\frac{\pi m x}{\ell}\right) \left(A_m \cos\left(\frac{c\pi m}{\ell}t\right) + B_m \sin\left(\frac{c\pi m}{\ell}t\right)\right) \quad (*)$$

If we impose

1. $u(x, 0) = \phi(x)$, then we have that

$$\phi(x) = \sum_{n \geq 1} A_n \sin\left(\frac{\pi n x}{\ell}\right) = \phi(x)$$

so if ϕ has a Fourier series representation as above for a specific A_m , we have found our A_m .

2. $u_t(x, 0) = \psi$, then we have that

$$\psi(x) = \sum_{n \geq 1} \sin\left(\frac{\pi n x}{\ell}\right) \cdot \frac{c\pi n}{\ell} B_n.$$

These both follow just by plugging in $(*)$ into these equations in the place of u . In summary, if everything works out nicely, then we have solved initial + dirichlet boundary value problem for the wave, provided that these sums work nicely.

The only consideration remaining is to show that the constant from separating variables is negative. So recall that in the separation of variables, we got the eigenvalue problem

$$\begin{cases} X'' = kX \\ X(0) = 0 = X(\ell). \end{cases}$$

Let k be a constant. We said that our constant k should be negative. Why? Well if our constant is 0, then our solution is linear, and therefore to satisfy the boundary conditions, it must be identically 0. If the constant is positive, then we end up with hyperbolic sine and cosine.

$$X(x) = ae^{\gamma x} + be^{-\gamma x}.$$

But if $X(0) = a + b = 0$ and $X(\ell) = ae^{\gamma\ell} + be^{-\gamma\ell} = 0$, then we have that

$$ae^{2\gamma\ell} + b = 0,$$

so $e^{2\gamma\ell} = 1$, which contradicts $\gamma > 0$. Thus, our only possible solutions are $\gamma < 0$ and the trivial case of $\gamma = 0$.

Next we'll solve Neumann boundary conditions. The book covers Robin and Mixed conditions in great detail. These are $u_x(0, t) = 0$ and $u_x(\ell, t) = 0$. This boils down to basically the same thing, just with a different eigenvalue problem.

§15 Day 15: Boundary and IVPs and Fourier Series (Nov. 3, 2025)

Happy birthday to my Dad (not Fabio's). Midterm in class on Wednesday (Nov. 5 i think?) if you weren't aware. Usual room and all that. Also a review session TODAY (Nov. 3) from 5 to 7. We get to meet the one and only Matt.

Last time we used "formal" calculations to write down solutions for the problems

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 & 0 < x < \ell, t \in \mathbb{R} \\ u(0, t) = 0 = u(\ell, t) & t \in \mathbb{R} \\ u(x, 0) = \phi(x) & u_t(x, 0) = \psi(x) \quad 0 < x < \ell \end{cases} \quad (\text{W})$$

which is the wave on an interval with Dirichlet boundary conditions (the second line), and

$$\begin{cases} u_t = k u_{xx} \\ u(0, t) = 0 = u(\ell, t) \\ u(x, 0) = \phi(x) \end{cases} \quad (\text{H})$$

which is the heat equation on an interval. We can consider boundary conditions other than the Dirichlet boundary conditions as well.

To solve the Dirichlet boundary conditions, using separation of variables and solving the eigenvalue problem

$$\begin{cases} -X'' = \lambda X & 0 < x < \ell \\ X(0) = 0 = X(\ell) \end{cases}$$

yields an eigenfunction X for the eigenvalue λ of the system.

Via this method, we arrived at the solution for the Dirichlet boundary conditions

$$u(x, t) = \sum_{n \in \mathbb{Z}} \left[A_n \cos\left(\frac{cn\pi t}{\ell}\right) + B_n \sin\left(\frac{cn\pi t}{\ell}\right) \right] \underbrace{\sin\left(\frac{n\pi x}{\ell}\right)}_{X_n(x)}.$$

Using the same method as last time, we can also solve the heat equation with

$$u(x, t) = \sum_{n \in \mathbb{Z}} A_n e^{-k\left(\frac{n\pi}{\ell}\right)^2 t} \underbrace{\sin\left(\frac{n\pi x}{\ell}\right)}_{X_n(x)}.$$

Notice that in these two solutions, our X_n is the same. This is because in the first step, look at $u(x, t) = X(x)T(t)$ to get that when we plug into $u_t = k u_{xx}$,

$$XT' = kX''T \implies \frac{X''}{X} = \frac{T'}{kT} = -\lambda,$$

where $\lambda \in \mathbb{R}_{>0}$ is a constant. We justified why this constant must be negative last time. Since the $\frac{X''}{X} = -\lambda$ is the same in both, this is why our X_n is the same between questions.

But what can λ possibly be? Well we need X to satisfy

$$\begin{cases} -X'' = \lambda X \\ X(0) = 0 = X(\ell) \end{cases}$$

for u to satisfy our boundary value problem. Thus, $\lambda = \left(\frac{n\pi}{\ell}\right)^2$. This implies $X(x) = \sin\left(\frac{n\pi}{\ell}x\right) = \sin\left(\sqrt{\lambda_n}x\right)$. We also have to solve for $T(t)$, but

$$T' = -k\lambda T \implies T(t) = Ae^{-k\lambda t} = Ae^{-k\left(\frac{n\pi}{\ell}\right)^2 t}.$$

This yields our solution of (H) above.

Now, let's solve the Schrödinger equation for fun. Recall that this is $u_t = iu_{xx}$. "It's like the heat, with $k \rightarrow i$." He then shouted loudly " k STILL HAS TO BE REAL!" which scared me. Nevertheless, if we just naively plug in to the formula for the heat solution, we get a solution to the Schrödinger equation

$$u(x, t) = \sum_{n \in \mathbb{Z}} A_n e^{-i\left(\frac{n\pi}{\ell}\right)^2 t} \sin\left(\frac{n\pi x}{\ell}\right).$$

This solution is found by literally replacing every k with an i in our proof of the solution of the heat equation.

We can get uniqueness using the energy for the wave and heat equation on an interval. Of course for the Schrödinger equation we have to be more careful because it's complex valued. The Schrödinger equation has conserved mass

$$M(t) = \frac{1}{2} \int_0^\ell |u(x, t)|^2 dx.$$

Next week we will need to find the coefficients A_m and B_m , which will lead us to Fourier series.

§16 Day 16: Boundary Value Problems with a view towards Fourier Series (Nov. 10, 2025)

We did separation of variables to find solutions to initial and boundary value problems for the heat, wave, and Schrödinger equation. Our boundary conditions can be

1. Dirichlet, where $u(x, 0) = u(\ell, 0) = 0$
2. Neumann, where $u_x(0, t) = u_x(\ell, t) = 0$
3. Robin (we will skip—but he might give up problems on it).

We also have initial conditions of $u(x, 0) = \phi(x)$, where $0 \leq x \leq \ell$. We did Dirichlet boundary conditions last time, so let's do Neumann boundary conditions. In this case, we will consider the wave equation $u_{tt} = c^2 u_{xx}$.

Example 16.1. Suppose $u_x(0, t) = u_x(\ell, t) = 0$. Now, let's look for a separable solution. $u(x, t) = X(x)T(t)$. Plugging into our PDE yields

$$\frac{X''(x)}{X(x)} = -\lambda, \quad \frac{T''(t)}{c^2 T(t)} = -\lambda,$$

where λ is a positive constant. Thus,

$$\begin{cases} -X'' = \lambda X & 0 < x < \ell \\ X'(0) = 0 = X'(\ell) \end{cases} \quad (\text{EP})$$

If $\lambda = \beta^2$, then the solution to this ODE (without the boundary conditions) is

$$X(x) = A \cos(\beta x) + B \sin(\beta x),$$

for some constants A, B . From here, differentiating yields

$$\begin{cases} X'(0) = \beta B \cos(0) - A\beta \sin(0) = 0 \\ X'(\ell) = \beta B \cos(\beta \ell) - A\beta \sin(\beta \ell) = 0, \end{cases}$$

so therefore $B = 0$, so as long as $A \neq 0$, $\beta \ell = n\pi$ for some $n \in \mathbb{N}$. Thus, $\beta = \frac{n\pi}{\ell}$. As usual, always think of X as an eigenvector, and λ as an eigenvalue.

It follows that the eigenvalue problem (EP) has solutions

$$\lambda_n = \left(\frac{n\pi}{\ell}\right)^2, \quad X_n(x) = \cos\left(\frac{n\pi}{\ell}x\right).$$

We will solve for T now. The eigenvalue problem for T is

$$-T'' = \lambda c^2 T,$$

and so solving gives

$$u(x, t) = \sum_{n \geq 1} \cos\left(\frac{n\pi}{\ell}x\right) \left[A_n \cos\left(\frac{n\pi c}{\ell}t\right) B_n \sin\left(\frac{n\pi c}{\ell}t\right) \right].$$

But something is missing here. This is correct, but not fully complete. This is because we only considered $\lambda > 0$. This was justified for the Dirichlet boundary conditions, because the other cases of λ yielded no solutions. But let's ask if the same is the case here. When $\lambda < 0$, then we have $X'' = aX$ for $a > 0$, but then it's impossible to satisfy

our boundary conditions $X'(0) = X'(\ell) = 0$ (if X is nontrivial). On the other hand, if $\lambda = 0$, then a linear function satisfies $X'' = 0$. To make it satisfy our boundary conditions $X'(0) = X'(\ell) = 0$, we have to have X be constant. This yields an ode $T'' = 0$, which would give $T(t)$ is linear. Thus, our general solution would be

$$u(x, t) = \left(\sum_{n \geq 1} \cos\left(\frac{n\pi}{\ell}x\right) \left[A_n \cos\left(\frac{n\pi c}{\ell}t\right) B_n \sin\left(\frac{n\pi c}{\ell}t\right) \right] \right) + A_0 + B_0 t.$$

Example 16.2 (Mixed Boundary Conditions). The mixed boundary conditions are

$$X(0) = 0 = X'(\ell).$$

These are distinct from Robin boundary conditions, which are $au(0, t) + bu_x(0, t) = 0$. For Schrödinger, Heat, and Wave, you always end up with the same ODE for X , so you should know the general solution.

$$\begin{cases} -X''(x) = \lambda X(x) \\ X(0) = 0 = X'(\ell). \end{cases}$$

But then this is solved by (assuming $\lambda = \beta^2$)

$$X(x) = A \cos(\beta x) + B \sin(\beta x).$$

Now, this implies that $A = 0$, and so if $X'(\ell) = 0$, this implies

$$0 = X'(\ell) = B\beta \cos(\beta\ell),$$

so assuming $B \neq 0$, we have $\beta\ell = n + \frac{\pi}{2}$ for some $n \in \mathbb{N}$. Thus,

$$X(x) = B \sin\left(\left(n + \frac{1}{2}\right) \frac{\pi}{\ell} x\right).$$

Robin is a bit harder he claims.

This will have to be done on the midterm, in the problems, etc.

We saw that we can solve Heat/Wave/Schrödinger equations by separating variables and solving the Eigenvalue problem, and we arrive at the solution u as a series of $\sin(\beta_n x)$ and/or $\cos(\beta_n x)$ times T_n . But we still haven't solved the IVP part ($u(x, 0) = \phi(x)$). We will do this next time.

§17 Day 17: Fourier Series (Nov. 12, 2025)

Before we've solved our 1+1 dimensional wave, heat, and Schrödinger boundary value problems by writing our solutions as a series of sin, cos, and so on. For example,

$$\begin{cases} u_t = ku_{xx} \\ \text{boundary conditions at } x = 0 \text{ and } x = \ell \end{cases}$$

is solved by

$$u(x, t) = \sum_{n \geq 0} e^{-\lambda_n kt} \left[A_n \cos(\sqrt{\lambda_n} x) + B_n \sin(\sqrt{\lambda_n} x) \right],$$

where λ_n is such that $-X'' = \lambda_n X$ has nontrivial solutions which satisfy our boundary conditions.¹⁶

Now, we will focus on the initial value problem. Let's forget about the PDE for now, and just consider how to solve the initial value problem

$$u(x, 0) = \sum_{n \geq 0} \left[A_n \cos(\sqrt{\lambda_n} x) + B_n \sin(\sqrt{\lambda_n} x) \right],$$

for a given sequence of λ_n . We will look at a theoretical/general approach to the series using real and also functional analysis.

The sine series (dirichlet boundary conditions gave us this) is

$$\sum_{n \geq 1} A_n \sin\left(\frac{n\pi}{\ell} x\right) = \phi(x).$$

If this is true, what should our A_n be? Well multiply both sides by $\sin\left(\frac{m\pi}{\ell} x\right)$ and integrate from 0 to ℓ .

$$\int_0^\ell \phi(x) \sin\left(\frac{m\pi}{\ell} x\right) dx = \sum_{n \geq 1} A_n \int_0^\ell \sin\left(\frac{n\pi}{\ell} x\right) \sin\left(\frac{m\pi}{\ell} x\right) dx.$$

Our claim is that

$$\int_0^\ell \sin\left(\frac{n\pi}{\ell} x\right) \sin\left(\frac{m\pi}{\ell} x\right) dx = \frac{\ell}{2} \delta_{nm}.$$

This can be proven by showing that

$$2 \sin(x) \sin(y) = \cos(x - y) - \cos(x + y),$$

so integrating this will yield 0 since it is periodic and even. You can prove this using the polarization identity for sin and cos. Since

$$\begin{aligned} \int_0^\ell \phi(x) \sin\left(\frac{m\pi}{\ell} x\right) dx &= \sum_{n \geq 1} A_n \int_0^\ell \sin\left(\frac{n\pi}{\ell} x\right) \sin\left(\frac{m\pi}{\ell} x\right) dx \\ &= \sum_{n \geq 1} A_n \delta_{n,m} \cdot \frac{\ell}{2} \\ &= A_m \frac{\ell}{2}. \end{aligned}$$

¹⁶Depending on our boundary conditions, we might be able to solve $X'' = \lambda_n X$ with $\lambda_n > 0$.

It follows that

$$A_m = \frac{2}{\ell} \int_0^\ell \phi(x) \sin\left(\frac{m\pi}{\ell}x\right) dx,$$

so we have found our coefficients A_n . We can do the same this with cosine using a similar product identity. If

$$\phi(x) = \sum_{n \geq 0} B_n \cos\left(\frac{n\pi}{\ell}x\right),$$

then by almost the exact same argument,

$$B_m = \frac{2}{\ell} \int_0^\ell \cos\left(\frac{m\pi}{\ell}x\right) \phi(x) dx.$$

Example 17.1. Suppose that $\phi(x) \equiv 1$. Then

$$A_n = \frac{2}{\ell} \int_0^\ell \sin\left(\frac{n\pi}{\ell}x\right) dx = \frac{2}{\ell} \left[-\cos\left(\frac{n\pi}{\ell}x\right) \frac{\ell}{n\pi} \right]_0^\ell = \frac{2}{n\pi} (1 - \cos(n\pi)).$$

Thus,

$$1 = \sum_{n \text{ odd}} \frac{4}{\pi n} \sin\left(\frac{\pi n}{\ell}x\right) = \sum_{n \geq 0} \frac{4}{\pi(2n+1)} \sin\left(\frac{\pi(2n+1)}{\ell}x\right).$$

In the problem set we use Fourier series methods to solve some interesting questions in number theory, such as the Basel problem.

Example 17.2. Set $\phi(x) = x$. Then we want to find the sine series for ϕ . This is just

$$\begin{aligned} A_n &= \frac{2}{\ell} \int_0^\ell \sin\left(\frac{n\pi}{\ell}x\right) x dx \\ &= \frac{2}{n\pi} \left[-\cos\left(\frac{n\pi}{\ell}x\right) \cdot x \right]_0^\ell + \frac{2}{n\pi} \int_0^\ell \cos\left(\frac{n\pi}{\ell}x\right) dx \\ &= \frac{2}{n\pi} [-\cos(n\pi)\ell] + \frac{2}{n\pi} \left[\sin(n\pi) \cdot \frac{\ell}{n\pi} \right] \\ &= \frac{-2\ell}{n\pi} \cos(n\pi) + \frac{2}{n\pi} \cdot \frac{\ell}{n\pi} \sin(n\pi) \\ &= \begin{cases} \frac{-2\ell}{n\pi} & n \text{ even} \\ \frac{2\ell}{n\pi} & n \text{ odd.} \end{cases} \\ &= (-1)^{n+1} \frac{2\ell}{n\pi} \end{aligned}$$

So the sine series is $A_n = \frac{(-1)^{n+1}2\ell}{n\pi}$, and the cosine series is $\frac{-4\ell}{n^2\pi^2}$, when n is odd, and 0 otherwise. This second one you have to verify for yourself, but it's a similar calculation.

Suppose that we are given a $\phi(x)$ on $[-\ell, \ell]$. Then we will express it as the full series

$$\phi(x) = \frac{A_0}{2} + \sum_{n \geq 1} A_n \cos\left(\frac{\pi n}{\ell}x\right) + B_n \sin\left(\frac{\pi n}{\ell}x\right).$$

To find A_m and B_m , we apply the same trick, integrating against $\cos\left(\frac{m\pi}{\ell}x\right)$ and $\sin\left(\frac{m\pi}{\ell}x\right)$ on $-\ell < x < \ell$. With the previous claims plus

$$\int_{-\ell}^\ell \sin\left(\frac{\pi a x}{\ell}\right) \cos\left(\frac{\pi b x}{\ell}\right) dx = 0,$$

for all $a, b \in \mathbb{Z}$. This leads us to a general formula for our coefficients that

$$\phi(x) = \frac{A_0}{2} + \sum_{n \geq 1} A_n \cos\left(\frac{\pi n x}{\ell}\right) + B_n \sin\left(\frac{\pi n x}{\ell}\right),$$

where

$$A_n = \frac{1}{\ell} \int_{-\ell}^{\ell} \cos\left(\frac{n\pi x}{\ell}\right) \phi(x) dx,$$

$$B_n = \frac{1}{\ell} \int_{-\ell}^{\ell} \sin\left(\frac{n\pi x}{\ell}\right) \phi(x) dx.$$

Example 17.3. Consider $\phi(x) = x$ on the interval $[-\ell, \ell]$ rather than $[0, \ell]$. Then we need the full Fourier series. Then

$$x = \frac{A_0}{2} + \sum_{n \geq 1} A_n \cos\left(\frac{\pi n}{\ell} x\right) + \sum_{n \geq 1} B_n \sin\left(\frac{\pi n}{\ell} x\right).$$

But ϕ is odd, so all of the A_n cancel out, including $\frac{A_0}{2}$. Now, since ϕ is odd and $\sin$ is odd, we have that their product is even, so

$$\int_{-\ell}^{\ell} \sin\left(\frac{n\pi x}{\ell}\right) \phi(x) dx = 2 \int_0^{\ell} \sin\left(\frac{n\pi x}{\ell}\right) \phi(x) dx.$$

But since our formula for B_n in the full Fourier series differs by a factor of 2 from our B_n in the Sine series, we have that the coefficients of $\phi(x) = x$ on $[0, \ell]$ is the same as the coefficients of $\phi(x) = x$ on $[-\ell, \ell]$. In other words,

$$B_n = (-1)^{n+1} \frac{2\ell}{n\pi}.$$

Another way to think about it is graphically, where since ϕ is odd and our sine series is odd, keeping our coefficients the same will yield ϕ on our full interval.

I just remembered that the subsection command exists, so expect the notes to henceforth be a bit more organized.

He talked a bit about extending ϕ using even, odd, and periodic extensions.

§18 Day 18: Fourier Series Part 2. (Nov. 12, 2025)

Solving PDEs on bounded domains (intervals for now) with boundary conditions led to Fourier Series.

Example 18.1. We would like to represent a function $f(x) = \sum_{n \in \mathbb{Z}} c_n X_n(x)$ where X_m solves the Eigenvalue Problem

$$\begin{cases} -X_n'' = \lambda_n X_n \\ \text{Boundary Conditions} \end{cases}.$$

Remark 18.2. If we want to write $f(x) = \sum_{n=0}^{\infty} c_n \sin\left(\frac{n\pi}{\ell}x\right)$, then we can do this by integrating against $\sin\left(\frac{m\pi}{\ell}x\right)$ so long as we can switch the integral with our sum. This will yield

$$c_n = \frac{2}{\ell} \int_0^{\ell} f(x) \sin\left(\frac{\pi n}{\ell}x\right) dx.$$

Furthermore, if this works, then it's the unique coefficient. This used the orthogonality of our sine functions.

If we are given f which is a function on $[0, \ell]$ and define

$$c_n = \int_0^{\ell} f(x) X_n(x) dx,$$

where X_n satisfies our eigenvalue problem. Is it the case that

$$f(x) = \sum_n c_n X_n(x)?$$

Well there's a limit here in our sum, so we have to answer with respect to which function space are we considering the convergence? From here we can ask which class of functions we can pick f from to make it work. Notice that it's also possible that our sum converges, but not to f . Let's start with a version that is a bit abstract.

§18.1 Orthogonality and Symmetric Boundary Conditions

Given two functions f, g on $[a, b]$, define an inner product for functions

$$(f, g) = \int_a^b f(x)g(x)dx.$$

[He did indeed use the round parentheses here. Come on Fabio.] If we are using Complex Valued functions, we should use the conjugate on the second function to make it a Hermitian inner product. Refer back to 247. We say $f \perp g$ if $(f, g) = 0$. Refer back to 247.

Question 18.3. What kind of functions are f and g ? If f and g are continuous, then this will be an inner product, but if we just require integrable, then we might have that there exists $f \neq 0$ but $(f, f) = 0$, which would contradict our inner product axioms. [This means that as it stands, (f, f) is a pre-inner product (which is defined as you'd guess)].

An example of such a function is the indicator function on a singleton. But for the sake of our theory, instead of worrying about this, we will think of functions as being parts of equivalence classes, where

$$f \sim g \iff (f - g, f - g) = 0.$$

This turns out to be equivalent to $\|f - g\|_{L^2} = 0$, which is fun. After modding out by the self-orthogonal vectors, we have that $(\cdot, \cdot)$ is an inner product. Fabio defined it in class as f and g differing only on a set with measure zero, but I like this definition more because it is more direct in my humble opinion.

Definition 18.4. We say that a Hilbert space H is a vector space with an inner product $\langle \cdot, \cdot \rangle$ such that H is complete with respect to the norm $\|f\| = \langle f, f \rangle^{\frac{1}{2}}$.¹⁷ Completeness here is defined such that Cauchy sequences with respect to $\|\cdot\|$ converge in H .

Hilbert spaces have lots of geometric properties. They are very similar to finite dimensional space. They are “closest to finite dimensional”. The problem is that in infinite dimensions we lose compactness of the unit closed ball. In this general context, define $v \perp w$ if $\langle v, w \rangle = 0$. Let us go over properties of Hilbert spaces.

Lemma 18.5. 1. Cauchy-Schwarz holds for Hilbert Spaces.

$$|\langle u, v \rangle| \leq \|u\| \|v\|$$

2. The Parallelogram Identity holds for Hilbert Spaces.

$$\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2).$$

3. Pythagorean Theorem holds for Hilbert Spaces.

$$u \perp v \implies \|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

Proposition 18.6. If $(w_n)_{n \geq 1}$ is an orthonormal sequence—namely, $\langle w_n, w_m \rangle = \delta_{n,m}$, then for all v ,

$$\sum \langle v, w_n \rangle^2 \leq \|v\|^2.$$

Corollary 18.7. If we look at the series $\sum \langle v, w_n \rangle w_n$, then this converges in H .

Proof. This is because our sum is absolutely convergent. □

¹⁷Isaac wants to point out that “erm technically[.]” I will not repeat any such remarks here. You, dear reader, had the choice to take 436 if you wanted to.

§19 Day 19: Fourier Series Part 3. (Nov. 12, 2025)

Intended midterm average was 60-65. In the past his averages have been 55-60. Our average was around 51-52. He will curve depending on the results of the other midterms.

Last time we talked about a bit of functional analysis. Hilbert spaces, inner products, and some basic properties.

Example 19.1. If $X \subseteq \mathbb{R}^n$, then $L^2(X)$ space is the functions

$$L^2(X) = \left\{ f : \int_X |f|^2 dx < \infty \right\}.$$

The norm is induced by the inner product from last time

$$(f, g) = \int_X f \cdot \bar{g} dx,$$

where the $\bar{\cdot}$ is the complex conjugate. We take the conjugate when our codomain is the complex numbers (or something like it where integrals make sense), to make this a Hermetian inner product. If our codomain is a real vector space, we can exclude it.

To show that L^2 is a Hilbert space, we need to show completeness. Of course this requires that we either take the Lebesgue measure, or we do a hack by taking closure of compactly supported functions or something. Ultimately these things aren't of large importance to this class.

Example 19.2.

$$\ell^2 = \left\{ (a_n)_{n \in \mathbb{Z}} : \sum_{n \in \mathbb{Z}} |a_n|^2 < \infty \right\},$$

where the norm is given by

$$\|(a_n)\|_{\ell^2} = \left(\sum_{n \in \mathbb{Z}} |a_n|^2 \right)^{\frac{1}{2}}.$$

This norm is induced by the inner product

$$((a_n), (b_n)) = \sum_{n \in \mathbb{Z}} \sum_{n \in \mathbb{Z}} a_n \cdot \bar{b}_n.$$

Our remarks about complex vector spaces are the same as before.

Example 19.3. $\mathbb{R}^n$ with the dot product and usual norm.

Example 19.4. Consider continuous functions on $[0, 1]$ with the norm

$$\|f\| = \left(\int_0^1 |f(x)|^2 dx \right)^{\frac{1}{2}}.$$

We claim that this is not a Hilbert space, because it's not complete. For exmple,

$$d(x^n, x^m) = \left(\int_0^1 |x^n - x^m|^2 dx \right)^{\frac{1}{2}} \rightarrow 0,$$

as $n, m \rightarrow \infty$. But these converge to a discontinuous function (it's 0 everywhere except for at 1, where it's 1). On the other hand, with the uniform norm (L^∞ , or sup norm), they are not Cauchy, so they do not converge.

One should remark that if we take this sequence in the L^2 space, then these do converge, and they converge to 0. This is because the indicator function on $\{1\}$ is in the same equivalence class as the 0 function in the L^2 space.

Example 19.5. Consider continuous functions on $[0, 1]$ with the L^∞ norm. We call this space $C^0([0, 1])$. Then this space *is* complete, because the uniform limit of continuous functions is continuous.

Example 19.6. For L^p spaces, they are not Hilbert spaces (when $p \neq 2$), but they are complete normed vector spaces, so they are Banach spaces. Of course,

$$\|f\|_{L^p} = \left(\int |f(x)|^p dx \right)^{\frac{1}{p}}.$$

To show this, we can show that the parallelogram identity does not hold for the L^p norm ($p \neq 2$).

§19.1 Orthogonality

We are trying to write $f(x) = \sum_{n \geq 0} a_n X_n(x)$, where X_n are eigenfunctions of $-\partial_x^2$.

Proposition 19.7. Two eigenfunctions of $-\partial_x^2$, $-X_i'' = \lambda_i X_i$, $i = 1, 2$ with $\lambda_1 \neq \lambda_2$, then we claim that $X_1 \perp X_2$ with respect to $(\cdot, \cdot)$.

Proof.

$$\begin{aligned} (\lambda_1 - \lambda_2) \int_a^b X_1 X_2 &= \int_a^b \lambda_1 X_1 X_2 - \int_a^b X_1 (\lambda_2 X_2) \\ &= \int_a^b -X_1'' X_2 + \int_a^b X_1 X_2'' \\ &= [-X_1' X_2]_a^b + \int_a^b X_1' X_2' \\ &\quad + [X_1 X_2']_a^b - \int_a^b X_1' X_2' \\ &= 0, \end{aligned}$$

so long as our boundary conditions ensure that

$$[X_1 X_2' - X_1' X_2]_a^b = 0.$$

□

Definition 19.8. Boundary conditions (for the eigenvalue problem $-X'' = \lambda X$) for which

$$[X_1 X_2' - X_1' X_2]_a^b = 0$$

are called symmetric boundary conditions.

Example 19.9. 1. Dirichlet $X(a) = 0 = X(b)$

2. Neumann $X'(a) = 0 = X'(b)$

3. Robin [Check in the Homework]

4. Periodic $X(a) = X(b)$, $X'(a) = X'(b)$

Remark 19.10. We may have 2 or more eigenfunctions with the same eigenvalue. For example, $-X'' = \lambda X$ on $[0, 2\pi]$,

$$X(0) = X(2\pi), X'(0) = X'(2\pi)$$

have eigenfunctions of $\sin(nx)$ and $\cos(nx)$ with eigenvalue n^2 for all n . Also, any linear combination. That being said, fortunately $\sin(nx)$ and $\cos(nx)$ are already orthogonal. In general, by Gram-Schmidt you can make these eigenfunctions orthogonal. We know that the eigenspace is finite dimensional because it's a homogeneous second order ODE, so the general solution is a 2D vector space.

Corollary 19.11. 1. With symmetric boundary conditions, eigenfunctions of $-X'' = \lambda X$ with different eigenvalues are orthogonal.

2. The eigenvalues of $-X'' = \lambda X$ are real.

3. We can make eigenfunctions with the same value orthogonal.

Proof. We already proved the first and third part. The second part follows doing the same proof as the other part, but on

$$(\lambda - \bar{\lambda}) \int X \bar{X}.$$

□

As a consequence, with symmetric boundary conditions, if

$$f(x) = \sum_{n \geq 0} a_n X_n(x)$$

(which we will call a generalized Fourier series) then it follows that

$$a_n = \frac{(f, X_n)}{(X_n, X_n)},$$

where our X_n are an orthogonal set (which we are sure we can find).

Theorem 19.12. For $-X'' = \lambda X$ with symmetric boundary conditions, there exists infinitely many eigenvalues $\lambda_1 \leq \lambda_2 \leq \dots$, with $\lambda_n \nearrow \infty$.

Question 19.13. When we write $f(x) = \lim_{N \rightarrow \infty} \sum_{n=1}^N a_n X_n(x)$ where

$$a_n = \frac{(f, X_n)}{(X_n, X_n)},$$

with respect to which norm does this limit converge? We are interested in the different modes of convergence of the series of functions

$$\sum_{n=1}^N a_n X_n(x)$$

and the convergence to what. That is, it may not converge to f even if it does converge.

§19.2 Notions of Convergence

Definition 19.14. We say that $\sum f_n(x)$ converges pointwise to f in (a, b) if

$$\lim_{N \rightarrow \infty} \left| \sum_{n=1}^N f_n(x) - f(x) \right| = 0.$$

Definition 19.15. We say $\sum f_n(x)$ converges uniformly to f on $[a, b]$ if

$$\lim_{N \rightarrow \infty} \sup_{[a, b]} \left| \sum_{n=0}^N f_n(x) - f(x) \right| = 0.$$

Definition 19.16. We say $\sum f_n$ converges to f in $L^2([a, b])$ if

$$\lim_{N \rightarrow \infty} \left\| \sum_{n=0}^N f_n(x) - f(x) \right\|_{L^2([a, b])} = 0.$$

Remark 19.17. Uniform Convergence implies Pointwise convergence and L^2 convergence.

Remark 19.18. Uniform convergence is strictly stronger than Pointwise convergence, as x^n converges pointwise but not uniformly.

Remark 19.19. Pointwise convergence does not imply L^2 convergence, which we can see by taking functions like $g_n = \chi_{(\frac{n-1}{n}, 1)} \cdot \sqrt{n}$. [This is my own example. If you round the edges you can make it continuous or smooth if you want. Fabio used triangles, which was basically the same idea.]

Remark 19.20. L^2 convergence also does not imply pointwise convergence. Consider

$$g_n = \chi_{(\sin(n - \frac{1}{n}), \sin(n + \frac{1}{n}))},$$

where as usual, χ is the indicator function. Then by Diophantine approximation, we have that this sequence will be nonzero at every point infinitely many times (indeed, even if we set it to be $(\sin(n - \frac{1}{n^2}), \sin(n + \frac{1}{n^2}))$,¹⁸ it would diverge on a comeagre set—a set whose complement has measure 0), but its L^2 norm also clearly goes to 0, as $\sin$ is a contraction mapping, so we have that the diameter of our interval goes to 0, and therefore the L^2 norm of our sequence goes to 0. As such, we are done.

Once again, this was my example. But in this case, Fabio did not give an example (or at least he didn't write one down. He may have said one out loud.)

¹⁸On the other hand, if instead of squaring it we take an exponent of α with $\alpha > 2$, then it has measure 0 and Hausdorff (fractal) dimension less than 2. Wenyu proved this fact in class in MAT335.

§20 Day 20: Convergence of Fourier Series (Nov. 24, 2025)

We did some formal non-rigorous calculations with fourier series already to solve boundary and initial value problems. We also talked about the theory behind it with orthogonality and that stuff. We also discussed in the more general context of Hilbert spaces. So now we will consider questions of convergence.

Our plan is to state and prove three theorems about convergence. One about uniform convergence, one about L^2 convergence, and one about pointwise convergence. This is going to be the most theoretical part so far.

Theorem 20.1. Assume that f on $[a, b]$ is such that f , f' , and f'' are all continuous. Then the Fourier series

$$\sum_{n=1}^N A_n X_n(x) \rightrightarrows f$$

on $[a, b]$.

Remark 20.2. Here, as in the previous lecture, X_n is defined as solutions to the ODE $-X'' = \lambda X$ with some symmetric boundary conditions, and

$$A_n = \frac{(f, X_n)}{(X_n, X_n)}.$$

We can get the same conclusion with different assumptions, but for that we need to do something slightly different.

Proof. Use Theorem 1.4 and the homework. □

Theorem 20.3. Let $f \in L^2([a, b])$. Namely, we want f to have L^2 norm less than infinity. Then

$$\sum_{n=1}^N A_n X_n(x) \rightarrow f$$

in $L^2([a, b])$.

Theorem 20.4. Let $f \in C([a, b])$ with f piecewise continuous. Then the classical fourier series (with sine and cosine) converges pointwise to f . And if f is piecewise continuous, then at any jump x_0 , the Fourier series converges to

$$\frac{\lim_{x \rightarrow x_0^+} f(x) + \lim_{x \rightarrow x_0^-} f(x)}{2}.$$

§20.1 The L^2 -Theory

Theorem 20.5 (Least Square). Let X_n be an orthogonal set of eigenfunctions and $\|f\|_{L^2} < \infty$. Then for N fixed,

$$\left\| f - \sum_{n=1}^N c_n X_n(x) \right\|_{L^2}$$

is going to be minimized by the Fourier coefficients

$$c_i = A_i = \frac{(f, X_n)}{(X_n, X_n)}.$$

Proof. This will give us some implications. Set $E_N = \left\| f - \sum_{n=1}^N c_n X_n \right\|_{L^2}^2$. Expanding this yields

$$E_N = \|f\|_{L^2}^2 - 2 \sum_{n=1}^N c_n (f, X_n) + \left\| \sum_{n=1}^N c_n^2 (X_n, X_n) \right\|_{L^2},$$

where we can bring the squares inside the norm by the pairwise orthogonality of our X_n . Now, we want to complete the squares.

$$0 \leq E_N = \|f\|_{L^2}^2 + \underbrace{\sum_{n=1}^N (X_n, X_n) \left[c_n - \frac{(f, X_n)}{(X_n, X_n)} \right]^2}_{\text{non-negative}} - \sum_{n=1}^N \frac{(f, X_n)^2}{(X_n, X_n)}.$$

where the term with the underbrace is the only term which varies with respect to our c_n . As such, by minimizing each of the individual terms of the sum, we minimize the series, which is accomplished by our hypotheses—when

$$c_n = \frac{(f, X_n)}{(X_n, X_n)}.$$

□

Corollary 20.6. With $c_n = A_n$, we get

$$\sum A_n^2 (X_n, X_n) \leq \|f\|_{L^2}^2.$$

Proof. Rearrange the final line of our previous proof to get this. (X_n, X_n) being multiplied and not divided is *not* a typo. □

We will continue on Wednesday.

§21 Day 21: Convergence of Fourier Series cont. (Nov. 24, 2025)

Last time we proved that our A_n really are the best coefficients. This is in the sense that

$$\sum_{n \geq 1} |A_n|^2 (X_n, X_n) \leq \|f\|_{L^2}^2.$$

If $(X_n, X_n) = 1$, then

$$\sum_{n \geq \mathbb{N}} |A_n|^2 \leq \|f\|_{L^2}^2, \quad A_n = (f, X_n).$$

So, we will keep this in mind. We also got Bessel's inequality

$$\begin{aligned} 0 \leq E_N &= \left\| f(x) - \sum_{n=1}^N A_n X_n(x) \right\|_{L^2}^2 \\ &= \|f\|_{L^2}^2 - \sum_{n=1}^N \frac{|(f, X_n)|^2}{(X_n, X_n)}. \end{aligned}$$

But this is just the error between our function and the partial fourier sum (measured in L^2). This will be zero if our right hand sum actually converges to the norm squared (rather than just having our inequality above). Namely,

$$\sum_{n=1}^{\infty} |A_n|^2 (X_n, X_n) = \|f\|_{L^2}^2.$$

This is called Parseval's identity, and when it holds, we have L^2 convergence of our Fourier series to f .

Theorem 21.1. Let w_n be an orthonormal sequence in a Hilbert space H . The following are equivalent:

1. Finite linear combinations

$$\sum_{n=1}^N b_n w_n$$

are dense ($b_n \in \mathbb{C}$) in the Hilbert space.

2. If $\langle u, w_n \rangle = 0$ for all n then $u = 0$.
3. Parseval's Identity.

$$\|u\|_H^2 = \sum_{n=1}^{\infty} |\langle u, w_n \rangle|^2.$$

4. (w_n) is an orthonormal basis for H .

Fabio didn't actually go through the proofs explicitly. It'll be easy to find this proof though.

It follows that to complete the proof of L^2 convergence in our setting, you can verify any of the above properties. A shortcut to prove property 1 is to use the Stone-Weierstrass theorem (from 357) to show that the trig polynomials are dense in the continuous functions with respect to the L^∞ norm. From here, the continuous functions are dense in L^2 with respect to the L^2 norm, and convergence in the L^∞ norm of a sequence our trig polynomials implies convergence (to the same limit) in the L^2 norm, therefore the trig polynomials are dense in the L^2 norm.

Theorem 21.2. Let f be a C^1 functions on $\mathbb{R}$ of period 2π . Define

$$(S_N f)(x) = \frac{A_0}{2} + \sum_{n=1}^N A_n \cos(nx) + \sum_{n=1}^N B_n \sin(nx).$$

Then $S_N f(x) \rightarrow f(x)$ pointwise.

Proof. Step 1 is to take S_N and write it as a convolution of a kernel with f .

$$\begin{aligned} (S_N f)(x) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) dy \\ &\quad + \sum_{n=1}^N \left(\frac{1}{\pi} \int_{-\pi}^{\pi} f(y) \cos(ny) dy \right) \cos(nx) \\ &\quad + \sum_{n=1}^N \left(\frac{1}{\pi} \int_{-\pi}^{\pi} f(y) \sin(ny) dy \right) \sin(nx) \\ (S_N f)(x) &= \frac{1}{2\pi} \left(\int_{-\pi}^{\pi} f(y) dy \right. \\ &\quad + \sum_{n=1}^N \left(\int_{-\pi}^{\pi} f(y) \cos(ny) dy \right) 2 \cos(nx) \\ &\quad \left. + \sum_{n=1}^N \left(\int_{-\pi}^{\pi} f(y) \sin(ny) dy \right) 2 \sin(nx) \right) \\ (S_N f)(x) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) \left(1 + \sum_{n=1}^N 2 \cos(ny) \cos(nx) + 2 \sin(ny) \sin(nx) \right) dy \\ (S_N f)(x) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) \left(1 + 2 \sum_{n=1}^N \cos(n(x-y)) \right) dy \end{aligned}$$

We notice that this kinda looks like a convolution, so we are going to set

$$K_N(x-y) = 1 + 2 \sum_{n=1}^N \cos(n(x-y)).$$

We claim

$$K_N(\theta) = \frac{\sin\left(\left(N + \frac{1}{2}\right)\theta\right)}{\sin\left(\frac{\theta}{2}\right)}.$$

This can be proven via residues! According to Fabio at least. He seems to have opted to

do it directly.

$$\begin{aligned}
 K_N(\theta) &= 1 + 2 \sum_{n=1}^N \cos(n\theta) \\
 &= 1 + 2 \operatorname{Re} \left(\sum_{n=1}^N e^{in\theta} \right) \\
 &= \sum_{n=-N}^N e^{in\theta} \\
 &= e^{-iN\theta} \sum_{n=0}^{2N} e^{in\theta} \\
 &= e^{-iN\theta} \frac{(1 - e^{i(2N+1)\theta})}{1 - e^{i\theta}} \quad (\text{Geometric Series}) \\
 &= \frac{e^{-iN\theta} - e^{i(N+1)\theta}}{1 - e^{i\theta}} \cdot \frac{e^{-i\frac{\theta}{2}}}{e^{-i\frac{\theta}{2}}} \\
 &= \frac{e^{-i(N+\frac{1}{2})\theta} - e^{i(N+\frac{1}{2})\theta}}{e^{-i\frac{\theta}{2}} - e^{i\frac{\theta}{2}}} \\
 &= \frac{\sin((N + \frac{1}{2})\theta)}{\sin(\frac{\theta}{2})}
 \end{aligned}$$

This function will be nice everywhere except possibly at $\theta = 0$. But the limit as we approach 0 is just

$$\lim_{\theta \rightarrow 0} K_N(\theta) = \lim_{\theta \rightarrow 0} \frac{\sin((N + \frac{1}{2})\theta)}{\sin(\frac{\theta}{2})} = \lim_{\theta \rightarrow 0} \frac{(N + \frac{1}{2})\theta + o(\theta)}{\frac{\theta}{2} + o(\theta)} = 2N + 1,$$

so the limit exists, and equals our original sum.

Now, we want to calculate $S_N f(x) - f(x) \rightarrow 0$ for a fixed x . But notice that

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} K_N(\theta) d\theta = 1, \quad (*)$$

by remarking that

$$K_N(\theta) = \sum_{n=-N}^N e^{in\theta},$$

thus the integral cancels out all the terms except $n = 0$, which will integrate to 2π . “This kernel is nice, but not *too* nice.” Rewrite

$$S_N f(x) = \int_{-\pi}^{\pi} K_N(\theta) f(x + \theta) \frac{d\theta}{2\pi}.$$

This is by the periodicity of f . From here,

$$S_N f(x) - f(x) = \int_{-\pi}^{\pi} K_N(\theta) [f(x + \theta) - f(x)] \frac{d\theta}{2\pi}$$

which is because of the integral on line (*).

$$= \int_{-\pi}^{\pi} \sin\left((N + \frac{1}{2})\theta\right) \frac{f(x + \theta) - f(x)}{\sin(\frac{\theta}{2})} \frac{d\theta}{2\pi}$$

If we set

$$g_x(\theta) = \frac{f(x+\theta) - f(x)}{\sin\left(\frac{\theta}{2}\right)},$$

(x is fixed) we want to show that

$$\int_{-\pi}^{\pi} \sin\left(\left(N + \frac{1}{2}\right)\theta\right) g_x(\theta) d\theta \rightarrow 0$$

as $N \rightarrow \infty$. But we know that g is continuous in θ , and its limit as $\theta \rightarrow 0$, of g_x is just $2f'(x)$, so we have that it can be extended to continuous everywhere. Thus, we will try to use some recent knowledge. Recall Bessel's inequality. If h has $\|h\|_{L^2}^2 < \infty$, and X_n is an orthogonal set.

$$\sum \frac{(h, X_n)^2}{(X_n, X_n)} \leq \|h\|_{L^2}^2.$$

If $|(X_n, X_n)| \leq C$ for some uniform bound C (independent of n), then we have that (h, X_n) is square summable, thus $(h, X_n) \rightarrow 0$ as $n \rightarrow \infty$.

Applying this, if we show g_x is L^2 , and that our $\sin\left(\left(N + \frac{1}{2}\right)\theta\right)$ have bounded inner product, then this would prove the theorem, because

$$\int_{-\pi}^{\pi} \sin\left(\left(N + \frac{1}{2}\right)\theta\right) g_x(\theta) d\theta = \left(\sin\left(\left(N + \frac{1}{2}\right)\theta\right), g_x(\theta)\right) \cdot \frac{1}{2\pi}.$$

Thus we just have to verify these.

1. We know that $\|g_x\|_{L^2} < \infty$ because $g(\theta) = \frac{f(x+\theta)-f(x)}{\sin(\frac{\theta}{2})}$ is continuous on $[-\pi, \pi]$, and therefore it is bounded and integrable, so it is L^2 .

2.

$$\left| \int_{-\pi}^{\pi} \sin^2\left(N + \frac{1}{2}\theta\right) d\theta \right| \leq 2\pi \left| \sup_{\theta \in [-\pi, \pi]} \sin(\theta) \right| = 2\pi,$$

so our functions are uniformly bounded.

3. For orthogonality, notice that our $\sin\left(\left(N + \frac{1}{2}\right)\theta\right)$ are the eigenfunctions of the eigenvalue problem

$$\begin{cases} X'' = -\lambda X \\ X(0) = 0 \\ X'(\pi) = 0 \end{cases}$$

and so since the different N correspond to different eigenfunctions, they are orthogonal.

Thus, we are done. We have shown pointwise convergence. $\square$

§22 Day 22: Laplace Equation (Dec. 1, 2025)

Fabio has gone to New York. Slacker. We will be looking at the Laplace equation with the amazing TA Oliver. This is $\Delta u = 0$.

Where does this arise? It gives us stationary solutions to the heat and wave equation. This is of course because if $u(x, t) : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}$ is harmonic in x and constant in t , then

$$\partial_t^k u + c \Delta u = 0.$$

(for all $c \in \mathbb{R}$). Both the heat and wave equation are of this form. Also the Schrödinger equation.

Example 22.1. Let's look at the Laplace equation in $[0, \pi]^2$.

$$\begin{cases} u_{xx} + u_{yy} = 0 \\ u(0, y) = u(\pi, y) = 0 \\ u(x, 0) = 0, u(x, \pi) = g(x) \end{cases}.$$

Then as usual we'll look for a separable solution.

$$u(x, y) = X(x)Y(y) \implies \frac{X''}{X} = -\frac{Y''}{Y} = \lambda,$$

so we do what we've always done, and write the eigenvalue problem

$$\begin{cases} X'' = \lambda X \\ X(0) = 0 = X(\pi) \end{cases}.$$

Using some (very difficult and hard to prove) facts about trig functions, $\lambda = n^2$ for some $n \in \mathbb{Z}$, so

$$X(x) = a_n \cos(nx) + b_n \sin(nx),$$

and since we know all the solutions for second order constant coefficient ODEs, we can also deduce from our vast repertoire of ODE knowledge that

$$Y(y) = c_n \sinh(ny) + d_n \cosh(ny)$$

Plugging in the boundary conditions of $X(0) = 0 = X(\pi)$ and $Y(0) = 0$, we can deduce $a_n = 0$ and $d_n = 0$, so

$$X(x) = b_n \sin(nx), \quad Y(y) = c_n \sinh(ny).$$

We can then write

$$u(x, y) = \sum_{n=1}^{\infty} A_n \sin(nx) \sinh(ny).$$

Our final boundary condition (and orthogonality of eigenfunctions with different eigenvalues) gives us that

$$A_n \sinh(n\pi) = \frac{2}{\pi} \int_0^{\pi} g(x) \sin(nx) dx,$$

so since $\sinh(n\pi) \neq 0$, we are done.

Example 22.2. We're gonna do the problem on a disc.¹⁹

$$\begin{cases} \Delta u = 0 & B(0, 1) \subseteq \mathbb{R}^2 \\ u = g & \partial B(0, 1). \end{cases}$$

The natural way to do this is using polar coordinates. In polar coordinates,

$$\Delta = \partial_r^2 + \frac{1}{r}\partial_r + \frac{1}{r^2}\partial_\theta^2.$$

If we write

$$u = R(r)\Theta(\theta),$$

then

$$R''\Theta + \frac{1}{r}R'\Theta + \frac{1}{r^2}R\Theta'' = 0.$$

Then

$$-\frac{\Theta''}{\Theta} = \frac{r^2 R'' + r R'}{R} = \Lambda.$$

What sign should λ be? Well we'll use the Θ term to exploit that since

$$\Theta'' = -\lambda\Theta$$

and Θ is 2π -periodic, then we either have that Θ is constant and $-\lambda = 0$, or $\Theta(\theta) = a_n \cos(n\theta) + b_n \sin(n\theta)$ and $-\lambda$ is negative. We'll look at the latter case.

Notice that

$$r^2 R'' + r R' - n^2 R = 0,$$

so this is a Cauchy-Euler differential equation, and therefore setting as ansatz $R = r^\alpha$,

$$r^\alpha [\alpha(\alpha - 1) + \alpha - n^2] = 0$$

thus $\alpha^2 - n^2 = 0$, so we have that $\alpha = \pm n$. Thus, this ODE will be solved by

$$R(r) = c_n r^n + d_n r^{-n}.$$

On the other hand, since we want this to be continuous on the disc, we need $d_n = 0$, as r^{-n} isn't defined at $r = 0$. It follows that

$$u(r, \theta) = A_0 + \sum_{n=1}^{\infty} r^n (A_n \cos(n\theta) + B_n \sin(n\theta)).$$

Since $g(\theta) = A_0 + \sum A_n \cos(n\theta) + B_n \sin(n\theta)$, we just need A_n and B_n to be the Fourier coefficients of g .

Example 22.3. Let's look at the Neumann boundary conditions.

$$\begin{cases} \Delta u = 0 & B(0, 1) \\ \frac{\partial u}{\partial n} = h & \partial B(0, 1). \end{cases}$$

Our deductions are the same to get up to the general solution

$$u(r, \theta) = A_0 + \sum_{n=1}^{\infty} r^n (A_n \cos(n\theta) + B_n \sin(n\theta)),$$

¹⁹"I wanna do something that's not on a *rectangle*." Oliver, your prejudices are showing.

so since we know (hello 257) $\frac{\partial u}{\partial n} = \nabla u \cdot n$ on the circle (and the normal vector at a point x on a circle of radius 1 is just x itself). Then we can deduce that

$$\frac{\partial u}{\partial n} = \sum_{n=1}^{\infty} n [A_n \cos(n\theta) + B_n \sin(n\theta)] = h(\theta).$$

Integrating yields our Fourier coefficients as usual. We can't determine A_0 by this formula which is confirmation of our non-uniqueness. Also, $\hat{h}(0) = 0$ (the 0th Fourier coefficient for h is 0). It follows that

$$0 = \frac{1}{\pi} \int_0^{2\pi} h(\theta) d\theta.$$

Example 22.4. Let's do another wacky and fun domain.

$$\begin{cases} \Delta u = 0 & 1 < r < 2 \quad 0 < \theta < \pi \\ u(r, 0) = 0 = u(r, \pi) \\ u(1, \theta) = \sin \theta \\ u(2, \theta) = 0. \end{cases}$$

This will look like a rainbow (waow). $\Theta(\theta) = a_n \cos(n\theta) + b_n \sin(n\theta)$ and $R(r) = c_n r^n + d_n r^{-n}$. In this case, we don't have any reason to cancel out the r^n and r^{-n} .

Now, $u(r, 0) = 0 = u(r, \pi)$ implies $\Theta(0) = 0 = \Theta(\pi)$, so $a_n = 0$. Thus,

$$u(r, \theta) = \sum_{n=1}^{\infty} (C_n r^n + D_n r^{-n}) \sin(n\theta).$$

There won't be a constant. When we set $r = 1$, we get that

$$\sum_{n=1}^{\infty} (C_n + D_n) \sin(n\theta) = \sin(\theta),$$

therefore

$$\begin{cases} C_1 + D_1 = 1 \\ C_n + D_n = 0 \quad n > 1. \end{cases}$$

When $r = 2$, then

$$\sum_{n=1}^{\infty} (C_n 2^n + D_n 2^{-n}) \sin(n\theta) = 0$$

and therefore

$$C_n 2^n + D_n 2^{-n} = 0, \quad n \geq 1.$$

For each n , this is a linear system of two equations in two unknowns, so we can solve it. I won't write out that process, taking on faith that you have taken 240.

$$C_1 = \frac{-1}{3}, D_1 = \frac{4}{3}, \quad C_n = 0, D_n = 0, \quad n > 1.$$

Therefore

$$u(r, \theta) = \left(\frac{-r}{3} + \frac{4}{3r} \right) \sin(\theta).$$

Happy Holidays everyone. It's Dynamical Systems December for me.

§23 Day 23: Welcome Back (Jan. 5, 2026)

My new year's resolution is to take 351. I'm succeeding so far.

Midterm 2 will cover problems from the problem sets plus a little more. It will cover everything since the beginning of the class, but it will skew towards later stuff. We need to set a date. Midterm 3 is like midterm 1. **Midterm 2 is on January 28th, and Midterm 3 is on March 4th.**

§23.1 Course Outline

In the second semester we'll cover

1. Laplace (2, 3 spacial dimensions)
 - a) Strong max principle, Poisson's Formula, Green's Functions
 - b) Solving on various domains with special geometry
2. Heat, Wave with 2, 3 spacial dimensions
3. Schrödinger, Hydrogen Atom (woah)
4. Fourier Transform
5. (This one might not be placed here chronologically) Do a bit more on conservation laws. We'll go back to first order 1 dimensional quasilinear problems (e.g. Burger's Equation).

Last semester we ended with Fourier Series, applying to solve Boundary/Initial value problems, etc. We also discussed the theory of convergence. **Assigned reading: Section 5.6 of Strauss.**

§23.2 Laplacian and Harmonic Functions

Laplace's equation is

- $u_{xx} + u_{yy} = 0$ in 2d
- $u_{xx} + u_{yy} + u_{zz}$ in 3d.

We will write this as $\Delta u = 0$ (and add a subscript $\Delta_3 u$ if the dimension is unclear. Poisson's equation is

$$-\Delta u = f(x),$$

where the minus sign is added so that this is a positive operator. We'll look at this on domains in $\mathbb{R}^d$ ($d = 2, 3$ and some things will work for arbitrary dimensions) and we'll also provide boundary conditions. Typically Dirichlet ($u = g$ on ∂D) or Neumann ($\frac{\partial u}{\partial n} = h$ on ∂D). In the first few weeks we showed uniqueness and such for these boundary conditions. Review if you forget.

Definition 23.1. A harmonic function u is a solution of $\Delta u = 0$.

Where do Laplace/Poisson's equations appear?

- In electrostatics, Maxwell's equations are $\nabla \times E = 0$ and $\nabla \cdot E = 4\pi e$. If the domain of E is simply connected, then it will be the gradient of some function $E = -\Delta\phi$. But then we know that $-\Delta\phi = 4\pi e$. This is a poisson equation.

- In fluid flow you have a velocity field v . If we have a potential flow, then $\nabla \times v = 0$.²⁰ Thus, $v = -\nabla\phi$. An incompressible flow satisfies $\nabla \cdot v = 0$ as well, because of conservation of mass (use the Divergence theorem to show that on any domain mass will be conserved). If we have these two, then $-\Delta\phi = 0$.
- Apparently there's some connection to complex analysis. Not that I've heard of it or anything. If $f : \mathbb{C} \rightarrow \mathbb{C}$ and $f(z) = u(z) + iv(z)$ where $u, v : \mathbb{C} \rightarrow \mathbb{R}$, then u and v are harmonic when f is holomorphic due to Cauchy-Riemann. He wrote this out more explicitly, I figure you can review your 354 notes if you want.

²⁰I have no clue what a potential flow is.

§24 Day 24: Harmonic Functions (Jan. 7, 2026)

A harmonic function is a function with $\Delta u = 0$. We'll mostly work in 2d and 3d, but some things are more general. We'll go over the weak maximum principle, which is actually review.

Theorem 24.1. Let D be a bounded connected domain in $\mathbb{R}^d$ and let u be harmonic in D . Let u be continuous on $\bar{D}$. Then the max and min of u are assumed on ∂D , and nowhere else—unless u is constant.

Notice that as usual, we need u to be differentiable twice in each its variables (or else Δu is not well defined) so we assume u to be twice continuously differentiable.

Proof (Weak Max/Min Principle). To do this, look at $v = u + \varepsilon|x|^2$, for $x \in \mathbb{R}^d$. Then $\Delta v = \Delta u + \varepsilon \cdot 2d > 0$. Therefore v cannot have negative second derivatives, so v cannot have a max in $\text{int } D$. We know that

$$\max_D u \leq \max_D v \leq \max_{\partial D} v,$$

so therefore as $\varepsilon \rightarrow 0$, $\max_{\partial D} v \rightarrow \max_{\partial D} u$, thus we have that $\max_D u \leq \max_{\partial D} u$. $\square$

Remark 24.2. Proof for min can be found by taking $-u$.

Remark 24.3. Same proof above for the max works if you only know that $\Delta u \geq 0$. Usually we write this as $-\Delta u \leq 0$. Same thing.

Let's work towards proving the strong max principle in 2d. We will need Poisson's formula. This is independently important, but also needed for this. Before we can consider Poisson's formula, we need to look at the Laplacian in polar coordinates. "I just thought of doing it like this. Then we'll go back to following the book."

Consider that $\Delta u = u_{xx} + u_{yy}$, and

$$\begin{cases} x &= r \cos \theta \\ y &= r \sin \theta \end{cases}$$

Then the laplacian in polar coordinates is

$$\Delta = \partial_r^2 + \frac{1}{r} \partial_r + \frac{1}{r^2} \partial_\theta^2.$$

To derive this, notice that

$$\begin{cases} r &= \sqrt{x^2 + y^2} \\ \theta &= \arctan\left(\frac{y}{x}\right) \end{cases}$$

so therefore

$$\frac{\partial}{\partial x} = \frac{\partial r}{\partial x} \frac{\partial}{\partial r} + \frac{\partial \theta}{\partial x} \frac{\partial}{\partial \theta}$$

Then (recall 257),

$$\frac{\partial r}{\partial x} = \cos \theta \quad \frac{\partial \theta}{\partial x} = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \left(-\frac{y}{x^2}\right) = -\frac{y}{x^2 + y^2},$$

and thus, plugging in,

$$\frac{\partial}{\partial x} = \cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta},$$

and repeating for y gives

$$\frac{\partial}{\partial y} = \sin \theta \frac{\partial}{\partial r} + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta}.$$

It follows by simplifying that

$$\partial_x(\partial_x) + \partial_y(\partial_y) = \partial_r^2 + \frac{1}{r}\partial_r + \frac{1}{r^2}\partial_\theta^2,$$

which tells us our formula for the radial laplacian.

Now, we want to find a radial harmonic function $f = v(r)$ such that $\Delta f = 0$. Then

$$f = f(x, y) = v(\sqrt{x^2 + y^2}).$$

This implies

$$(\partial_r^2 + \frac{1}{r}\partial_r)v = 0.$$

Now, if $v'' + \frac{1}{r}v' = 0$, then $(v' = g)$

$$\begin{aligned} g' &= -\frac{1}{r}g \\ \frac{g'}{g} &= -\frac{1}{r} \\ (\log g)' &= (-\log r)' \\ \log g &= -\log r + C \\ g &= \frac{C}{r} \\ v(r) &= C_1 \log r + C_2. \end{aligned}$$

Notice that v is not smooth at $r = 0$.

Theorem 24.4 (Dirichlet Problem in a Disc). Suppose that

$$\begin{cases} \Delta u = 0 & B = \{x^2 + y^2 < a^2\} \\ u = f & \partial B = \{x^2 + y^2 = a^2\} \end{cases}$$

Then

$$u(r, \theta) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta') P_a(r, \theta - \theta') d\theta',$$

where

$$P_a(r, \psi) = \frac{a^2 - r^2}{a^2 + r^2 - 2ar \cos(\psi)},$$

Proof. First we need to solve $\Delta u = 0$ using polar separation of variables. Then we will impose the boundary conditions $u = f$ on ∂B .

Let's pass to $u(r, \theta)$ and look for a separable solution $u(r, \theta) = X(r)Y(\theta)$. Then

$$\partial_r^2 \frac{1}{r} \partial_r + \frac{1}{r^2} \partial_\theta^2 (X(r)Y(\theta)) = 0$$

yields that

$$X''Y + \frac{1}{r}X'Y + \frac{1}{r^2}Y''X = 0,$$

so we have to separate variables. If $Y \neq 0$, then

$$\frac{r^2}{X} \left(X'' + \frac{1}{r}X' \right) (r) + \frac{Y''}{Y} = 0.$$

Since these two terms are dependent on different variables, they must both be constant.

$$\frac{Y''}{Y} = C \quad \frac{r^2}{X} \left(X'' + \frac{1}{r} X' \right) = C$$

for some constant C . Using the periodicity of Y , we have that $C = -n^2$ for some $n \in \mathbb{N} \cup \{0\}$. We've done this a million times.

Let's solve our eigenvalue problems.

$$Y'' = -n^2 Y, \quad \frac{r^2}{X} \left(X'' + \frac{1}{r} X' \right) = n^2.$$

The solutions to Y we've done a million times.

$$Y(\theta) = A_n \cos(n\theta) + B_n \sin(n\theta).$$

For X ,

$$r^2 X'' + r X' - n^2 X = 0,$$

so this is a Cauchy-Euler equation, and thus if our Ansatz is $X(r) = r^\alpha$, then

$$\alpha(\alpha - 1)r^2 \cdot r^{\alpha-2} + \alpha r \cdot r^{\alpha-1} - n^2 r^\alpha = 0.$$

Thus, we have that

$$\alpha(\alpha - 1) + \alpha - n^2 = 0 \implies \alpha^2 - n^2 = 0 \implies \alpha = \pm n.$$

On the other hand, if $\alpha = 0$, then this gives $\log r$ and 1 as our solutions, which we showed solves our equation above. Thus, our solutions are

$$A_0 + B_0 \log r, \quad A_n r^n + \frac{B_n}{r^n}.$$

But since the function we are looking for is continuous, we know that $B_0 = 0$ and $B_n = 0$ for all $n \in \mathbb{N}$. If you exclude the center, then you can have these other solutions.

For $n \geq 1$, our solutions are

$$u_n(r, \theta) = [A_n \cos(n\theta) + B_n \sin(n\theta)] r^n,$$

and for $n = 0$,

$$u_0(r, \theta) = \frac{A_0}{2} + \sum_{n \geq 1} [A_n \cos(n\theta) + B_n \sin(n\theta)] r^n,$$

which solve $\Delta u = 0$.

Applying the boundary conditions at $r = a$ give that

$$u(a, \theta) = f(\theta),$$

and thus

$$u(a, \theta) = \frac{A_0}{2} + \sum_{n \geq 0} a^n [A_n \cos(n\theta) + B_n \sin(n\theta)],$$

so from our Fourier theory (for 2π -periodic functions), we know that

$$a^n A_n = \frac{1}{\pi} \int_0^{2\pi} f(\theta') \cos(n\theta') d\theta',$$

when $n \geq 0$, and

$$a^n B_n = \frac{1}{\pi} \int_0^{2\pi} f(\theta') \sin(n\theta') d\theta',$$

when $n \geq 1$. This gives us our specific solution

$$u(r, \theta) = \frac{A_0}{2} + \sum_{n \geq 1} [A_n \cos(n\theta) + B_n \sin(n\theta)],$$

where A_k and B_k are defined in terms of the above formulas.

We can now use the formula to get a more compact expression.

$$\begin{aligned} u(r, \theta) &= \frac{1}{2\pi} \int_0^{2\pi} f(\theta') d\theta' + \sum_{n \geq 1} \left(\frac{r}{a}\right)^n \cdot \\ &\quad \left[\int_0^{2\pi} f(\theta') \cos(n\theta') \cos(n\theta) d\theta' + \int_0^{2\pi} f(\theta') \sin(n\theta') \sin(n\theta) d\theta \right] \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(\theta') \left[1 + 2 \sum_{n \geq 1} \left(\frac{r}{a}\right)^n (\cos(n\theta) \cos(n\theta') + \sin(n\theta) \sin(n\theta')) \right] d\theta' \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(\theta') \left[1 + 2 \sum_{n \geq 1} \left(\frac{r}{a}\right)^n \cos(n(\theta - \theta')) \right] d\theta' \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(\theta') \left[1 + \sum_{n \geq 1} \left(\frac{r}{a}\right)^n e^{in(\theta - \theta')} + e^{-in(\theta - \theta')} \right] d\theta' \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(\theta') \left[1 + \frac{\frac{r}{a} e^{i(\theta - \theta')}}{1 - \frac{r}{a} e^{i(\theta - \theta')}} + \frac{\frac{r}{a} e^{-i(\theta - \theta')}}{1 - r e^{-i(\theta - \theta')}} \right] d\theta' \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(\theta') \left[1 + \frac{r e^{i(\theta - \theta')}}{a - r e^{i(\theta - \theta')}} + \frac{r e^{-i(\theta - \theta')}}{a - r e^{-i(\theta - \theta')}} \right] d\theta' \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(\theta') \left[1 + \frac{2ar \cos(\theta - \theta')}{a^2 + r^2 - 2ar \cos(\theta - \theta')} \right] d\theta' \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(\theta') \left[\frac{a^2 - r^2}{a^2 + r^2 - 2ar \cos(\theta - \theta')} \right] d\theta' \end{aligned}$$

This is Poisson's formula, which we know from complex analysis. If we set

$$P_a(r, \psi) = \frac{a^2 - r^2}{a^2 + r^2 - 2ar \cos(\psi)},$$

then we have that

$$u(r, \theta) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta') P_a(r, \theta - \theta') d\theta'.$$

□

Remark 24.5. Let's think about this in a more geometric fashion. If $\vec{x} = (x, y)$, then

$$u(\vec{x}) = \frac{1}{2\pi} \int_{|x'|=a} u(x') \frac{a^2 - |\vec{x}|^2}{|\vec{x} - x'|^2} dx',$$

which you can find by plugging in (or working backwards from the formula).

We will start from here next time.

§25 Day 25: Poisson's Formula (Jan. 12, 2026)

The plan is

1. Recap of Poisson's Formula
2. Theorem on the consequences of Poisson
3. Solve Dirichlet's Problem on some simple 2D geometry other than the disk
4. We will do stuff in higher dimensions using Green's formulas and Identities. We will do Green's functions eventually, but this will not appear before the midterm.

Last time we solved the Dirichlet problem in the disk using Poisson's formula. This formula in particular is

$$u(r, \theta) = \frac{1}{2\pi} \int_0^{2\pi} P_a(r, \phi - \theta) f(\phi) d\phi,$$

where we have

$$P_a(r, \theta) = \frac{a^2 - r^2}{a^2 + r^2 - 2ar \cos(\theta)}.$$

This solves it for sufficiently nice functions f .

$$u(x) = \frac{a^2 - |x|^2}{2\pi a} \int_{|x'|=a} \frac{1}{|x - x'|^2} f(x') dx', \quad (*)$$

when $x \in D_a$ and $x' \in \partial D_a$.

Theorem 25.1. Let $f \in C(\partial D_a)$. Then $(*)$ is the unique solution to the Dirichlet problem in the disk, and

1. If $x \in D$ and $x_0 \in \partial D$, then $\lim_{x \rightarrow x_0} u(x) = f(x_0)$.
2. $u \in C^\infty(D)$.

Notice that 2 is the real exciting bit. 1 is basically just restating that $(*)$ solves our equation.

Proof. We essentially already proved this in the context of the heat equation. To prove 2, notice that for any fixed $x \in D$ there's some ball $B_\ell(x) \subseteq D$ such that $u(x)$ is a well-defined function

$$\begin{aligned} \partial_{x_1} u(x) &= \partial_{x_1} \left(\frac{a^2 - |x|^2}{2\pi a} \int_{|x'|=a} \frac{1}{|x - x'|^2} f(x') dx' \right) \\ &= \partial_{x_1} \left(\frac{a^2 - |x|^2}{2\pi a} \right) \int_{|x'|=a} \frac{1}{|x - x'|^2} f(x') dx' + \frac{a^2 - |x|^2}{2\pi a} \partial_{x_1} \left(\int_{|x'|=a} \frac{1}{|x - x'|^2} f(x') dx' \right) \end{aligned}$$

But notice that our domain is bounded, and the function on the inside is smooth with all derivatives bounded, so we can pull our derivative inside the integral.

$$= \partial_{x_1} \left(\frac{a^2 - |x|^2}{2\pi a} \right) \int_{|x'|=a} \frac{1}{|x - x'|^2} f(x') dx' + \frac{a^2 - |x|^2}{2\pi a} \left(\int_{|x'|=a} \partial_{x_1} \frac{1}{|x - x'|^2} f(x') dx' \right),$$

and this will be C^∞ .

Now, consider

$$\begin{aligned}\Delta u(x) &= \Delta \int_{|x'|=a} \frac{a^2 - |x|^2}{2\pi a} \frac{1}{|x - x'|^2} f(x') dx' \\ &= \int_{|x'|=a} \Delta \left(\frac{a^2 - |x|^2}{2\pi a} \frac{1}{|x - x'|^2} \right) f(x') dx' \\ &= 0,\end{aligned}$$

and we can check this is 0 directly by just computing the partial derivatives. Let us prove the limit equals what we hope. This proof turns out to be very similar to that of the heat equation. This is because our Poisson kernel approaches a constant multiple of the Dirac delta function as our radius approaches a , so convolving with our function yields the desired solution. On Wednesday we will cover this more explicitly. $\square$

§26 Day 26: Poisson's Formula Cont. (Jan. 14, 2026)

We proved Poisson's Formula last time.

Remark 26.1. Harmonic functions in D are smooth in D . They are also analytic, which we will not prove.

Theorem 26.2. Given f continuous,

$$u(x) = \frac{a^2 - |x|^2}{2\pi a} \int_{|x'|=a} \frac{f(x')}{|x - x'|^2} ds$$

is the unique solution to the Dirichlet problem in the disc, where the ds is integrating with respect to the arclength. This is

$$u(x) = \frac{a^2 - r^2}{2\pi} \int_0^{2\pi} \frac{f(\theta')}{r^2 - a^2 - 2ar \cos(\theta - \theta')} d\theta'.$$

in polar coordinates. Specifically, they have the properties that $-\Delta u = 0$ and

$$\lim_{x \rightarrow x_0} u(x) = f(x_0),$$

where $x \in D$ and $x_0 \in \partial D$.

Proof. To show that $\lim_{x \rightarrow x_0} u(x) = f(x_0)$, fix θ_0 and consider

$$u(r, \theta_0) = \int_0^{2\pi} P_a(r, \theta - \theta') f(\theta') d\theta'.$$

We ask whether as $r \rightarrow a$ this is $f(\theta_0)$. First we note that $P_a \geq 0$ for $P > 0$ and $r < a$. Further, we have that

$$\int_0^{2\pi} P_a(r, \psi) d\psi = 1,$$

and we can see this by looking at how the Poisson kernel was derived (it was a sum of sine and cosines, so just look at the only terms which don't cancel. One can also do it directly, which was actually on one of my midterms in 157.

Notice that if $|\psi| > \delta > 0$, then

$$\begin{aligned} P_a(r, \psi) &= \frac{a^2 - r^2}{(a - r)^2 + 2ar(1 - \cos \psi)} \\ &= \frac{a^2 - r^2}{(a - r)^2 + ar \sin^2\left(\frac{\pi}{2}\right)} \\ &\leq \frac{a^2 - r^2}{(a - r)^2 + ar \sin^2\left(\frac{\delta}{2}\right)} \rightarrow 0. \end{aligned}$$

Thus, it's gonna approximate a Delta function as $r \rightarrow a$, [as I remarked last time.] The formal proof is that

$$\begin{aligned} u(r, \theta_0) - f(\theta_0) &= \frac{1}{2\pi} \int_0^{2\pi} P_a(r, \theta_0 - \theta') [f(\theta') - f(\theta_0)] d\theta' \\ &= \frac{1}{2\pi} \int_{|\theta' - \theta_0| < \delta} P_a(r, \theta_0 - \theta') [f(\theta') - f(\theta_0)] d\theta' \\ &\quad + \frac{1}{2\pi} \int_{|\theta' - \theta_0| \geq \delta} P_a(r, \theta_0 - \theta') [f(\theta') - f(\theta_0)] d\theta' \end{aligned}$$

and we have that the former limit goes to 0 since $f(\theta') - f(\theta_0) \rightarrow 0$ and the latter goes to 0 as $r \rightarrow a$. $\square$

The Poisson kernel is the fundamental solution (sometimes called the Green's function) for the Dirichlet problem in the disk

$$\Delta u|_{D_a} = 0, \quad u|_{\partial D_a} = f.$$

Theorem 26.3. If u is harmonic in a disc D and continuous in $\bar{D}$, then the value of u at the center of D is the average of u on ∂D .

Proof. By translation invariance, let D be centered at the origin. Then recall Poisson's formula.

$$\begin{aligned} u(x) &= \frac{a^2 - |x|^2}{2\pi a} \int_{|x'|=a} \frac{u(x')}{|x - x'|^2} ds \\ u(0) &= \frac{a^2}{2\pi a} \int_{|x'|=a} \frac{u(x')}{|x'|^2} ds \\ &= \oint_{|x'|=a} u(x') ds, \end{aligned}$$

where f is defined to be the average value. □

Theorem 26.4. If u is harmonic in a domain $D \subseteq \mathbb{R}^2$ connected, then u can only attain a max on ∂D unless it is constant.

Proof. This will follow by our mean value theorem. Suppose that $x \in D$ is a max, where $D \subseteq \mathbb{R}^2$ is connected. Then there exists some $\varepsilon > 0$ such that $D_\varepsilon(x) \subseteq D$. As such,

$$\oint_{|x-x'|=\varepsilon} u(x') dx' \leq \oint_{|x-x'|=\varepsilon} u(x) dx',$$

and therefore we have that for all $x' \in D$ with $|x - x'| = \varepsilon$, $u(x') = u(x)$ (they cannot be greater, so they must be identically equal). But then on $x_0 \in D_\varepsilon(x)$ we have that $u(x_0) = u(x)$, so we have that the set $u^{-1}(\{u(x)\})$ is open. But since it is the preimage of a single point, it must be a closed set, so since it is both open and closed and D is connected, we have that $u^{-1}(\{u(x)\}) = D$, so u is constant. □

Remark 26.5. The reason we didn't show it for the heat is because it was more complicated to get domains on which the heat is the average, but once you have that property it's the same proof.

Recall that Harmonic functions are C^∞ smooth. We know through Poisson's formula, where we assumed u is merely continuous on the boundary. We are assuming that $u \in C^2$ when we right

Remark 26.6. We can make sense of the equation $\Delta u = 0$ even for non- C^2 functions. One way to do this is to look at (for $w : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$) the Dirichlet energy

$$\int_D |\nabla w|^2 dw,$$

and to minimize it subject to $u = f$ on ∂D . This is analogous to the principle of least action in classical mechanics.

We'll solve the Dirichlet problem on some special geometries. We've done the disc, so we can next do it on the wedge and the annuli. On the rectangle is in the homework. It can be solved using proper separation of variables and Fourier series. Really for any of these we just want them to be rectangles in some coordinate system.

Example 26.7 (Dirichlet Problem in a Wedge). Let $D = \{0 < \theta < \beta, 0 < r < \theta\}$. Then we want to solve $\Delta u = 0$. Let's suppose that our boundary conditions are

$$u(r, 0) = 0 = u(r, \beta) = \frac{\partial}{\partial r} u(a, \theta) = f(\theta).$$

Let's start by separating variables.

$$u(r, \theta) = R(r)\Theta(\theta) \implies \frac{R''}{R} + \frac{1}{r} \frac{R'}{R} + \frac{1}{r^2} \frac{\Theta''}{\Theta} = 0.$$

Thus, we have that

$$\begin{cases} \frac{\Theta''}{\Theta} = -\lambda \\ \Theta(0) = \Theta(\beta) = 0 \end{cases}$$

and $r^2 R'' + r R' - \lambda R = 0$. Then

$$\Theta(\theta) = c_1 \sin(\sqrt{\lambda}\theta) + c_2 \cos(\sqrt{\lambda}\theta),$$

but the cosine term cancels out from our boundary conditions, so

$$0 = \Theta(\beta) = c_1 \sin(\sqrt{\lambda}\beta),$$

and thus $\sqrt{\lambda}\beta = n\pi$ and thus $\sqrt{\lambda} = \frac{n\pi}{\beta}$, and also $\Theta(\theta) = c_n \sin\left(\frac{n\pi}{\beta}\theta\right)$.

Solving for R (Cauchy-Euler), we have that

$$r^2 R'' + r R' - \left(\frac{n\pi}{\beta}\right)^2 R = 0,$$

and so trying out $R(r) = r^\gamma$ gives

$$\gamma \cdot (\gamma - 1) r^\gamma + \gamma \cdot r^\gamma - \left(\frac{n\pi}{\beta}\right)^2 r^\gamma = 0$$

which yields

$$\gamma^2 - \left(\frac{n\pi}{\beta}\right)^2 = 0,$$

then

$$\gamma = \pm \frac{n\pi}{\beta} = \pm \sqrt{\lambda}.$$

After all this, we arrive at the general solution

$$u(r, \theta) = \sum_{n \geq 1} A_n r^{\frac{n\pi}{\beta}} \sin\left(\frac{n\pi}{\beta}\theta\right),$$

and since we need $\partial_r u(a, \theta) = f(\theta)$, we want

$$\partial_r u(a, \theta) = \sum_{n \geq 1} A_n \frac{n\pi}{\beta} a^{\frac{n\pi}{\beta}-1} \sin\left(\frac{n\pi}{\beta}\theta\right).$$

And so we can get our fourier coefficients by integrating against $f(\theta)$ here in the usual way.

$$A_n \frac{n\pi}{\beta} a^{\frac{n\pi}{\beta}-1} = \frac{2}{\beta} \int_0^\beta f(\theta) \sin \frac{n\pi}{\beta} \theta d\theta.$$

We do 3D stuff next time

§27 Day 27: Dirichlet's Problem (Jan. 19, 2026)

We did Dirichlet's problem

$$\begin{cases} \Delta u = 0 & \text{in } D \\ u = f & \text{on } \partial D \end{cases}$$

using separation of variables and Fourier series on “simple” geometries. Simple geometries are

1. Disc (Poisson)
2. Rectangle (and Triangle)
3. Annuli (using radial solutions to $\Delta u = 0$)
4. Wedge (in class).

For the wedge in particular,

$$\begin{cases} \Delta u = 0 & \text{interior of wedge} \\ \frac{\partial u}{\partial n} = f & \text{radial part of wedge} \\ u = 0 & \text{sides of wedge} \end{cases}$$

gave

$$u(r, \theta) = \sum_{n \geq 1} A_n r^{\frac{n\pi}{\beta}} \sin\left(\frac{n\pi}{\beta} \theta\right),$$

where

$$A_n = \frac{2}{n\pi a^{\frac{n\pi}{\beta}-1}} \int_0^\beta f(\theta) \sin\left(\frac{n\pi}{\beta} \theta\right) d\theta.$$

In general we should make sure that our boundary conditions make sense. For example, if we have Neumann boundary conditions on the side of our domain, then we should have at least that u is C^1 .

As another example, for a disc D , let's solve

$$\begin{cases} \Delta u = 0 & \text{in } D^c \\ u = f & \text{on } \partial D \end{cases}$$

Then our solutions will be linear combinations of r^n , r^{-n} , and $\log r$, for $n \geq 0$. But if we wanna impose an additional condition (e.g. boundedness) then this would eliminate all the r^n and $\log r$ terms.

The plan is to cover the Laplacian in 3D in spherical coordinates (read from the book, and Fabio will discuss how to remember it—it is just a bunch of chain rule, he says). From there, we will find radial solutions.

§27.1 Green's Functions

We are hoping to solve Dirichlet's problem

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = g & \text{on } \partial\Omega \end{cases}$$

(or Neumann— $\frac{\partial u}{\partial n} = g$). We want to find some special function $G(x, y)$ (for $x, y \in \Omega$, and which depends on Ω) that we can use to get a representation formula for our solution.

Let us first review Green's identities.

1.

$$\int_{\Omega} (\nabla u \nabla v + v \cdot \Delta u) dV = \int_{\partial\Omega} v \frac{\partial u}{\partial n} dS$$

2. Next time.

We can use this to prove the mean value theorem in 3D. Let $D = \{|x| < a\} \subseteq \mathbb{R}^3$. Then $\partial D = \{|x| = a\} = S$. Let $\Delta u = 0$. Note that the outward unit is $\frac{x}{|x|}$ when $x \in \mathbb{R}^3$. By the divergence theorem (Green's 1st theorem with $v \equiv 1$), we get

$$\int_S \frac{\partial u}{\partial n} dS = 0.$$

Thinking of u in spherical coordinates (where $u = u(r, \theta, \phi)$), we get

$$\int_0^{2\pi} \int_0^\pi u_r(a, \theta, \phi) a^2 \sin \phi d\phi d\theta = 0.$$

Then we have

$$\frac{1}{4\pi} \int_0^{2\pi} \int_0^\pi u_r(a, \theta, \phi) \sin \phi d\phi d\theta = 0$$

therefore

$$0 = \partial_r|_{r=a} \left(\frac{1}{4\pi} \int_0^{2\pi} \int_0^\pi u(r, \theta, \phi) \sin \phi d\phi d\theta \right)$$

and so since this is true for any a , this gives that

$$\frac{1}{4\pi} \int_0^{2\pi} \int_0^\pi u(r, \theta, \phi) \sin \phi d\phi d\theta$$

is a constant in r , so if we evaluate at $r = 0$, this gives the value for all r . Specifically, at $r = a$,

$$u(0) = \oint_S u(a, \theta, \phi) dS.$$

§28 Day 28: Green's Identities (Jan. 21, 2026)

Recall Green's identities.

1. For $u, v : D \subseteq \mathbb{R}^d \rightarrow \mathbb{R}$ (where we're assuming u and v are sufficiently nice to make the below work)

$$\int_D v \Delta u + \int_D \nabla u \cdot \nabla v = \int_{\partial D} v \frac{\partial u}{\partial n} dS.$$

In this case, dS is the surface element and n is the outward unit normal, so this right-hand integral is a surface integral.

2. The second identity can be given by exchanging u and v in the above then subtracting to get that

$$\int_D v \Delta u - u \Delta v = \int_{\partial D} v \frac{\partial u}{\partial n} - u \frac{\partial v}{\partial n} dS.$$

By applying the divergence theorem to $v \nabla u - u \nabla v$ we get this.

Our applications are

1. The mean value property for harmonic functions in 3-dimensions (and arbitrary dimensions).
2. Dirichlet Principle
3. Uniqueness for $\Delta u = f$ with boundary conditions (although this was already in a homework at some point).

Let's look at application 2 for now.

Theorem 28.1 (Dirichlet's Principle). Consider the functional E on a fixed D such that

$$w \mapsto \int_D |\nabla w|^2 dx.$$

which we call the Dirichlet energy of w in D . Actually, typically the Dirichlet energy is half of this, but Fabio forgot about it and only mentioned later.

Among the w in D which satisfy $w = h$ on ∂D (for some fixed h), then if there exists a harmonic function on our boundary conditions, it minimizes the energy E .

Proof. Consider u harmonic with $u = h$ on ∂D and any other w such that $w = h$ on ∂D . Then we want to show that

$$E[w] \geq E[u].$$

Let us consider $v = u - w$. Then we want to expand $E[w]$.

$$\begin{aligned} E[w] &= E[v - u] \\ &= \int_D |\nabla v - \nabla u|^2 \\ &= \int_D |\nabla v|^2 - 2 \nabla v \cdot \nabla u + \int_D |\nabla u|^2 \\ &\geq -2 \int_D \nabla u \cdot \nabla v + E[u] \end{aligned}$$

Note that $\int_D \nabla u \cdot \nabla v = 0$ because by Green's first identity, we have that

$$\int_D \nabla u \cdot \nabla v = - \int_D \Delta u v + \int_{\partial D} \frac{\partial u}{\partial n} v,$$

and the first integral vanishes because u is harmonic, and the second vanishes because $v \equiv 0$ on ∂D . It follows that $E[w] \geq E[u]$ by subbing in to the above. $\square$

Remark 28.2. We don't know a priori that a minimizer exists. By Wenyu's complex analysis class we have nice general criteria for when one exists fortunately—when the barriers exist and etc.

Let us do something similar but slightly different in conception.

Theorem 28.3. If $u : D \rightarrow \mathbb{R}$ with $u|_{\partial D} = h$ has minimal Dirichlet energy among all functions w with the same boundary conditions then u is harmonic. In other words, no non-harmonic function is a minimizer.²¹

Proof. Let $E[u] = \int_D |\nabla u|^2$ and let u be minimal among functions with the same boundary conditions. Set $u = \varepsilon v$ for a small ε , where v is smooth and $v = 0$ on ∂D .

$$E[u] \leq E[u + \varepsilon v] = E[u] + \varepsilon^2 E[v] + 2\varepsilon \int_D \nabla u \cdot \nabla v.$$

But then think of this as a function of ε . It has a minimum at $\varepsilon = 0$ so

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \left(E[u] + \varepsilon^2 E[v] + 2\varepsilon \int_D \nabla u \cdot \nabla v \right) = 0.$$

Then we have that

$$\int_D \nabla u \cdot \nabla v = 0 = - \int_D \Delta u v + \int_{\partial D} \frac{\partial u}{\partial n} v.$$

But we have that $v \equiv 0$ on ∂D so we have that

$$\int_D \Delta u v = 0,$$

and since this holds for all smooth v (also compactly supported), we must have that $\Delta u \equiv 0$. This is because we can vary v (setting it to be bump functions or such) to show that on any region its integral is 0. □

Remark 28.4. When we took $\frac{d}{d\varepsilon} E[u + \varepsilon v]$, this is called taking the derivative of our operator E . This is an idea taken from the calculus of variations. Beautiful.

§28.1 Representation Formula for Harmonic Functions

Take u harmonic and

$$v = -\frac{1}{4\pi} \frac{1}{|x - x_0|}.$$

This is motivated by solving “ $\Delta v = 0$ ” in 3 dimensions. Then

$$\int_D u \Delta v = \int_{\partial D} u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} dS.$$

Obviously this doesn't make sense when $x = x_0$. On the other hand, we can show that this is $u(x_0)$ on the left hand side in a natural way.²² Thus, it acts like a δ_{x_0} , a delta function.

Theorem 28.5. Let u be harmonic in D . Then

$$u(x_0) = \int_{\partial D} u \frac{\partial}{\partial n} \left(-\frac{1}{4\pi} \frac{1}{|x - x_0|} \right) dS - \int_{\partial D} \left(-\frac{1}{4\pi} \frac{1}{|x - x_0|} \right) \frac{\partial u}{\partial n} dS.$$

²¹Notice that this is stronger than our previous proposition—it shows that in the absence of a solution to the Dirichlet problem with our boundary conditions, there is no minimizer.

²²Which we could do formally with Radon-Nikodym

We are saying (and will show rigorously)

$$\int_D u(x) \Delta \left(-\frac{1}{4\pi} \frac{1}{|x - x_0|} \right) dx = u(x_0).$$

Of course this doesn't quite make sense. If we look for radial solutions of $\Delta u = 0$ in 3 dimensions, we get $(\partial_r^2 + \frac{2}{r} \partial_r) u = 0$, so one possible solution is

$$u = \frac{1}{r} = \frac{1}{|x|},$$

which is just singular at $r = 0$. We call $-\frac{1}{4\pi r}$ our fundamental solution for $\Delta u = 0$. This is because

$$\begin{cases} \Delta \left(-\frac{1}{4\pi r} \right) = 0 & r \neq 0 \\ \Delta \left(-\frac{1}{4\pi r} \right) = \delta & r = 0 \end{cases}$$

[Fabio put quotation marks around the second line. I don't know how to tex that.] In actuality, our second line should mean that

$$\int_{\partial D} u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} dS = u(x_0),$$

when u is harmonic.

Remark 28.6. What's a δ function? Well actually δ_x is not a function. It's a linear functional. Specifically, δ_x has $f \mapsto f(x)$. We denote linear functionals by using $\int$ so we might write

$$\delta(f), \quad \int f \cdot \delta, \quad \langle \delta, f \rangle.$$

We can think of $\int f \delta$ as being an integral with respect to a certain measure δ . This is the point measure. But we are moving more in the direction of the physics way of thinking about things.

Consider some smooth bump function with $\int_{\mathbb{R}} \varphi = 1$. Then consider

$$\lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}} \frac{1}{\varepsilon} \varphi \left(\frac{x}{\varepsilon} \right) f(x) dx.$$

This is going to be equal to (for nice functions f)

$$\lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}} \varphi(y) f(\varepsilon y) dy \rightarrow f(0).$$

thus $(\varphi_\varepsilon, f) \rightarrow f(0)$, therefore the *action* of φ_ε converges even though the function does not (it seems to go to 0 everywhere except at 0 where it goes to ∞).

This is the physics version of a δ function. What we really care about is the action on nice enough functions, not the actual function. We say δ is the weak limit of φ_ε —the limit, not as a function, but in terms of how it acts on other functions.

§29 Day 29: A Lemma About Representation (Feb. 2, 2026)

Going towards a representation formula for solutions of

$$\begin{cases} -\Delta u = f & \text{in } D \\ u = g \text{ or } \frac{\partial u}{\partial n} = g & \text{on } \partial D \end{cases}$$

and explicit formulae in some “simple” geometries (e.g. half space or the 3d ball).

We looked at Green’s identities previously. We will do a lemma about representation in 3d, and then go over Green’s functions.

Let’s start in 3d. Some facts we want are that $\frac{1}{r}$ is a radial solution of $\Delta u = 0$ in $\mathbb{R}^3$ (in $2d$ we have $\log r$). For this, we need to be able to write the laplacian in spherical coordinates (r, θ, ϕ) . Let’s restrict to only radial. Then the laplacian is $\partial_r^2 + \frac{2}{r}\partial_r$. In n dimensions, it is $\partial_r^2 + \frac{n-1}{r}\partial_r$. In general,

$$\Delta_{\mathbb{R}^3} = \partial_r^2 + \frac{2}{r}\partial_r + \frac{1}{r^2}\Delta_{S^2}$$

where Δ_{S^2} is the laplacian in θ, ϕ .

Find a radial function on $\mathbb{R}^3$ such that $\Delta u = 0$ then

$$u'' + \frac{2}{r}u' = 0$$

so setting $v = u'$ we get that $v' + \frac{2}{r}v = 0$ and therefore

$$\begin{aligned} \frac{v'}{v} &= -\frac{2}{r} \\ (\log v)' &= -\frac{2}{r} \\ \log v &= -2\log r + C_1 \\ v &= \frac{C_1}{r^2} \\ u &= \frac{C_1}{r} + C_2 \end{aligned}$$

which tells us that these really are solutions.

Lemma 29.1. If u is harmonic in D , then at any point $x_0 \in D$, we have that

$$u(x_0) = \int_{\partial D} -u(x) \frac{\partial}{\partial n} \frac{1}{|x - x_0|} + \frac{\partial u}{\partial n}(x) \frac{1}{|x - x_0|} \frac{dS_x}{4\pi}. \quad (R)$$

Proof. Recall Green’s second identity.

$$\int_D u \Delta v - v \Delta u = \int_{\partial D} u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n}.$$

Since $\Delta u = 0$, this gives

$$\int_D u \Delta v = \int_{\partial D} u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n}, \quad (*)$$

so from here we want to use $v_{x_0}(x) = -\frac{1}{4\pi|x-x_0|}$. With this v , the right hand side of (R) and $(*)$ match. But this doesn’t really make sense, because it’s not differentiable everywhere. But if we can establish that

$$\int_D u(x) \Delta \left(-\frac{1}{4\pi|x-x_0|} \right) dx = u(x_0),$$

in some manner, we would be done. As usual, when we can't do something globally, we just ignore the bad regions and see if things work out. In the spirit of this method, define $D_\varepsilon = D \setminus B_\varepsilon(x_0)$.

$$\int_{D_\varepsilon} u \Delta v = \int_{\partial D} u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} dS + \int_{\partial B_\varepsilon(x_0)} u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} dS$$

This works fine because we aren't integrating over the singular point. Suppose without loss of generality that $x_0 = 0$. Then all that is left to show is that

$$\int_{\partial B_\varepsilon(0)} u \frac{\partial}{\partial n} \frac{1}{|x|} - \frac{1}{|x|} \frac{\partial u}{\partial n} \frac{dS}{4\pi} \xrightarrow{\varepsilon \rightarrow 0} u(0)$$

(compared to our original formula our orientation is inverted which is why this formula seems to be negative of that in the question). Let's look at only the second term of this integral. Then

$$\int_{\partial B_\varepsilon(0)} \frac{1}{|x|} \frac{\partial u}{\partial n}(x) dS_x \xrightarrow{\varepsilon \rightarrow 0} 0,$$

as the surface area goes to 0 at the rate of $O(\varepsilon^2)$ and $\frac{1}{|x|} = \frac{1}{\varepsilon}$, and u is a nice function so its derivative is bounded. For the first term, we need only show that

$$\int_{\partial B_\varepsilon(0)} u \frac{\partial}{\partial n} \frac{1}{|x|} \frac{dS}{4\pi} \xrightarrow{\varepsilon \rightarrow 0} u$$

But then this is just

$$\int_{\partial B_\varepsilon(0)} u(x) \frac{1}{|x|^2} \frac{dS}{4\pi} = \frac{1}{4\pi\varepsilon^2} \int_{\partial B_\varepsilon(0)} u(x) dS = \oint_{\partial B_\varepsilon(0)} u(x) dS \xrightarrow{\varepsilon \rightarrow 0} u(0).$$

Up to the aforementioned considerations about the orientation, this proves the lemma. $\square$

§30 Day 30: Toward's Green's Function (Feb. 4, 2026)

Lemma 30.1 (Representation Formula). $u \in \mathcal{C}^2$ and $x_0 \in D$, $\Delta u = 0$ then

$$u(x_0) = \int_D v(x)(-\Delta)u(x)dx + \int_{\partial D} u \frac{\partial}{\partial n} v - v \frac{\partial u}{\partial n} dS_x$$

where

$$v_{x_0}(x) = \frac{1}{4\pi} \frac{1}{|x - x_0|}$$

Proof. We did it with $\Delta u = 0$. As an exercise we want to add $\Delta u \neq 0$ to that proof. $\square$

We used

$$v_{x_0}(x) = -\frac{1}{4\pi|x - x_0|}$$

as a very important function. This is because intuitively it has the property that

$$\int_D -\Delta v u dx = u(x_0),$$

even though this is not well defined, so in some sense $v(x, x_0)$ solves $\Delta_x v_{x_0}(x) = \delta_{x_0}(x)$.

Remark 30.2. We saw a similar idea with the heat kernel where it approximates the delta function. Also we saw it with the Poisson kernel.

We'll use our representation formula to solve

$$\begin{cases} -\Delta u = 0 & \text{in } D \\ u = h & \text{on } \partial D. \end{cases}$$

for a general domain D .

Definition 30.3. A Green's function for $-\Delta$ in a domain D at a point $x_0 \in D$ is a function defined such that $\forall x \in D$ such that

1. $G \in \mathcal{C}^2$ with $\Delta_x G = 0$ in D
2. $G(x) = 0$ on ∂D
3. $G(x) + \frac{1}{4\pi|x-x_0|}$ is finite at x_0 and $\mathcal{C}^2$ everywhere, and harmonic everywhere.

Remark 30.4. Varying x_0 we get a two parameter function $G(x, x_0)$ $x \neq x_0$. This will be notation used henceforth.

Theorem 30.5. If $G(x, x_0)$ is a Green's function for a domain D , then

$$u(x_0) = \int_{\partial D} \frac{\partial G}{\partial n}(x, x_0) u(x) dS_x.$$

solves the problem

$$\begin{cases} \Delta u = 0 & \text{in } D \\ u = h & \text{on } \partial D \end{cases}$$

If we have the alternative condition that

$$-\Delta u = f,$$

this adds

$$\int_D f(x) G(x, x_0) dx$$

to our solution.

Proof. Recall the representation formula above. Define H by $G(x, x_0) = v_{x_0}(x) + H(x)$. Then by our properties of Greens' functions, we have that $\Delta H = 0$ everywhere (specifically, (i) gives it everywhere away from x_0 and (iii) tells us at x_0). From Green's second identity on u and H , we have that

$$0 = \int_{\partial D} u \frac{\partial H}{\partial n} - H \frac{\partial u}{\partial n} dS_x.$$

Therefore, if we add this to our representation formula we get

$$\begin{aligned} u(x_0) &= \int_{\partial D} u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} dS_x + \int_{\partial D} u \frac{\partial H}{\partial n} - H \frac{\partial u}{\partial n} dS_x \\ &= \int_{\partial D} u(x) \frac{\partial G_{x_0}(x)}{\partial n}(x) - G_{x_0}(x) \frac{\partial u}{\partial n}(x) dS_x \end{aligned}$$

and the second term cancels by property 2. $\square$

In a bit we'll find Green's function for some domain (half-space, ball). Before this let's show that

Lemma 30.6. For Green's function $G(x, x_0)$, $x, x_0 \in D$, $x \neq x_0$, then

$$G(x, x_0) = G(x_0, x).$$

Proof. Consider $u(x) = G(x, a)$ and $v(x) = G(x, b)$ for $a, b \in D$. Then Green's second identity tells us that

$$\begin{aligned} 0 &= \int_{D \setminus (B_\varepsilon(a) \cup B_\varepsilon(b))} u \Delta v - v \Delta u dx \\ &= \int_{\partial(B_\varepsilon(a) \cup B_\varepsilon(b))} u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} dS \end{aligned}$$

because $G|_{x \in \partial D} = 0$

$$= \int_{\partial B_\varepsilon(a)} u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} dS + \int_{\partial B_\varepsilon(b)} u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} dS$$

Recall that u is singular at $x = a$ and v is singular at $x = b$. Then,

$$0 = \lim_{\varepsilon \rightarrow 0} \int_{\partial B_\varepsilon(a)} -v \frac{\partial u}{\partial n} dS + \int_{\partial B_\varepsilon(b)} u \frac{\partial v}{\partial n} dS.$$

Now, we ask where do these different pieces converge to as $\varepsilon \rightarrow 0$? Well

$$\lim_{\varepsilon \rightarrow 0} \int_{\partial B_\varepsilon(a)} -v \frac{\partial u}{\partial n} dS \xrightarrow{\varepsilon \rightarrow 0} v(a).$$

This is because we're basically just taking the average on a sphere of the values of our v . This is the same proof as last class. $\square$

Let's solve Dirichlet's problem in half-space. We'll use Green's function method.

$$\begin{cases} \Delta u = 0 & \text{in } z > 0 \\ u = h(x, y) & \text{on } z = 0 \end{cases}$$

$(x, y) \in \mathbb{R}^2$ coord on the boundary $P = (x_0, y_0, z_0)$ a point in $z > 0$. We're looking for a Green's function. We want $G(x, x_0)$ defined at all $x \neq x_0$ in $z > 0$.

1. $G \in C^2$ when $x \neq x_0$
2. $G(x, x_0) = 0$ when $x \in \partial D$
3. $G(x, x_0) + \frac{1}{4\pi|x-x_0|}$ is C^2 and harmonic.

If we let x_0^* be the reflection over the z axis of x_0 , then we can write

$$-\frac{1}{4\pi|x-x_0|} + \frac{1}{4\pi|x-x_0^*|}.$$

Such a reflection is gonna be anti-symmetric over the z -axis, so we will have 0 on the boundary. Also we've already shown it satisfies property one. For property three, notice that this is

$$G(x, x_0) + \frac{1}{4\pi|x-x_0|} = \frac{1}{4\pi|x-x_0^*|}$$

which we know is C^2 and harmonic away from x_0^* , which is outside of our domain D .

Therefore, the solution to the problem is

$$u(x_0) = \int_{z=0} \frac{\partial G}{\partial n}(x, x_0) \cdot h(x) dS_x$$

writing this out explicitly, we have that

$$\left. \frac{\partial G}{\partial n} \right|_{z=0} = \underbrace{2}_{G \text{ is odd}} \cdot \left(\frac{1}{2} \frac{2(z-z_0)}{4\pi|x-x_0|^3} \right) = \frac{-z_0}{2\pi|x-x_0|^3}.$$

Plugging back in, when $x_0 \in \{z > 0\}$,

$$u(x_0) = \frac{-z_0}{2\pi} \int_{z=0} \frac{1}{|x_0-x'|^3} h(x') dx'.$$

Next time we'll do the ball in $\mathbb{R}^3$. The exercise is to apply Green's functions method to the half-plane.

Remark 30.7. So far, all the procedures and ideas are general, except the only specific thing we used is

$$v_{x_0}(x) = -\frac{1}{4\pi|x-x_0|}$$

as the starting point. This was because we are in 3d. In 2d we'd do the same thing but with a different v_{x_0} . Specifically, in 2d, our fundamental solution is going to look something like $\log|x-x_0|$. This is because in 3d the radial solution was the above v . This might need a coefficient and such. Next time try applying it to the disk to get the Poisson kernel.

§31 Day 31: Green's Stuff (Feb. 9, 2026)

Regarding the midterm: “When you get a high grade you think ‘Oh I’m good’...don’t think you’re okay.”

Today we’ll finish the discussion on Green’s functions, then do the Ball in $\mathbb{R}^3$. Next we’ll do waves in 2d and heat in higher dimensions. This stuff will be on the midterm. Then we’ll see what we do. It won’t be on the midterm but it may be on the final. Later: first order quasilinear and fourier analysis and also some Schrödinger. Plus we’ll do the hydrogen atom. yay.

I couldn’t really focus today because 425 was on my mind. Please consult Strauss for this content if they are unclear. But most of the class was spent on a question which we had already answered.

Our aim is to solve the question

$$\begin{cases} -\Delta u = f & \text{in } D \\ u = h & \text{on } \partial D \end{cases}$$

We proved the following: Given a Green’s function $G(x, x_0)$ $x, x_0 \in D$

Theorem 31.1 (Representation Formula). The solution of this problem is

$$u(x_0) = \int_{\partial D} \frac{\partial G}{\partial n}(x, x_0) h(x) dS + \int_D G(x, x_0) f(x) dV.$$

Fabio then recapped our previous lectures to a student.

It reduces to solving for H

$$\begin{cases} -\Delta H = 0 & \text{in } D \\ H = \frac{1}{4\pi|x-x_0|} & x \in \partial D \end{cases}$$

1. We can do this explicitly in some geometries.
2. In general we have methods to solve this problem. Perron’s method and minimization of

$$\int_D |\nabla u|^2$$

Let’s solve the Dirichlet problem in the ball.

Theorem 31.2. The solution of

$$\begin{cases} -\Delta u = 0 & \text{in } B_a = \{x \in \mathbb{R}^3 : |x| < a\} \\ u = h & \text{on } \partial B_a \end{cases}$$

is the function

$$u(x_0) = \frac{a^2 - |x_0|^2}{4\pi a} \int_{|x|=a} \frac{h(x)}{|x - x_0|^3} dS_x.$$

We have that

$$\frac{\partial G}{\partial n}(x, x_0) = \frac{a^2 - |x_0|^2}{4\pi a} \frac{1}{|x - x_0|^3}$$

Proof. Start with

$$-\frac{1}{4\pi|x - x_0|}$$

and correct by a term H to get something zero on the boundary ∂B_a . To do this we're gonna reflect across the sphere. $x_0^* = \frac{x_0}{|x_0|^2}a^2$ then write

$$G(x, x_0) = -\frac{1}{4\pi|x - x_0|} + \frac{1}{4\pi|x - x_0^*|}.$$

This gives us our Green's function. Well not really. It doesn't vanish along the boundary. So we need to add a new term to the denominator to make it satisfy the properties we want. We will cover this next time. $\square$

§32 Day 32: Green's Function in Ball, Wave Equation (Feb. 11, 2026)

Theorem 32.1 (Green's Function on 3d Ball). Solution of the problem

$$\begin{cases} \Delta u = 0 & \text{in } B_a \\ u = h & \text{on } \partial B_a \end{cases}$$

is given by

$$u(x_0) = \frac{a^2 - |x_0|^2}{4\pi a} \int_{|x|=a} \frac{h(x)}{|x - x_0|^3} dS_x$$

for each $x_0 \in B$.

Proof. Reflect x_0 along ∂B_a . $x_0^* = \frac{x_0}{|x_0|^2} a^2$ so as usual start with

$$-\frac{1}{4\pi|x - x_0|} + \text{correction term}$$

so that the correction term makes this function zero on ∂B_a . Let's try

$$-\frac{1}{4\pi|x - x_0|} + \frac{1}{4\pi|x - x_0^*| \cdot \bullet}$$

where $\bullet$ is an unknown term added to make our function vanish along the boundary. We will get zero on the boundary if $|x - x_0| = |x - x_0^*| \cdot \bullet$, when $|x| = a$.

Ensure that the length from y to $\frac{x_0}{|x_0|} \cdot a$ equals the length from x_0 to x . But then

$$y = \frac{x}{a}|x_0|,$$

so therefore we can compute that

$$\begin{aligned} |x - x_0| &= \left| y - \frac{x_0}{|x_0|} a \right| \\ &= \left| \frac{x}{a}|x_0| - \frac{x_0}{|x_0|} a \right| \\ &= \left| \frac{x}{a}|x_0| - x_0^* \frac{|x_0|}{a} \right| \\ &= \frac{|x_0|}{a} |x - x_0^*|. \end{aligned}$$

It follows that our $\bullet$ term is

$$\bullet = \frac{|x_0|}{a}.$$

There's other ways to get this, but the diagram was clearly the best way to derive this.

[I will include the diagram in a bit. Until then just keep in mind that there was some magical geometry he used.]

Since now we have that

$$G(x, x_0) = -\frac{1}{4\pi|x - x_0|} + \frac{1}{4\pi|x - x_0^*|} \cdot \frac{a}{|x_0|},$$

we just need to find the normal derivative of our Green's function at $|x| = a$ to conclude. Namely,

$$\begin{aligned}
 \frac{\partial}{\partial n} \left(-\frac{1}{4\pi|x-x_0|} + \frac{1}{4\pi|x-x_0^*|} \frac{a}{|x_0|} \right) \Big|_{|x|=a} &= \left(\frac{x-x_0}{4\pi|x-x_0|^3} - \frac{x-x_0^*}{4\pi|x-x_0^*|^3} \frac{a}{|x_0|} \right) \frac{x}{a} \\
 &= \left(\frac{x-x_0}{4\pi|x-x_0|^3} - \frac{x-x_0^*}{4\pi|x-x_0^*|^3} \frac{a^3}{|x_0|^3} \frac{|x_0|^2}{a^2} \right) \frac{x}{a} \\
 &= \frac{x}{a} \cdot \frac{1}{4\pi|x-x_0|^3} \left(x-x_0 - (x-x_0^*) \frac{|x_0|^2}{a^2} \right) \\
 &= \frac{x}{a} \left(\frac{x - x \frac{|x_0|^2}{a^2}}{4\pi|x-x_0|^3} \right) \\
 &= \frac{a - \frac{|x_0|^2}{a}}{4\pi|x-x_0|^3},
 \end{aligned}$$

which brings us to the formula given in the statement of the problem. $\square$

§32.1 Wave Equation

Let's do the 3d wave equation first, then we'll go to 2d. This may seem weird because we've done the 1d wave and now we're jumping straight to 3d, but this is natural.

To remind you, the 1d wave equation is $u_{tt} - c^2 u_{xx} = 0$. In 3d this is $u_{tt} - c^2 \Delta_x u = 0$, where $u(x, t)$, with $x \in D \subseteq \mathbb{R}^3$, $t \in \mathbb{R}$.

There are a couple different nice symmetries of this equation:

1. Time translation
2. Space translation
3. Symmetry under Lorentz transformations
4. Space rotation symmetry
5. Scaling of spacetime
6. Time-reversible ($t \mapsto -t$). This is quite important!

We looked at the domain of dependence and of influence and of the finite speed of propagation. Fabio restated this, but check out Day 9 of my notes for finite speed of propagation in $d+1$ dimensions plus the proof. As a word of warning, I think my notes from that day may have a typo. He says it will most likely be in the midterm.

To prove that harmonic functions are smooth we could've done something different, but we proved it by getting a formula. But if they're general properties we should be able to prove them independently. Likewise we should be able to prove finite speed independently of a formula.

Remark 32.2. In 1d we proved D'Alembert's formula for the solution of the IVP. With that we proved finite speed. Let's get a formula on $\mathbb{R}^{3+1}$ to do the same.

Theorem 32.3 (Kirchoff's Formula). The solution to the problem

$$\begin{cases} u_{tt} - c^2 \Delta u = 0 & x \in \mathbb{R}^3, t_0 \in \mathbb{R} \\ u(x, 0) = \phi(x) & u_t(x, 0) = \psi(x) \end{cases}$$

is given by

$$u(x_0, t_0) = \frac{1}{4\pi c^2 t_0} \iint_S \psi(x) dS_x + \frac{\partial}{\partial t_0} \left(\frac{1}{4\pi c^2 t_0} \iint_S \phi(x) dS_x \right),$$

where S is the sphere with center x_0 and radius ct_0 . Namely

$$S = \{y : |y - x_0| = ct_0\}.$$

Really what's happening is that we're finding the solution in the same sort of manner as D'Alembert's—we draw the backwards cone until it hits the data at $t = 0$, and then we integrate along this sphere to get what our function should be equal to).

Remark 32.4. Really, our domain S should be written as S_{t_0} . Later we will rewrite it differently by taking our domain to be constant then rescaling the initial data instead. This is equivalent.

Proof. We want to try to reduce this to the case of a one spatial dimension problem. In that case we could use D'Alembert. One way to think about it is that the equation is invariant under rotations, so we might rotate around the center of the sphere and take the average to get roughly what the function should be doing. This is called the method of spherical average.

Define $\hat{u}(t, r; x_0)$ to be our averaged function, where t is the time, r is the radius, and x_0 is the center of our function. Then

$$\bar{u}(x, r; x_0) = \frac{1}{4\pi r^2} \int_{|x-x_0|=r} u(x, t) dS.$$

Think for now $\bar{u} = \bar{u}(t, r)$ and think of the x_0 as just a parameter. What happens if $r \searrow 0$? Well naturally

$$\lim_{r \rightarrow 0} \bar{u} = u(x_0, t).$$

Now, let's derive the PDE in (t, r) for $\bar{u}$. We claim that if u solves the PDE, then $\bar{u}$ solves

$$c^2 \left(\partial_r^2 \bar{u} + \frac{2}{r} \partial_r \bar{u} \right) = \bar{u}_{tt}.$$

Let us now prove this claim. For now let's assume that $x_0 = 0$ (by translation invariance this doesn't lose us any information). Then letting $w \in S^2$,

$$\bar{u}(t, r) = \frac{1}{4\pi r^2} \int_0^{2\pi} \int_0^\pi u(rw, t) r^2 \sin \theta \, d\varphi \, d\theta = \frac{1}{4\pi} \int_0^{2\pi} \int_0^\pi u(rw, t) \sin \theta \, d\varphi \, d\theta$$

so we have that

$$\begin{aligned} \partial_r \bar{u}(t, r) &= \frac{1}{4\pi r^2} \int_0^{2\pi} \int_0^\pi w \cdot \nabla_x u(rw, t) r^2 \sin \theta \, d\varphi \, d\theta \\ &= \frac{1}{4\pi r^2} \int_{B_r} \Delta_x u \, dV \\ &= \frac{1}{4\pi r^2} \int_{B_r} c^{-2} u_{tt} \, dV \end{aligned}$$

but we have that

$$r^2 \partial_r \bar{u} = \frac{1}{4\pi} \int_{B_r} c^{-2} u_{tt} \, dV,$$

so therefore

$$\begin{aligned}\partial_r (r^2 \partial_r \bar{u}) &= \partial_r \left(\frac{1}{4\pi} \int_0^r \int_{S_{r'}} c^{-2} u_{tt} dS dr' \right) \\ &= \frac{1}{4\pi} \int_{S_r} c^{-2} u_{tt} dS \\ &= r^2 c^{-2} \bar{u}_{tt}.\end{aligned}$$

Therefore we've got that

$$c^2 \partial_r (r^2 \partial_r \bar{u}) = r^2 \bar{u}_{tt},$$

which is our desired PDE, which can be seen by expanding.

Thus, we have that

$$\bar{u}_{tt} - c^2 \left(\partial_r^2 \bar{u} + \frac{2}{r} \partial_r \bar{u} \right) = 0.$$

Let $v(r, t) = r \bar{u}(r, t)$. Then we have that

$$\partial_{tt} v = r \partial_{tt} \bar{u}, \quad \partial_r^2 v = \partial_r (\bar{u} + r \bar{u}_r) = 2 \bar{u}_r + r \bar{u}_{rr},$$

thus multiplying by r in the PDE above,

$$r \bar{u}_{tt} - c^2 (r \partial_r^2 \bar{u} + 2 \partial_r \bar{u}) = 0$$

which implies therefore that

$$v_{tt} - c^2 \partial_r^2 v = 0.$$

Since $r > 0$, we need to use reflection before we can apply D'Alembert's formula. Let's use odd reflection in r . We need to be careful with regularity at $r = 0$ of the reflection. It might not be smooth and then we'll have issues with non-differentiability at the origin. There won't be according to Fabio, but we still have to check.

Our data is

$$\begin{cases} v(0, t) = 0 \\ v(r, 0) = r \bar{u}(r, 0) = r \bar{\phi}(r) \\ v_t(r, 0) = r \bar{u}_t(r, 0) = r \bar{\psi}(r). \end{cases}$$

Then D'Alembert's tells us that

$$v(r, t) = \frac{1}{2c} \left[\int_{r-ct}^{r+ct} s \bar{\psi}(s) ds \right] + \frac{\partial}{\partial t} \frac{1}{2c} \int_{ct-r}^{ct+r} s \bar{\phi}(s) ds.$$

This is the solution of the half line problem for $r < ct$. We only care about $r \searrow 0$ so we won't worry about $r \geq ct$.

Now let's get u .

$$\begin{aligned}u(t, x_0) &= \lim_{r \searrow 0} \bar{u}(r, t) \\ &= \lim_{r \searrow 0} \frac{1}{r} v(r, t) \\ &= \lim_{r \searrow 0} \frac{1}{2c} \frac{1}{r} \int_{ct-r}^{r+ct} s \bar{\psi}(s) ds + \frac{1}{r} \frac{1}{2c} \frac{\partial}{\partial t} \left[\int_{ct-r}^{ct+r} s \bar{\phi}(s) ds \right] \\ &= \frac{1}{2c} \partial_r|_{r=0} \left[\int_{ct-r}^{r+ct} s \bar{\psi}(s) ds + \frac{1}{r} \frac{1}{2c} \frac{\partial}{\partial t} \left[\int_{ct-r}^{ct+r} s \bar{\phi}(s) ds \right] \right] \\ &= \frac{1}{2c} \left[(r+ct) \bar{\psi}(r+ct) - (ct-r) \bar{\psi}(ct-r) \right]_{r=0} \\ &\quad + \frac{1}{2c} \frac{\partial}{\partial t} \left[(ct+r) \bar{\phi}(r+ct) + (ct-r) \bar{\phi}(ct-r) \right]_{r=0} \\ &= \frac{1}{2c} [ct \bar{\psi}(ct) + ct \bar{\psi}(ct)] + \frac{1}{2c} \frac{\partial}{\partial t} [ct \bar{\phi}(ct) + (ct) \bar{\phi}(ct)]\end{aligned}$$

so we are done. Averaging yields the other coefficients. □

§33 Day 33: Higher Dimensional Wave Equation (Feb. 23, 2026)

The midterm next wednesday will cover everything up to waves in dimension 3 and 2. Also of course waves in 1d. It will have the Laplacian, Elliptic Problems, Green's function stuff, and Harmonic functions. It'll have diffusion in 1d, and higher dimensions if we cover it.

Theorem 33.1 (Kirchoff's Formula). Consider the problem

$$\begin{cases} u_{tt} - \Delta u = 0 & t \in \mathbb{R}, \text{ and } x \in \mathbb{R}^3, \\ u(x, 0) = \phi(x) & x \in \mathbb{R}^3 \\ u_t(x, 0) = \psi(x) & x \in \mathbb{R}^3. \end{cases}$$

Then u is given by the formula

$$u(x_0, t) = \frac{1}{4\pi c^2 t} \int_S \psi(x) dS_x + \frac{\partial}{\partial t} \left(\frac{1}{4\pi c^2 t} \int_S \phi(x) dS_x \right),$$

where we have that

$$S = \{x : |x - x_0| = ct\}.$$

Namely, S is a sphere.

Proof. We looked at $v(r, t) = r\bar{u}(r, t; x)$, where $\bar{u}$ is the spherical average (and $r > 0$). We proved that v satisfies the 1d wave equation on the half line, so by odd reflection, we got a formula using D'Alembert.

$$v(r, t) = \frac{1}{2c} \int_{ct-r}^{ct+r} s\bar{\psi}(s) ds + \frac{\partial}{\partial t} \left[\frac{1}{2c} \int_{ct-r}^{ct+r} s\bar{\phi}(s) ds \right]$$

with $r \leq ct$.

Then from here observe that (wlog $x = 0$)

$$u(t, x) = \lim_{r \rightarrow 0} \frac{v(r, t)}{r} = \partial_r v|_{r=0}.$$

From here, we know that

$$\begin{aligned} u(t, x_0) &= \frac{1}{2c} [(r+ct)\bar{\psi}(r+ct) - (ct-r)\bar{\psi}(ct-r)]_{r=0} \\ &\quad + \frac{1}{2c} \frac{\partial}{\partial t} [(ct+r)\bar{\phi}(r+ct) + (ct-r)\bar{\phi}(ct-r)]_{r=0} \\ &= \frac{1}{2c} [ct\bar{\psi}(ct) + ct\bar{\psi}(ct)] + \frac{1}{2c} \frac{\partial}{\partial t} [ct\bar{\phi}(ct) + (ct)\bar{\phi}(ct)] \\ &= t\bar{\psi}(ct) + \frac{\partial}{\partial t} [t\bar{\phi}(ct)] \end{aligned}$$

Now, notice that indeed

$$\bar{\psi}(ct; 0) = \oint_{|x|=ct} \psi(x) dS_x = \frac{1}{4\pi (ct)^2} \int_S \psi(x) dS_x,$$

which plugging in (both $\bar{\psi}$ and $\bar{\phi}$) would match our formula above. We just need to translate stuff, but since we have invariance by space translations, it doesn't mess stuff up to only consider it centered at 0. $\square$

This proof is nice because it reduces the 3d to the 1d version. So we'll do some comments of the proof and the outcomes.

Remark 33.2. We solved our 3d wave equation by using the 1d formula. This can be done in all odd dimensions, but not in even dimensions. On the other hand, we can use the $2k + 1$ d formula to get the $2k$ d formula. We'll do the case of $3d \rightarrow 2d$ now. We'll leave comparison and other properties of our solution until later—next time. This will all (potentially) be on the midterm.

§33.1 2d Wave: Method of Descent

$$\begin{cases} u_{tt} - \Delta u = 0 & (x, y) \in \mathbb{R}^2 \\ u(x, 0) = 0 \\ u_t(x, 0) = \psi \end{cases}$$

Think of $\psi = \psi(x, y)$ as a function on $\mathbb{R}^3$ constant in z . Then (set $(0, 0, 0)$ to be our point of reference)

$$u(0, 0, t) = \frac{1}{4\pi c^2 t} \int_{|(x, y, z)|=ct} \psi(x, y) dS_{x, y, z}.$$

Let us calculate this surface integral. To do this, parameterize the sphere by

$$z = \pm \sqrt{(ct)^2 - x^2 - y^2}.$$

The $\pm$ corresponds to the top and bottom halves of the sphere. But the value of their integrals will be the same because ψ is independent of z . Then

$$\begin{aligned} u(0, 0, t) &= \frac{2}{4\pi c^2 t} \int_{\text{top}} \psi(x, y) dS \\ &= \frac{1}{2\pi c^2 t} \int_{x^2 + y^2 \leq c^2 t^2} \psi(x, y) \sqrt{1 + |\nabla_{x, y} f(x, y)|^2} dx dy \end{aligned}$$

because ψ is independent of z . Then we just want to calculate this square root term.

$$\sqrt{1 + |\nabla f|^2} = \sqrt{\frac{|(x, y)|^2}{c^2 t^2 - x^2 - y^2}} = \sqrt{\frac{x^2 + y^2}{c^2 t^2 - x^2 - y^2}} = \sqrt{\frac{c^2 t^2}{c^2 t^2 - x^2 - y^2}}.$$

Plugging it in yields our final formula.

$$u(0, 0, t) = \frac{1}{2\pi c} \int_{x^2 + y^2 \leq c^2 t^2} \psi(x, y) \frac{1}{\sqrt{c^2 t^2 - x^2 - y^2}} dx dy.$$

§34 Day 34: More on Higher Dimensional Wave Equation (Feb. 23, 2026)

Our plan is to review the formulas we proved for the wave equation on $\mathbb{R}^3$ and $\mathbb{R}^2$, discuss consequences and compare (also with 1d), discuss sources and non-homogeneous problem for wave, and finally discuss diffusion, and Schrödinger for $d \geq 1$. Midterm will cover whatever we manage to do today.

Theorem 34.1. Kirchoff's formula (in 3d) for

$$\begin{cases} (\partial_t^2 - \Delta) u = 0, & x \in \mathbb{R}^3, t \in \mathbb{R} \\ u(x, 0) = \phi \\ u_t(x, 0) = \psi \end{cases}$$

will be (for solutions u)

$$u(x, t) = \frac{1}{4\pi c^2 t} \int_{|x-y|=ct} \psi(y) dS_y + \frac{\partial}{\partial t} \left(\frac{1}{4\pi c^2 t} \int_{|x-y|=ct} \phi(y) dS_y \right)$$

By using the above for a function which was constant with respect to z , we derived the 2d formula.

Theorem 34.2. When we have 2d data, the solution is given by

$$u(x, t) = \frac{1}{2\pi c} \int_{|x-y| \leq ct} \frac{\psi(y)}{c^2 t^2 - |x-y|^2} dy + \frac{\partial}{\partial t} \left(\frac{1}{2\pi c} \int_{|x-y| \leq ct} \frac{\phi(y)}{\sqrt{c^2 t^2 - |x-y|^2}} dy \right)$$

where the $\sqrt{\cdot}$ terms are due to us integrating over the sphere.

Remark 34.3. We see from both formulas that if ϕ, ψ vanish outside $B_R(0)$, then the solution vanishes outside of the ball $B_{R+ct}(0)$. By applying linearity of the wave equation we can get that only modifying a solution inside our ball $B_R(0)$ will only affect our solution in the balls $B_{R+ct}(0)$, which means that the data propagates with finite speed.

We showed earlier that if ϕ, ψ vanish *inside* $B_R(0)$, then on the cone $B_{R-ct}(0)$ our function is identically zero, which we can easily show is equivalent. This was by integrating the energy on our cone with three pieces.

Let's look more carefully at the formula in 3d. Suppose that $\varphi = 0$. Then

$$u(x, t) = \frac{1}{4\pi c^2 t} \int_{|x-y|=ct} \psi(y) dS_y.$$

But we can see from the formula that this is only influence by changes to our data inside the cone. Notice also that we're only looking at $|x-y|=ct$, so any data in the interior of this circle will not affect our solution. In other words, when $|x| < ct - R$, we have that our solution is zero.

This may seem strange, but imagine it like a sound. You turn on a speaker you make an initial perturbation in the air which will travel outwards with finite speed of propagation. Eventually the sound reaches any point, but then since there was only a compact region of the air displaced by your speaker initially, the waves will eventually flow past you and the sound will stop.

Looking at the formula, if $|x| > R + ct$ the wave has not yet reached us. On the other hand, if $|x| < R - ct$, then the wave has already passed us. This property is the sharp Huygen's.

On the other hand, in 2d, we don't have this sharp Huygen's (the notion that eventually the initial data will pass us) because when integrating over the sphere, this sphere will always pass through the whole disc. Formulaically, we see that indeed the 2d system integrates on $|x - y| \leq ct$.

Finally, in the 1d case, while with respect to the initial displacement the sharp Huygens is satisfied, with respect to the initial velocity it is not.

$$\frac{1}{2} (\phi(x + ct) + \phi(x - ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(y) dy.$$

The former will eventually vanish with compact support, whereas the latter will not in general.

But isn't this strange physically? Think of the decay of the waves.

$$\lim_{t \rightarrow \infty} u(x, t) = 0$$

everywhere when our data is compactly supported, which we might expect. On the other hand, in 2d, we have an inequality that

$$|u(x, t)| \leq \frac{C}{t},$$

which Fabio asks you to check, and in 1d (the strangest case) we might even just have that $|u(x, t)|$ does not decay at all. For example, imagine if $\psi \not\equiv 0$ and $\psi \geq 0$ with compact support. Then every point will just eventually be constant and nonzero.

Well an analogy is an infinitely large sheet in outer space (for 2d) and an infinitely long rope in outer space (for 1d). Push the center of either. This may provide an intuition.

We also would hope that the energy is conserved, and if our solutions are going to zero at a controlled rate, then how is the energy conserved? Well we have that the area where our solution is changing cancels out the decay; although our solution is decaying in magnitude, it's growing in the area where it is changing, and these happen to precisely cancel out to conserve the energy.

§34.1 Sources / Non-Homogeneous Problem

First, let us use Duhamel's.

$$\begin{cases} u_{tt} - \Delta u = f(x, t) \\ u(x, 0) = 0 = u_t(x, 0). \end{cases}$$

If we don't have that the initial conditions are zero we can use linearity.

Now, how do we find a formula for u ? Well there's a few ways. The first is Duhamel's Principle. Then we have that by fixing an x (which will be implicit below),

$$\begin{cases} v_{tt} - \Delta_x v = 0 & t > s \\ v_t(t = s) = f(s, x) & t = s \\ v(t = s) = 0 & t = s \end{cases}$$

gives a solution at x by

$$u(t) = \int_0^t v(t, s) ds.$$

This is because

$$u_t(t) = v(t, t) = 0 \quad u_{tt}(t) = v_t(t, t) = f(s, x).$$

Now, we can solve this to get an explicit solution. Specifically, since v satisfies a usual homogeneous wave IVP, we can compute it explicitly.

$$u(x, t) = \int_0^t \frac{1}{4\pi c^2(t-s)} \int_{|x-y|=c(t-s)} f(y, s) dy ds$$

In the homework we do some manipulation to make it nicer and write it as an integral over a cone.

Now there's another way, which Fabio called the operator method, then said "I don't think this is a good name for it." Consider the problem

$$\begin{cases} u_{tt} + A^2 u = f(x, t) \\ u(x, 0) = g \\ u_t(x, 0) = h \end{cases}$$

where we consider A to be an operator (or a matrix). Then what's a formula for u ?

Well let's imagine a solution to the homogeneous problem when $f = 0$. Then we that $u(t)$ is a linear combination of $\cos(At)$ and $\sin(At)$. Then we get

$$u(t) = \cos(At)g + \frac{\sin(At)}{A}h.$$

Now, we get a particular solution without data. This is given by doing as our Ansatz

$$u(t) = c_1(t) \cos(At) + c_2(t) \sin(At).$$

When we differentiate, we know that differentiating our cos and sin terms solves the ODE, therefore

$$\partial_t u = \underbrace{c_1' \cos(At) + c_2' \sin(At)} + c_1 (-\sin(At)) A + c_2 \cos(At) A.$$

We choose c_1 and c_2 such that the underbraced term cancels, therefore

$$\partial_t^2 u = c_1' (-\sin At) A + c_1 (-\cos(At)) A^2 + c_2' \cos(At) A + c_2 (-\sin(At)) A^2.$$

As such, we get the 2×2 system

$$\begin{cases} c_1' \cos At + c_2' \sin At = 0 \\ c_1' (-\sin At) A + c_2' (\cos At) A = f \end{cases}$$

this can be solved explicitly and

$$c_1' = \frac{-\sin At}{A} f \quad c_2' = \frac{\cos At}{A} f.$$

Therefore, by solving these two, we get that

$$\begin{aligned} u(t) &= \int_0^t \frac{-\sin As}{A} f(s) ds \cos At + \int_0^t \frac{\cos As}{A} f(s) ds \sin At \\ &= \int_0^t \frac{\sin((t-s)A)}{A} f(s) ds \end{aligned}$$

But this is only our particular solution. Summarizing, the general solution is

$$u(t) = \cos(At)g + \frac{\sin(At)}{A}h + \int_0^t \frac{\sin(A(t-s))}{A} f(s) ds$$

If this were an ODE this would be how you'd solve it. But does this match our earlier solution? Well notice that by the above, $\frac{\sin(At)}{A}h$ is going to be a solution to the homogeneous problem with data $u_t = h$ and $u = 0$. As such, the expression $\frac{\sin(At)}{A}f$ must be a solution to the homogeneous problem with data $u_t = f$, which we have an explicit formula for. Subbing into the integral above yields

$$\int_0^t \frac{\sin(A(t-s))}{A} f(s) ds = \int_0^t \frac{1}{4\pi c^2 t^2} \int_{|x-y|=c(t-s)} f(y, s) dy ds,$$

which is exactly what we wanted from above.

We reduced our PDE, which is infinite dimensional, to a second order ODE by treating the operator in the x variable as just a number or maybe a matrix. We will do this more in a few weeks via the Fourier transform.

§35 Day 35: Diffusion/Schrödinger in Higher Dimensions (Mar. 2, 2026)

We already did $d = 1$, so let's do $d > 1$. For the heat equation,

$$\begin{cases} u_t = k\Delta u & x \in \mathbb{R}^d, t > 0 \\ u(x, 0) = \phi(x) \end{cases}$$

Theorem 35.1. The solution is given by

$$u(x, t) = \frac{1}{(4\pi kt)^{d/2}} \int_{\mathbb{R}^d} e^{-\frac{|x-y|^2}{4kt}} \phi(y) dy. \quad (\text{H})$$

Remark 35.2. The formula (H) in 1d was already proven a long long time ago. This was including the fact that

$$\lim_{t \searrow 0} u(x, t) = \phi(x)$$

when ϕ is continuous and bounded.

Remark 35.3. There's two ways to figure out the formula for $d > 1$.

1. When $d = 1$, we had that

$$\frac{e^{-\frac{(x_1-y_1)^2}{4\pi kt}}}{(4\pi kt)^{\frac{1}{2}}} = K_t(x_1 - y_1)$$

the heat kernel at time t , and then from here, the problem was solved by

$$\int_{-\infty}^{\infty} K_t(x_1 - y_1) \phi(y_1) dy_1.$$

Notice that K_t itself solved $\partial_t K_t = k\partial_x^2 K_t$ for $t > 0$. If we consider

$$\vec{K}_t(x_1, \dots, x_d) = K_t(x_1)K_t(x_2) \cdots K_t(x_d),$$

then $\vec{K}_t$ solves

$$(\partial_t - k\Delta_{\mathbb{R}^d}) \vec{K}_t \equiv 0,$$

and this is because $\Delta_{\mathbb{R}^d}$ is by definition just

$$\partial_{x_1}^2 + \cdots + \partial_{x_d}^2,$$

so therefore

$$\partial_t \vec{K}_t = \sum_{i=1}^d K_t(x_1) \cdots \partial_t (K_t(x_i)) \cdots K_t(x_d),$$

and likewise

$$\Delta_{\mathbb{R}^d} \vec{K}_t = \sum_{i=1}^d K_t(x_1) \cdots \partial_{x_i}^2 (K_t(x_i)) \cdots K_t(x_d),$$

so factoring yields that it satisfies this. Set for convenience $x = (x_1, \dots, x_d)$. Is this going to be enough to solve our heat equation by convolution? If we consider

$$\int_{\mathbb{R}^d} \frac{e^{-\frac{|x-y|^2}{4\pi kt}}}{(4\pi kt)^{d/2}} \phi(y) dy = U(x, t)$$

can we show it solves (H) ?

We first need to show that $\lim_{t \searrow 0} U(x, t) = \phi(x)$. There's a nice way to do this. First, we could just repeat the proof from 1d, then notice that our kernel approximates a delta function. On the other hand, we can just use the fact that when ϕ is bounded, then it can be well approximated by

$$\phi_1(x_1) \cdots \phi_d(x_d),$$

so the fact that $K_t(x_i)$ approximates a delta function as $t \searrow 0$ is sufficient.

2. In the pset we write as an ansatz that

$$u(x, t) = \frac{1}{t^\alpha} v\left(\frac{x}{t^\beta}\right),$$

and therefore we plug in, find convenient $\alpha = \frac{n}{2}$ and $\beta = \frac{1}{2}$, etc.

From the fundamental solution (the convolution formula) we see that our solutions have

1. Infinite smoothing
2. Infinite speed of propagation
3. Decay at a rate of $t^{-d/2}$ when t is much larger than 1, assuming that

$$\int_{\mathbb{R}^d} |\phi| < \infty.$$

4. Max principle.

Let's look at the Schrödinger equation quickly. This is

$$ihu_t = \frac{h^2}{2m} \Delta u + V(x)u,$$

for $u: (x, t) \subseteq U \times I \rightarrow \mathbb{C}$, where h is Planck's constant, m is the mass, and V is the potential. If we assume the potential is zero, and we rescale to avoid units, we have that

$$-iu_t = \Delta u \implies u_t = i\Delta u.$$

If we give the data $u(x, 0) = \phi(x)$, this is solved by the same thing as the heat equation except the constant k which was positive becomes an i . We will prove this rigorously.

§36 Day 36: Schrödinger in Higher Dimensions (Mar. 9, 2026)

We skipped 9.3 in Strauss, but you should read it. We looked at $u_t = k\Delta u$ for $k > 0$. We found the fundamental solution $x \in \mathbb{R}^d$, $t > 0$ whose Kernel is

$$S_t(x) = \frac{1}{(4\pi kt)^{d/2}} e^{-\frac{|x|^2}{4kt}}.$$

Then we moved on to Schrödinger very briefly. We will expound upon this. Specifically, the plan is

1. Use analogy to the formula with the heat.
2. Do some comparison to the heat equation. There are many differences.
3. Solve the harmonic oscillator.

The Schrödinger equation with one electron is

$$-i\hbar u_t = \frac{\hbar^2}{2m} \Delta u + \frac{\ell^2}{r} u,$$

where $\hbar$ is Planck's constant, and ℓ is a constant, and r is the radius. We're going to change coordinates here to get the equation

$$-iu_t = \frac{1}{2} \Delta u + V(x)u.$$

Here, V represents the potential somehow.

Let $u: I \times U \rightarrow \mathbb{C}$. We say $u = \operatorname{Re}(u) + i\operatorname{Im}(u)$. Then if we do $u_t = k\Delta u$ for $k \in \mathbb{C}$ with $\operatorname{Re} k > 0$, this is the proof of the heat equation we had before. So instead, let's look at

$$u_t = \frac{i + \varepsilon}{2} \Delta u \quad \varepsilon > 0.$$

This is not entirely rigorous, but look at this formula and take the limit as $\varepsilon \rightarrow 0$ to get our fundamental solution. The solution to

$$\begin{cases} u_t = \frac{i}{2} \Delta u = 0 \\ u(0, x) = f(x) \end{cases}$$

is given by

$$u(t, x) = \frac{1}{(2\pi it)^{d/2}} \int_{\mathbb{R}^d} e^{-\frac{|x-y|^2}{2it}} f(y) dy.$$

The meaning of $\sqrt{i}$ here is just $\lim_{\varepsilon \rightarrow 0} \sqrt{i + \varepsilon} = \frac{1+i}{\sqrt{2}}$.

Remark 36.1. This is very different from the heat equation. This is mainly due to the i in the denominator of our exponent of e . Let's see some ways in which it's very different for fun.

Proposition 36.2. • The Schrödinger equation is time reversible. If $u(t, x)$ is a solution, then $\bar{u}(-t, x)$ is a solution.

- Conservation of total mass. So long as the integral is defined,

$$\int_{\mathbb{R}^d} |u(t, x)|^2 dx \equiv c.$$

What's the physical meaning of Schrödinger? Well $|u(t, x)|^2$ represents the probability density to find our particle at position x at time t . I don't know this might be wrong, I'm not a physicist. Naturally then this mass should be conserved for this to make sense, otherwise the probability of the particle existing will change which is ridiculous.

- We also see that energy and the momentum are conserved.

$$\int_{\mathbb{R}^d} |\nabla u(t, x)|^2 dx, \quad \operatorname{Im} \left(\int_{\mathbb{R}^d} \nabla u(t, x) \cdot \bar{u}(t, x) dx \right).$$

in the homework or exam, he might ask “look at u_t plus more terms, is the energy or momentum conserved, what happens?”

Proof. Let's start with the mass. If M is our mass, then

$$\begin{aligned} \frac{d}{dt} M &= \int_{\mathbb{R}^d} \partial_t u \bar{u} + u \partial_t \bar{u} dx = 2 \int_{\mathbb{R}^d} \operatorname{Re}(\partial_t u \bar{u}) dx \\ &= 2 \operatorname{Re} \left(\int_{\mathbb{R}^d} \frac{i}{2} \Delta u \bar{u} dx \right) \\ &= 2 \operatorname{Re} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} i \nabla u \cdot \nabla \bar{u} dx \\ &= 0 \end{aligned}$$

Now let's check the momentum.

$$\begin{aligned} \partial_t \int_{\mathbb{R}^d} \partial_{x_1} u \bar{u} dx &= \int_{\mathbb{R}^d} \partial_{x_1} \partial_t u \bar{u} + \partial_{x_1} u \partial_t \bar{u} dx \\ &= \frac{1}{2} \int_{\mathbb{R}^d} \partial_{x_1} i \Delta \bar{u} - \partial_{x_1} u i \Delta \bar{u} dx \\ &= \frac{1}{2} \int_{\mathbb{R}^d} -\partial_{x_1} i \nabla u \cdot \nabla \bar{u} dx - \frac{1}{2} \int_{\mathbb{R}^d} \partial_{x_1} u i \Delta \bar{u} dx \\ &= \frac{1}{2} \int_{\mathbb{R}^d} i \partial_{x_1} u \Delta \bar{u} dx - \frac{1}{2} \int_{\mathbb{R}^d} \partial_{x_1} u i \Delta \bar{u} dx \\ &= 0. \end{aligned}$$

□

As mentioned above, we can look at what happens with $u_t = \frac{i}{2} \Delta u + iV(x)u$.

Let's search for oscillating waves. For the Schrödinger equation, we have solutions $u(x, t) = e^{ix\xi} e^{it\omega(\xi)}$, which are oscillating waves so long as $\omega: \mathbb{R}^d \rightarrow \mathbb{C}$ ends up having no imaginary part, this ends up being an oscillator. We want to find $\xi \in \mathbb{R}^d$ and suitable ω for this. So let's plug into our equation.

$$\partial_t u = i\omega(\xi) \cdot u \quad i\Delta u = i(-|\xi|^2) \cdot u.$$

So if we have $\omega(\xi) = -|\xi|^2$, we have an oscillating solution

$$u(x, t) = e^{ix\xi} e^{-it|\xi|^2}.$$

§37 Day 37: Properties of Heat and Schrödinger; Hydrogen Atom (Mar. 11, 2026)

Pop Quiz! For the following property, does the Heat/Schrödinger equation satisfy these

1. Time Reversible
2. Conserves $\int |u|^2$
3. Finite speed
4. Smooths initial data
5. There's waves which are oscillating in time when $u(x, 0) = e^{ix\xi}$ (where for the heat this is thought of as cosines and sines in our coordinates or such).
6. The function decays when $\int |u(x, 0)| dx < \infty$. Namely, $\lim_{t \rightarrow \infty} |u(x, t)| = 0$.

Recall that for the Schrödinger and Heat respectively our solutions are

$$u(x, t) = \frac{1}{(4\pi it)^{d/2}} \int_{\mathbb{R}^d} e^{-\frac{|x-y|^2}{4it}} u(y, 0) dy, \quad u(x, t) = \frac{1}{(4\pi kt)^{d/2}} \int_{\mathbb{R}^d} e^{-\frac{|x-y|^2}{4kt}} u(y, 0) dy.$$

Then I implore you to try this yourself without checking the below, but the answers are on the next page.

1. Heat is not time-reversible, Schrödinger is, so long as you take the conjugate of your solutions.
2. Heat does not conserve mass (it can decay), but Schrödinger does.
3. Neither has finite speed—just look at the formulas.
4. For heat it smooths. For Schrödinger it's more complicated. Part of the proof that heat smooths was that the heat kernel decays inside the integral. The Schrödinger equation doesn't, so something that looks like a saw wave might not be smooth for the Schrödinger. In other words, when we try differentiating, we have that the integral inside the integral of the heat solution will always exist, but for Schrödinger it might explode if our function does not decay. So we have to be careful.
5. For Schrödinger we found them last time. For heat, the initial data yields solutions which decay exponentially, so they are not periodic in time.
6. This is true for both of them. Just look at the equation.

§37.1 Schrödinger with Potential

The 1d model is

$$-iu_t = u_{xx} - x^2u.$$

In this case, our potential is $V = x^2$. Then trying to solve yields, for separable solutions $u(x, t) = T(t)X(x)$,

$$-i\frac{T'}{T} = \frac{X''}{X} - x^2 = \lambda,$$

for some constant λ . We also want that $\lim_{x \rightarrow \pm\infty} |u(x, t)| = 0$. If we want it to be a probability distribution (in the case of Schrödinger) this would be necessary.

Notice that if we had no $V(x)$ potential, then we would have

$$\frac{X''}{X} = \lambda$$

yields no solutions. None of our possible choices of λ work for this, they're either linear, constant, sine waves, hyperbolic, etc. In all these cases they do not decay in both directions. Thus the point is that a potential can allow us to produce separable solutions which are bounded. These are called bound states.

Let's solve the above equation.

$$-i\frac{T'}{T} = \frac{X''}{X} - x^2 = -\lambda.$$

Then we have that $X'' - x^2X - \lambda X = 0$. So then let's integrate against $X(x)$. $-\lambda$ is thought to be the energy of the state X .

Let's find solutions to $X'' + (\lambda - x^2)X = 0$. Not so easy to find, but if $\lambda = 1$ it's not so bad. In this case, we can find

$$X(x) = e^{-\frac{x^2}{2}}$$

by some ad hoc methods. For an arbitrary λ , we might try to find solutions X of the form

$$X(x) = w(x)e^{-\frac{x^2}{2}}$$

then we have that the equation for w is

$$w'' - 2xw' + (\lambda - 1)w = 0,$$

which we can get by just subbing in w to our above equation for X .

Let's try to solve this ODE via power series.

$$w(x) = \sum_{k \geq 0} a_k x^k,$$

then we have that

$$\sum_{k \geq 2} k(k-1)a_k x^{k-2} - \sum_{k \geq 1} 2ka_k x^k - \sum_{k \geq 0} (\lambda - 1)a_k x^k = 0.$$

Then

1. For our x^0 term, we have that $2a_2 = (1 - \lambda)a_0$
2. For our x^1 term, we have that $6a_3 = (3 - \lambda)a_1$
3. For our x^2 term, we have that $12a_4 = (5 - \lambda)a_2$.

So in general we get a recursion formula.

$$(k+2)(k+1)a_{k+2} = (2k+1-\lambda)a_k.$$

Since our a_0 and a_1 are free coefficients, a_0 determines our a_{2k} terms and a_1 determines the a_{2k+1} terms. We have that the expansion stops in one of these sequences if $2k - \lambda + 1 = 0$ which is the same as $\lambda = 2k + 1$.

Remark 37.1. If $a_0 = 0$ and $a_1 \neq 0$ this is an odd solution. Likewise, if $a_0 \neq 0$ and $a_1 = 0$, then this is an even solution.

Therefore, if $\lambda = 2k + 1$ and we have that $a_1 = 0$, then we get an even polynomial of degree k . On the other hand, if k is odd and $a_0 = 0$, then this is an odd polynomial with degree k .

As such, we have found some of the solutions. These are the only solutions, which is a problem on the homework. These are Hermite polynomials. Namely,

$$H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2, \quad H_3 = 8x^3 - 12x, \quad \dots$$

Then our solutions are

$$X_k(x) = e^{-\frac{x^2}{2}} H_k(x).$$

If we evaluate these, then

$$u_k(x, t) = e^{-\frac{x^2}{2}} H_k(x) e^{-i(2k+1)t}.$$

§37.2 Hydrogen Atom

Consider

$$iu_t = -\frac{1}{2}\Delta u - \frac{1}{r}u, \quad r = |x|.$$

where $x \in \mathbb{R}^3$. Let's look for separated solutions which are decaying. Namely,

$$\int_{\mathbb{R}^3} |u|^2 dx < \infty$$

Notice that we are taking minus the Laplacian, so the sign of the potential is going to be negative. This is because our Schrödinger equation with potential is

$$-iu_t = \frac{1}{2}\Delta u + V(x)u,$$

so we've multiplied everything by negative 1, and minus a positive function ($\frac{1}{r}$) is negative. Our potential also decays.

Separation of variables yields that when $u = T(t)v(x)$,

$$2i\frac{T'}{T} = -\frac{\Delta X}{X} - \frac{2}{r} = \lambda,$$

which is a constant. Let's look for $0 > \lambda = -\beta^2$. Then our solution is of the form

$$u_\lambda(x, t) = e^{2i\lambda t} v_\lambda(x),$$

where we need v_λ to solve the equation

$$-\Delta v - \frac{2}{r}v = -\beta^2 v.$$

Looking for radial $v_\lambda = v_\lambda(r)$ yields that

$$-v'' - \frac{2}{r}v' - \frac{2}{r}v = -\beta^2 v.$$

As an aside for now, why did we take λ to be negative? Well in this case then we have that

$$\int_{\mathbb{R}^3} |u|^2 dx < \infty \iff \int_0^\infty |v(r)|^2 r^2 dr < \infty.$$

So for r large, we have that $-v'' \sim -\beta^2 v$, so it would be decaying nicely at the rate of $v \sim e^{-\beta r}$.

Set $v(r) = w(r)e^{-\beta r}$ as an Ansatz. Then the equation for w is going to be

$$-w'' + \left(2\beta - \frac{2}{r}\right)w' + \frac{2}{r}(\beta - 1)w = 0.$$

This once again comes down to factoring.

$$w(r) = \sum_{k \geq 0} a_k r^k.$$

Before we plug in, let's rewrite it to make things a bit simpler when we plug in. Then this is

$$\frac{1}{2}rw'' - \beta rw' + w' - (\beta - 1)w = 0.$$

Since our exponents of r are sufficiently low on the higher terms, we can expect the power series method to work. If our r^k was too high on the higher order terms, we wouldn't get a recursion, so it wouldn't work. But the recursion in this case is

$$\sum_{k \geq 0} \left(\frac{1}{2}k(k-1) + k \right) a_k r^{k-1} + \sum_{k \geq 1} [-\beta(k-1) + (1-\beta)] a_{k-1} r^{k-1} = 0.$$

As before, this series eventually stops. But Fabio made an error, so I am not going to copy his conclusions at risk of spreading misinformation.

§38 Day 38: Fourier Transform (Mar. 16, 2026)

I was away today so Arky has typed up these notes in my stead. Thanks Arky!

– Emily

50% of the final will be stuff we've seen (from homework, class, etc.), though, some parts of the problems may be harder. The midterm will be graded by tomorrow.

We will mainly deal with the Fourier transform and its applications (to PDEs) in the last 2 weeks of class, with relevant textbook references in Strauss §12.3 (we will have different notation than Strauss, though, so be careful when interpreting) and §5 in **Stein & Shakarchi's Fourier Analysis** (the relevant parts are also uploaded onto Quercus). Our notation will follow that of the latter book.

The Fourier transform is an analogue of the Fourier series. For functions on all space (for now, we say $\mathbb{R}$, $\mathbb{R}^d$, etc.). To start, for sufficiently nice f , we can write the complex Fourier series

$$f(x) = \sum_{M=-\infty}^{\infty} c_M e^{iM\pi \frac{x}{\ell}}, \quad x \in (-\ell, \ell).$$

Here, we regard f as a 2ℓ -periodic function, where c_M is the Fourier coefficient given by

$$c_M = \frac{1}{2\ell} \int_{-\ell}^{\ell} f(y) e^{-iM\pi \frac{y}{\ell}} dy.$$

If we want, we may write the Fourier series with the coefficients substituted in as follows,

$$f(x) = \sum_{M=-\infty}^{\infty} \frac{1}{2\ell} \left(\int_{-\ell}^{\ell} f(y) e^{i\frac{M}{\ell}\pi y} dy \right) e^{i\frac{M}{\ell}\pi x},$$

where we wish to send $\ell \rightarrow \infty$ so that the interval $(-\ell, \ell)$ expands to the whole line, making each $\frac{M}{2\ell}$ closer to each other, so we may consider it a continuous variable.²³ Indeed, denote $k = \frac{M}{2\ell}$; we get may turn the summation into an integral,

$$f(x) = \sum_{M=-\infty}^{\infty} \frac{1}{2\ell} \left(\int_{-\ell}^{\ell} f(y) e^{-2\pi i k y} dy \right) e^{2\pi i k x} \xrightarrow{\ell \rightarrow \infty} \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(y) e^{-2\pi i k y} dy \right) e^{2\pi i k x} dk,$$

where $k = \frac{\pi}{2\ell}$ yields $\frac{dM}{2\ell} = dk$.²⁴ The above computation leads us to define

$$\hat{f}(k) := \int_{\mathbb{R}} f(x) e^{-2\pi i k x} dx$$

to be the Fourier transform of f . Note that the formal calculation says

$$f(x) = \int_{\mathbb{R}} \hat{f}(k) e^{2\pi i k x} dk.$$

We now make some definitions. An *integrable* function $f: \mathbb{R}^d \rightarrow \mathbb{C}$ is a function that satisfies $\|f\|_1 < \infty$, i.e.,

$$\int_{\mathbb{R}^d} |f(x)| dx < \infty.$$

²³Fabio's side remark: technically, we haven't done the theory for Fourier transform in higher dimensions yet, but the processes and constructions are exactly the same.

²⁴This is somewhat informal

In this manner, the Fourier coefficient is defined as $\hat{f}: \mathbb{R}^d \rightarrow \mathbb{C}$, where

$$\hat{f}(\xi) := \int_{\mathbb{R}^d} f(x) e^{-2\pi i x \cdot \xi} dx.$$

“Practice writing ξ ... if you want to write it on the final or homework properly. I’ve seen some things.” Note that the dot product written above is important! After all, we’re exponentiating $x \cdot \xi = x_1 \xi_1 + \cdots + x_d \xi_d$. We call ξ the *Fourier, frequency, or wave number variable*. In particular, if $\|f\|_1 < \infty$, then so is $\|\hat{f}\|_1 < \infty$, which we may see by writing

$$|\hat{f}(\xi)| \leq \int_{\mathbb{R}^d} |f(x)| |e^{-2\pi i x \cdot \xi}| dx = \|f\|_1.$$

In other words, if we denote $\hat{\mathcal{F}}$ as the map $f \mapsto \hat{f}$, then $\hat{f}$ being integrable implies that it is bounded. We’d like to integrate

$$\int_{\mathbb{R}^d} \hat{f}(\xi) e^{2\pi i x \cdot \xi} d\xi$$

to get $f(x)$. In principle, the above may not be defined yet under just the assumption that $\hat{f}$ is bounded.

Lemma 38.1 (Riemann–Lebesgue). If f is integrable, then $\hat{f}$ is bounded with $\hat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$.

Again, this is not enough for the above to be integrable, but it is a first step. We will skip the proof for now, since it is immediate from dominated convergence; see §3.1.4 on p.80 of Stein’s *Fourier Analysis* for this exact proof.

Definition 38.2 (Schwartz Space: Stein, p.134). A *Schwartz space* $\mathcal{S}(\mathbb{R}^d)$ is the set of all C^∞ smooth functions $\mathbb{R}^d \rightarrow \mathbb{C}$ such that all its derivatives *rapidly decrease* or *rapidly decay*, i.e.,

$$\sup_{x \in \mathbb{R}^d} |x|^k |f^{(\ell)}(x)| = c_{k,\ell} < \infty, \quad f^{(\ell)} := D_x^{(\ell)} f.$$

Note that we define $D_x^{(\ell)}$ to be the collections of derivatives of order ℓ . Next class, we will see that $\hat{\mathcal{F}}: \mathcal{S} \rightarrow \mathcal{S}$ (which you can read about in Stein §5.1.3!). To wrap up today’s lecture, we give a few examples of Schwartz functions on $\mathbb{R}^d$.

- (i) The Gaussian $\exp(-|x|^2)$,
- (ii) Any bump function with compact support (since bump functions are always taken to be smooth).

These are our two main examples.

§39 Day 39: Properties of Fourier Transform (Mar. 18, 2026)

Midterm average 64. There will be a wave question on the final.

Suppose $f: \mathbb{R}^d \rightarrow \mathbb{C}$. Then we say that

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} e^{-2\pi i x \xi} f(x) dx.$$

We are thinking of

$$f(x) = \int_{\mathbb{R}^d} e^{2\pi i x \xi} \hat{f}(\xi) d\xi.$$

Here x is physical space and ξ is frequency space. This integration is all just over $\mathbb{R}^d$ so

$$\int_{\mathbb{R}^d} f = \int_{\mathbb{R}^d} \operatorname{Re} f + i \int_{\mathbb{R}^d} \operatorname{Im} f.$$

There's no complex integration happening.

Remark 39.1. Some books don't include the factor of $2\pi i$ in the exponent of the Fourier transform, which can cause constants/factors to change.

Definition 39.2 (Schwartz Space). $S(\mathbb{R}^d)$ is the Schwartz space, which consists of all $f: \mathbb{R}^d \rightarrow \mathbb{C}$ such that

$$\sup_{x \in \mathbb{R}^d} |x^\alpha| |D^\beta f(x)| < c_{\alpha, \beta}.$$

So the Schwartz space is smooth functions which are decaying faster than any polynomial.

Lemma 39.3. If $\int_{\mathbb{R}^d} |f(x)| dx < \infty$ then $\hat{f}$ is bounded and continuous and

$$\lim_{|\xi| \rightarrow \infty} |\hat{f}(\xi)| = 0.$$

Proposition 39.4. Let $f \in S(\mathbb{R}^d)$. Then if we set

$$\hat{F}(f)(\xi) = \hat{f}(\xi) = \int_{\mathbb{R}^d} e^{-2\pi i x \xi} f(x) dx,$$

then we have the properties

1.

$$\widehat{f(\cdot + h)}(\xi) = e^{2\pi i h \cdot \xi} \hat{f}(\xi), \quad h \in \mathbb{R}^d$$

2.

$$\hat{F}(f(\cdot) e^{-2\pi i h \cdot x})(\xi) = \hat{f}(\xi + h), \quad h \in \mathbb{R}^d$$

3.

$$\widehat{f(\delta \cdot)}(\xi) = \delta^{-d} \hat{f}(\delta^{-1} \xi).$$

4. ($d = 1$)

$$\widehat{f'}(\xi) = 2\pi i \xi \hat{f}(\xi).$$

5. ($d = 1$)

$$\hat{F}(-2\pi i x f)(\xi) = \frac{d}{d\xi} \hat{f}(\xi)$$

Remark 39.5. [What's actually going on here in these propositions? Well it's establishing some sort of correspondence between physical space and frequency space. When you imagine physical/frequency space, a good intuition is to think about your intuition about sound waves or light waves. Physical space returns amplitude and phase at a given point, and frequency space returns the signal's frequency over all space. Immediately we can intuit property three if we recall from Grade 10 science class that wavelength and frequency of light are inversely related.]

	Physical space	Fourier/Frequency Space
1.	Translation	Modulation
2.	Modulate	Translation
3.	Rescaling of x	"Inverse" rescaling
4.	Derivative	Multiply by frequency
5.	Multiply by position x	Derivative in frequency.

1. Think of the physical meaning. When we translate in physical space this is the same as modulating in Fourier space. Moving over just changes the initial point of our cycling function $e^{2\pi i h \cdot \xi}$.
2. On the other hand, when we modulate in physical space, this is translation in Fourier space. There's some sort of duality between the Fourier transform and its inverse.
3. When you rescale x , this is the inverse scaling of frequencies in Fourier space. This can be derived from the formula, or you can think about it as frequency being inversely correlated with wavelength.
4. So long as our function is Schwartz, the algebra works out. In analogy, imagine the Fourier series, where differentiating multiplies by the frequency.
5. This is once again by duality.

I'm gonna leave the proof as an exercise for the reader, because Fabio mostly just skimmed through the algebra and it will be in Stein and Shakarchi.

Fabio then drew pictures, but this is a pain to embed, so I'm gonna link a desmos graph.

Example 39.6 (Some Pics).

1. <https://www.desmos.com/calculator/qcisryezot>
2. <https://www.desmos.com/calculator/b6cuk5f9sd>

Proposition 39.7. Property 4 gives " f regular $\implies \hat{f}$ decays". Assume that

$$\left(\frac{d}{dx}\right)^d f(x) \in L^1,$$

which in this case means L^1 integrable. Then we have that $\widehat{\left(\frac{d}{dx}\right)^d f}(\xi)$ is bounded with the bound

$$|\hat{f}(\xi)| \leq \frac{C}{|\xi|^d}.$$

Proof. Quite immediate.

$$\widehat{\left(\frac{d}{dx}\right)^d f(\xi)} = (2\pi i \xi)^d \hat{f}(\xi) = c|\xi|^d \hat{f}(\xi)$$

so when the former is bounded, so is the latter. $\square$

Towards an inversion formula. The aim is to show that

$$f(x) = \int_{\mathbb{R}^d} e^{2\pi i x \cdot \xi} \hat{f}(\xi) d\xi.$$

Lemma 39.8. If $f(x) = e^{-\pi|x|^2}$ then we have that it is its own Fourier transform. $\hat{f}(\xi) = e^{-\pi|\xi|^2}$.

Proof. This is in the problem set using ODEs. We also did this in complex analysis. $\square$

Theorem 39.9. $f \in S(\mathbb{R}^d)$ implies that $\hat{f} \in S(\mathbb{R}^d)$.

Proof ($d = 1$). $f \in S$ then we have that $\left| x^\alpha \left(\frac{d}{dx}\right)^\beta f \right| < c_{\alpha\beta}$, where $\alpha, \beta \in \mathbb{N}$. Likewise for $\hat{f}$, potentially varying our constants.

To show that $\hat{f} \in S$ it suffices that

$$\left(\frac{d}{dx}\right)^d x^\beta f \in L^1,$$

because this implies

$$\widehat{\left(\frac{d}{dx}\right)^\alpha x^\beta f(\xi)}$$

is bounded. So how do we show it's L^1 ? Well this follows because

$$\begin{aligned} \int_{\mathbb{R}} g(x) dx &= \int_{\mathbb{R}} \frac{1+x^2}{1+x^2} g(x) dx \\ &\leq c \sup \left| (1+x^2) \left(\frac{d}{dx}\right)^\alpha x^\beta f(x) \right| < \infty. \end{aligned}$$

$\square$

§40 Day 40: More on Fourier Transform (Mar. 23, 2026)

Recap: The Fourier Transform for $f \in L^1$ is

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} e^{-2\pi i x \cdot \xi} f(x) dx = \hat{\mathcal{F}}(f)$$

We are thinking of

$$\hat{\mathcal{F}}^{-1}(g)(x) = \int_{\mathbb{R}^d} e^{2\pi i x \cdot y} g(y) dy.$$

The most important property of the Fourier Transform is that in physical space frequency space the notions of localization and regularity are dual. That is to say that in physical space if your function is decaying at some rate, then the fourier transform will be correspondingly smooth, and likewise if in your physical space you have some smoothness, then there's corresponding decay in frequency space. Likewise going backwards from frequency space to Fourier space.

This was proven last time, but you should know the proof yourself and be able to recreate it. Do it right after lecture if necessary. Now because of this we have that $\hat{\mathcal{F}}$ maps Schwartz space into itself, and we also have that

$$\hat{\mathcal{F}}\left(e^{-\pi|x|^2}\right) = e^{-\pi|\xi|^2}.$$

We now want to prove the Fourier inversion formula for the Schwartz space. Namely,

$$\hat{\mathcal{F}}^{-1} \circ \hat{\mathcal{F}} = \text{id}, \quad \hat{\mathcal{F}} \circ \hat{\mathcal{F}}^{-1} = \text{id}.$$

Let us give one last reason to motivate this being true. There were 5 properties. One was like dilation and translation and such, but then we also had

$$\widehat{\partial_{x_1} f}(\xi) = 2\pi i \xi_1 \hat{f}.$$

Then we have also that

$$\widehat{\Delta f}(\xi) = -4\pi^2 |\xi|^2 \hat{f}(\xi).$$

So $\hat{\mathcal{F}}$ “diagonalizes” the laplacian. It can transform a PDE into an ODE. Let's suppose for the time being that we can actually invert. Namely,

$$\Delta f = \hat{\mathcal{F}}^{-1} \left(-4\pi^2 |\xi|^2 \hat{f}(\xi) \right),$$

then we can make sense of $a(\sqrt{-\Delta})f$. Namely, when $a: \mathbb{R} \rightarrow \mathbb{C}$ is a bounded function, then

$$a(\sqrt{-\Delta})f = \hat{\mathcal{F}}^{-1} \left(a(2\pi|\xi|) \hat{f}(\xi) \right)$$

Think of $\sqrt{-\Delta} \rightarrow |\xi|$ on the Fourier side. Then our equation corresponds to

$$\hat{\mathcal{F}} \left(a(\sqrt{-\Delta})f \right) = a(2\pi|\xi|) \hat{f}(\xi).$$

Examples: $e^{t\Delta}f$ with $t \geq 0$. This is going to be a solution of

$$\begin{cases} \partial_t u = \Delta u & t > 0 \\ u|_{t=0} = f \end{cases}$$

We did this a while back in a non-rigorous sense. So let's do it now formally. The meaning of $e^{t\Delta}f$ is just

$$\hat{\mathcal{F}} \left(e^{t\Delta}f \right) = e^{-t4\pi^2|\xi|^2} \hat{f}(\xi),$$

where we think of $a(y) = e^{-ty^2}$.

We computed $\hat{u}(t, \xi) = e^{-4\pi^2 t |\xi|^2} \hat{f}(\xi)$. Then we check that

$$\partial_t \hat{u}(t, \xi) = -4\pi^2 |\xi|^2 \hat{u}(\xi).$$

Take $\hat{\mathcal{F}}^{-1}$. Then we get

$$\partial_t u(t, x) = \Delta u(t, x).$$

Lemma 40.1.

$$\hat{\mathcal{F}} \left(e^{-\pi |x|^2} \right) (\xi) = e^{-\pi |\xi|^2}$$

and therefore we have that

$$\hat{\mathcal{F}} \left(e^{-\delta \pi |x|^2} \right) (\xi) = e^{-\pi \frac{|\xi|^2}{\delta}} \cdot \frac{1}{\sqrt{\delta}}.$$

Proposition 40.2. Let $f, g \in S$ then

$$\int \hat{f} g = \int f \hat{g}.$$

Proof.

$$\begin{aligned} \int \hat{f}(x) g(x) dx &= \int_x \left(\int_y e^{-2\pi i x y} f(y) dy \right) g(x) dx \\ &= \int_y \left(\int_x e^{-2\pi i x y} g(x) dx \right) f(y) dy && \text{(Fubini)} \\ &= \int \hat{g}(y) f(y) dy \end{aligned}$$

□

Remark 40.3. Even though we prove everything for Schwartz functions, typically we can extend all of these identities to make sense beyond just the restricted spots where we prove them to basically whenever they make sense. In this case, the claim is that the previous proposition holds even when $f, g \in L^1$.

Proposition 40.4.

$$f(0) = \int \hat{f}(\xi) d\xi.$$

More generally, when we plug in x , we have that

$$f(x) = \int \hat{f}(\xi) e^{2\pi i x \xi} d\xi$$

Proof. Let $f \in S$ be a Schwartz function and $g = e^{-\pi \delta |x|^2}$. Then

$$\begin{aligned} \int \hat{f} g &= \int \hat{f}(\xi) e^{-\pi \delta |\xi|^2} d\xi \\ &= \int f \hat{g} \\ &= \int f(x) e^{-\pi \frac{|x|^2}{\delta}} \frac{1}{\sqrt{\delta}} dx \end{aligned}$$

So now let us take the limit as $\delta \rightarrow 0$. Then

$$\int \hat{f}(\xi) e^{-\pi\delta|\xi|^2} d\xi \xrightarrow{\delta \rightarrow 0} \int \hat{f}(\xi) d\xi$$

whereas for our other integral it's slightly harder. But we know that actually if we set

$$g_\delta(x) = e^{-\pi\frac{|x|^2}{\delta}} \frac{1}{\sqrt{\delta}},$$

then g_δ is going to form approximate identities. Therefore,

$$\int f(x) e^{-\pi\frac{|x|^2}{\delta}} \frac{1}{\sqrt{\delta}} dx \xrightarrow{\delta \rightarrow 0} f(0).$$

□

§41 Day 41: Theory and Applications (of the Fourier Transform) (Mar. 25, 2026)

For some reason “Theory and Applications” as a title gives me the same vibes as “Crime and Punishment”. Our plan is to finish the Fourier Theory part then look at applications to PDE and the Heisenberg uncertainty principle.

Theorem 41.1. If $f \in S$ is a Schwartz function, then

$$f(x) = \int_{\mathbb{R}} e^{2\pi i x \xi} \hat{f}(\xi) d\xi,$$

where

$$\hat{f}(\xi) = \int_{\mathbb{R}} e^{-2\pi i x \xi} f(x) dx.$$

Proof. Last time we arrived at

$$f(0) = \int_{\mathbb{R}} \hat{f}(\xi) d\xi$$

using the identity

$$\int f \hat{g} = \int \hat{f} g$$

with the function $g = e^{-\pi \delta |\xi|^2}$. Behind this is also the fact that as $\delta \rightarrow 0$ this gives approximate identities. To conclude the theorem, let

$$f(x+y) = F(y) \implies F(0) = f(x).$$

Then from the above we have that

$$F(0) = \int_{\mathbb{R}} \hat{F}(\xi) d\xi \implies f(x) = \int_{\mathbb{R}} e^{2\pi i x \xi} \hat{f}(\xi) d\xi.$$

□

Let's now extend the Fourier Transform to $L^2(\mathbb{R}^d)$. Here,

$$L^2(\mathbb{R}^d) = \{f \text{ measurable} : \|f\|_{L^2} < \infty\}.$$

where

$$\|f\|_{L^2} = \left(\int_{\mathbb{R}^d} |f(x)|^2 dx \right)^{\frac{1}{2}}.$$

Notice that $f: \mathbb{R}^d \rightarrow \mathbb{C}$ potentially. We will prove $L^2(\mathbb{R}^d) = \bar{S}$ the topological closure with the topology given by the L^2 norm. We need a few steps to show this.

Lemma 41.2. Let $f, g \in S$. Then define

$$(f * g)(x) = \int_{\mathbb{R}^d} f(x-y)g(y) dy.$$

1. $f * g = g * f$
2. $f * g \in S$.
3. $\hat{\mathcal{F}}(f * g)(\xi) = \hat{f}(\xi) \cdot \hat{g}(\xi)$.

We call $f * g$ the convolution of f and g .

Proof. 1. Change of variables $y \rightarrow x - y$.

2. Homework 15.

3. We're gonna do this formally for the time being and make it rigorous later. I'll add a star to any of the lines that are dubious.

$$\begin{aligned}
 \hat{\mathcal{F}}(f * g)(\xi) &= \int_x e^{-2\pi i \xi x} \int_y f(x - y)g(y) dy dx \\
 &= \int_y g(y) \left(\int_x e^{-2\pi i \xi x} f(x - y) dx \right) dy \\
 &= \int_y g(y) e^{-2\pi i \xi y} \left(\int_x e^{-2\pi i \xi (x - y)} f(x - y) \right) dx dy \\
 &= \left(\int_y g(y) e^{-2\pi i \xi y} dy \right) \left(\int_x e^{-2\pi i \xi (x - y)} f(x - y) dx \right) \\
 &= \hat{g}(\xi) \hat{f}(\xi).
 \end{aligned} \tag{*}$$

□

Remark 41.3. The order of integration for multiple integrals doesn't matter when using the Lebesgue measure. Namely, given $F: \mathbb{R}^2 \rightarrow \mathbb{R}$ with $F \in L^1(\mathbb{R} \times \mathbb{R})$, then

$$\int \left(\int F(x, y) dx \right) dy = \int \left(\int F(x, y) dy \right) dx = \iint F(x, y) dx dy$$

This is Fubini's theorem. We also have a theorem of Tonelli. If $G(x, y) \geq 0$, then

$$\iint G(x, y) dx dy = \int \left(\int G(x, y) dx \right) dy = \int \left(\int G(x, y) dy \right) dx.$$

The remarkable part for Tonelli is that it need not be integrable. In this sense G being positive basically accounts for the condition on $\int |F|$. Namely it's basically saying "Fubini *plus* if one of these integrals are unbounded then all three are." This is by remarking that if we decompose F into the functions F^+, F^- which are both non-negative functions and $F = F^+ - F^-$, then we can integrate them individually then apply Tonelli. This kind of thing.

For our purposes, we apply the above with

$$F(x, y) = f(x - y)g(y), \implies G(x, y) = |F(x, y)| = |f(x - y)| \cdot |g(y)|.$$

It follows then that

$$\hat{\mathcal{F}}(f * g)(\xi) = \hat{f}(\xi) \hat{g}(\xi), \quad f, g \in L^1.$$

Remark 41.4. We are about to prove Plancherel's for Fourier transforms. The analogue for Fourier series was

$$\|f\|_{L^2([0, 2\pi])}^2 = 2\pi \sum |\hat{f}_n|^2.$$

The 2π factor can be remembered by setting $f \equiv 1$ and analyzing like this. Well Fabio says maybe. Doesn't remember if there was such a factor. Me neither to be fair.

Theorem 41.5 (Plancherel). If $f \in S$ then $\|f\|_{L^2} = \|\hat{f}\|_{L^2}$.

Proof. First note that

$$\hat{\mathcal{F}}(\bar{f}(-\cdot)) = \overline{\hat{\mathcal{F}}(f)}.$$

Namely, conjugates can be exchanged with the fourier transform, up to a minus sign in the argument of f . Now, let

$$h = f * \bar{f}(-\cdot).$$

We will use that $h(0) = \int \hat{h}(\xi) d\xi$ (proposition from last time) to prove the theorem.

$$h(0) = \int f(-y) \bar{f}(-y) dy = \|f\|_{L^2}^2$$

$$\hat{h}(\xi) = \hat{\mathcal{F}}(f * \bar{f}(-\cdot)) = \hat{f}(\xi) \overline{\hat{\mathcal{F}}(f)(\xi)} = |\hat{f}|^2(\xi)$$

so we have that

$$\int \hat{h}(\xi) d\xi = \int |\hat{f}(\xi)|^2 d\xi = \|\hat{f}\|_{L^2}^2.$$

□

Our final step is to define $\hat{f}$ for $f \in L^2$.

Lemma 41.6. S is dense in L^2 . Namely, for all $f \in L^2$ there exists a sequence $\psi_n \in S$ such that

$$\psi_n \xrightarrow{L^2} f.$$

Let us finally prove the original claim.

Proof. Given $f \in L^2$ let $\psi_n \in S$ as above with $\psi_n \rightarrow f$ in L^2 . Look at $\hat{\psi}_n$. We claim that $\hat{\psi}_n$ is Cauchy in L^2 . To do this, notice that by Plancherel,

$$\|\hat{\psi}_n - \hat{\psi}_m\| = \|\widehat{\psi_n - \psi_m}\| = \|\psi_n - \psi_m\|_{L^2}.$$

So since our ψ_n are Cauchy, our $\hat{\psi}_n$ are as well. Thus, we have that since L^2 is complete, $\hat{\psi}_n \rightarrow g$ for some g . We define $\hat{f} = g$. □

Proposition 41.7. The Fourier transform is a bijection on the Schwartz space and it's an isometry on L^2 .

Proof. For the first fact, the Fourier inversion formula shows that the Fourier transform is invertible. For the second fact, use the parallelogram law.

$$\langle f, g \rangle_{L^2} = \frac{1}{4} ((\|f + g\|^2 + \|f - g\|^2) + i(\|f + ig\|^2 + \|f - ig\|^2)).$$

Since the Fourier transform preserves the L^2 norm, it preserves the L^2 inner product, so it's an isometry. □

§41.1 Applications

We're gonna solve the heat, Schrödinger, wave, and poisson equations with the Fourier transform. For Poisson we'll do it on the half plane and also on a strip. Then we'll cover the Heisenberg uncertainty principle.

Example 41.8.

$$\begin{cases} u_t = \Delta u & t > 0, x \in \mathbb{R}^d \\ u(0, x) = f \end{cases}$$

Take the PDE $u_t = \Delta u$ and transform it in the x variable. Then

$$\hat{u}(t, \xi) = \int e^{-2\pi i x \xi} u(t, x) dx.$$

Then

$$\begin{cases} \hat{u}_t(t, \xi) = -(2\pi)^2 |\xi|^2 \hat{u}(t, \xi) \\ \hat{u}(0, \xi) = \hat{f}(\xi) \end{cases}$$

which gives us an ODE for $Z(t) = \hat{u}(t, \xi)$ for each $\xi \in \mathbb{R}^d$. But we know how to solve an ODE of this form.

$$\hat{u}(t, \xi) = e^{-(2\pi)^2 |\xi|^2 t} \hat{f}(\xi).$$

Above we cited the quite useful properties that

$$\hat{\mathcal{F}}(\nabla \odot f) = 2\pi i \xi \odot \hat{f}(\xi)$$

where $\odot$ is one of our bilinear operations. So like gradient, divergence, curl, go to multiplication, dot product, and cross product.

From here,

$$u(t, x) = \hat{\mathcal{F}}^{-1} \left(e^{-(2\pi)^2 |\xi|^2 t} \hat{f}(\xi) \right) = \left(e^{-(2\pi)^2 |\xi|^2 t} \hat{f}(\xi) \right)^\vee.$$

Fabio notated the inverse as a power of $\vee$.

But we know that

$$\begin{aligned} u(t, x) &= \left(\hat{K}_t(\xi) \cdot \hat{f}(\xi) \right)^\vee(x) \\ &= K_t * f(x). \end{aligned}$$

So we just need to find the inverse transform $\left(e^{-(2\pi)^2 |\xi|^2 t} \right)^\vee$.

$$\begin{aligned} \hat{K}_t(\xi) &= e^{-\pi |2\sqrt{\pi t} \xi|^2} \\ &= G(2\sqrt{\pi t} \xi) && (G = e^{-\pi |x|^2}) \\ \hat{\mathcal{F}}^{-1}(\hat{K}_t)(\xi) &= \hat{\mathcal{F}}^{-1}(G(\sqrt{4\pi t} \xi)) \\ &= G\left(\frac{x}{\sqrt{4\pi t}}\right) \frac{1}{(4\pi t)^{d/2}} \\ &= \frac{e^{-\frac{|x|^2}{4t}}}{(4\pi t)^{d/2}} \end{aligned}$$

§42 Day 42: More Applications of the Fourier Transform (Mar. 30, 2026)

We solved the heat before. The same procedure works for the Schrödinger equation.

Example 42.1.

$$\begin{cases} u_t = \Delta u & t > 0, x \in \mathbb{R}^d \\ u(0, x) = f \end{cases}$$

Take the PDE $u_t = \Delta u$ and transform it in the x variable. Then

$$\hat{u}(t, \xi) = \int e^{-2\pi i x \xi} u(t, x) dx.$$

Then

$$\begin{cases} \hat{u}_t(t, \xi) = -i(2\pi)^2 |\xi|^2 \hat{u}(t, \xi) \\ \hat{u}(0, \xi) = \hat{f}(\xi) \end{cases}$$

which gives us an ODE for $Z(t) = \hat{u}(t, \xi)$ for each $\xi \in \mathbb{R}^d$. This is like before.

$$\hat{u}(t, \xi) = e^{-i(2\pi)^2 |\xi|^2 t} \hat{f}(\xi).$$

From here,

$$u(t, x) = \hat{\mathcal{F}}^{-1} \left(e^{-i(2\pi)^2 |\xi|^2 t} \hat{f}(\xi) \right) = \left(e^{-i(2\pi)^2 |\xi|^2 t} \hat{f}(\xi) \right)^\vee.$$

As before though,

$$\begin{aligned} u(t, x) &= \left(\hat{K}_t(\xi) \cdot \hat{f}(\xi) \right)^\vee(x) \\ &= K_t * f(x). \end{aligned}$$

So we just need to find the inverse transform $\left(e^{-i(2\pi)^2 |\xi|^2 t} \right)^\vee$. Notice though that this time it was a formal calculation, because we had to take the inverse transform of the function

$$\hat{K}_t(x) = e^{-4\pi i t |x|^2},$$

which is smooth and bounded but not necessarily L^2 .

Example 42.2. The same kind of thing could solve

$$-\Delta u = f \quad -\Delta v + v = f$$

because in $\mathbb{R}^d$,

$$4\pi |\xi|^2 \hat{u} = \hat{f}, \quad (1 + 4\pi |\xi|^2) \hat{v} = \hat{f}$$

so therefore

$$\hat{u}(\xi) = \frac{\hat{f}}{4\pi |\xi|^2}, \quad \hat{v}(\xi) = \frac{\hat{f}}{1 + 4\pi |\xi|^2}.$$

Then we have that

$$u = \left(\frac{1}{4\pi^2 |\xi|^2} \right)^\vee * f, \quad v = \left(\frac{1}{1 + 4\pi^2 |\xi|^2} \right)^\vee * f$$

and the former is

$$u = \left(\frac{1}{4\pi |x|} \right) * f.$$

The latter he didn't write. Notice however the former function—namely,

$$\frac{1}{4\pi^2|\xi|^2},$$

is not L^2 . So we once again have a problem.

Proposition 42.3 (Exercises).

1. Is the Fourier transform of a radial function radial?
2. A function f is homogeneous of degree α if

$$f(\lambda x) = \lambda^\alpha f(x), \lambda > 0, x \in \mathbb{R}^d.$$

If f is homogeneous of degree α , then $\hat{f}$ is homogeneous of degree $-d - \alpha$.

Remark 42.4. If we know 1. then we can prove the first one easily. For the second one we would do the same kind of thing, finding radial solutions of our PDE, and so on.

Theorem 42.5 (Heisenberg's Uncertainty Principle). Assume that $\|f\|_{L^2} = 1$. Define $\bar{x} = \|xf\|_{L^2}$ and $\sqrt{2\pi}\bar{k} = \|\xi\hat{f}\|_{L^2}$. Then

$$\bar{x} \cdot \bar{k} \geq \frac{1}{2}$$

Proof. Notice that Cauchy-Schwarz gives that

$$\begin{aligned} \left| \int_{-\infty}^{\infty} xf(x)f'(x) dx \right| &\leq \left[\int_{-\infty}^{\infty} |xf(x)|^2 dx \right]^{1/2} \left[\int_{-\infty}^{\infty} |f'(x)|^2 dx \right]^{1/2} \\ &= \bar{x} \left[\int_{-\infty}^{\infty} |ik\hat{f}(k)|^2 \frac{dx}{2\pi} \right]^{1/2} \\ &= \bar{x}\bar{k} \end{aligned}$$

On the other hand, taking the original integral and integrating it by parts yields that

$$\int_{-\infty}^{\infty} xf(x)f'(x) dx = \left[\frac{1}{2}xf(x)^2 \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{1}{2}f(x)^2 dx = 0 - \frac{1}{2}$$

by f having L^2 norm 1. This implies that

$$\bar{x}\bar{k} \geq \frac{1}{2}.$$

□

§43 Day 43: More Applications of the Fourier Transform (Apr. 1, 2026)

What does $e^{-it\Delta}$ mean?

The answer is

$$e^{-it\Delta}: f \rightarrow u = e^{-it\Delta} f$$

such that

$$\begin{cases} \partial_t u = -i\Delta u \\ u(0, x) = f \end{cases} \quad x \in \mathbb{R}^d$$

set $u = U(t)f$. Then we want that

$$\partial_t u = -i\Delta u, \quad e^{-it\Delta} f|_{t=0} = f$$

So one answer is through the Fourier transform.

$$(e^{-it\Delta} f)(t, x) = \hat{\mathcal{F}}_{\xi \rightarrow x}^{-1} \left(e^{it4\pi^2|\xi|^2} \hat{f}(\xi) \right),$$

where we map the ξ variable to x (which is what this somewhat enigmatic subscript means). From the definition we can check that

$$\begin{aligned} \partial_t \hat{\mathcal{F}}^{-1} \left(e^{it4\pi^2|\xi|^2} \hat{f}(\xi) \right) &= \hat{\mathcal{F}}^{-1} \left(\partial_t e^{it4\pi^2|\xi|^2} \hat{f}(\xi) \right) \\ &= \hat{\mathcal{F}}^{-1} \left(e^{it4\pi^2|\xi|^2} i2\pi\xi \cdot 2\pi\xi \hat{f}(\xi) \right) \\ &= i\hat{\mathcal{F}}^{-1} \left(2\pi\xi \cdot 2\pi\xi e^{it4\pi^2|\xi|^2} \hat{f}(\xi) \right) \\ &= -i\Delta \hat{\mathcal{F}}^{-1} \left(e^{it4\pi^2|\xi|^2} \hat{f}(\xi) \right) \end{aligned}$$

where this last line is because $\hat{\mathcal{F}}(\nabla g) = 2\pi i\xi \hat{g}(\xi)$ and likewise $\hat{\mathcal{F}}(\nabla \cdot G) = 2\pi i\xi \hat{G}(\xi)$.

For a general function a , we can define $a(\Delta): f \rightarrow a(\Delta)f$ by doing

$$a(\Delta)f = \hat{\mathcal{F}} \left(a(2\pi i\xi) \hat{f}(\xi) \right)$$

Last class we proved Heisenberg's uncertainty principle. Namely, if $f: \mathbb{R} \rightarrow \mathbb{C}$ has $\|f\|_{L^2} = 1$, and

$$\bar{x} = \int_{\mathbb{R}} x^2 |f|^2 dx \quad \bar{k} = \int_{\mathbb{R}} \xi^2 |\hat{f}|^2 d\xi,$$

then

$$\bar{x} \cdot \bar{k} \geq \frac{1}{16\pi^2}.$$

We also get the same with

$$\bar{x} = \int (x - x_0)^2 |f|^2 dx \quad \bar{k} = \int (\xi - \xi_0)^2 |\hat{f}|^2 d\xi.$$

How? Well apply the original to $f(x+x_0)$ and we can shift $\bar{x}$. This won't change the value of $\bar{k}$. For shifting $\bar{k}$, we should modulate f . Namely, we should take $e^{-2\pi i\xi_0 x} f(x+x_0)$, which should shift $\bar{k}$. Fabio says to check this yourself.

But now one can ask when is $\bar{x}\bar{k} = \frac{1}{16\pi^2}$? Well we proved the inequality originally via Cauchy-Schwarz, so

$$\left| \int x f f' \right| = \|x f\|_{L^2} \|f'\|_{L^2}$$

implies that for some c

$$xf(x) = cf'(x),$$

and therefore we have that for some A, B

$$f(x) = Ae^{B\frac{x^2}{2}}.$$

But which choices of A, B yield $\|f\|_{L^2} = 1$? We want $\operatorname{Re} B < 0$ so that it decays, therefore by using our knowledge of the Gaussian integral, that

$$\int_{\mathbb{R}} e^{-\pi x^2} dx = 1,$$

we arrive at

$$\int_{\mathbb{R}} e^{-Bx^2} dx = \frac{\sqrt{\pi}}{\sqrt{2\operatorname{Re} B}}$$

which then yields

$$|A|^2 = \frac{\sqrt{2\operatorname{Re}(B)}}{\sqrt{\pi}}$$

to get that

$$1 = \int |A|^2 e^{-2\operatorname{Re}(B)x^2} dx.$$

Fabio says to check this as well because he's not sure of it.

We are going to solve

$$\begin{cases} -\Delta u = 0 \\ u(x, 0) = g(x) \\ u(x, b) = h(x) \end{cases}$$

on the domain of

$$0 < z < b \quad x \in \mathbb{R}^d$$

and rediscover D'Alembert's formula for

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 \\ u(0, x) = g(x) \\ u_t(0, x) = h(x). \end{cases}$$

For this second question, take the Fourier transform in x of

$$\begin{cases} \hat{u}_{tt} + (2\pi c\xi)^2 \hat{u}(\xi) = 0 \\ \hat{u}(0, \xi) = \hat{g}(\xi) \\ \hat{u}_t(0, \xi) = \hat{h}(\xi) \end{cases}$$

This PDE has turned into a second order ODE. Back when we did this a while ago we spoke about the “operator method” and said “treat the Laplacian as a matrix”. This is just that.

To solve our ODE, the general solution is

$$\hat{u}(t, \xi) = A(\xi) \cos(2\pi c\xi t) + B(\xi) \sin(2\pi c\xi t).$$

Imposing our initial conditions gives us that

$$\hat{u}(0, \xi) = A(\xi) \implies A = \hat{g}.$$

Likewise we also have that

$$\partial_t \hat{u}(0, \xi) = B(\xi)2\pi c\xi = \hat{h}(\xi) \implies B(\xi) = \frac{\hat{h}(\xi)}{2\pi c\xi}.$$

This implies that

$$\hat{u}(t, \xi) = \cos(2\pi c\xi t) \hat{g}(\xi) + \frac{\sin(2\pi c\xi t)}{2\pi c\xi} \hat{h}(\xi).$$

This is what we saw before. Rewriting this gives us

$$u(t, x) = \cos\left(c \frac{\partial_x}{i} t\right) g + \frac{\sin\left(c \frac{\partial_x}{i} t\right)}{-i \partial_x} h.$$

This is by our earlier proposition on $a(\nabla)$ and such. This rewriting is just a curiosity though, it's not necessary for the solution. We'll now show it's equivalent to D'Alembert. How would we show this? Well first we need to show

$$\begin{aligned} \hat{\mathcal{F}}^{-1}(\cos(2\pi c\xi t) \hat{g}(\xi)) &= \frac{1}{2} (g(x+ct) + g(x-ct)) \\ \cos(2\pi c\xi t) \hat{g}(\xi) &= \frac{1}{2} \hat{\mathcal{F}}(g(x+ct) + g(x-ct)) \\ &= \frac{1}{2} \hat{g}(\xi) e^{2\pi i c t \xi} + \frac{1}{2} \hat{g}(\xi) e^{-2\pi i c t \xi} \\ &= \cos(2\pi c t \xi) \hat{g}(\xi) \end{aligned}$$

the second part for sin you really should do, but he's not going to himself. Also we should do the first problem ourselves if we have time.

How do we define the Fourier Transform of a not nice function? If it's not L^2 or not Schwartz. If ψ is in the dual of the Schwartz functions—a bounded linear function on S —then we say that

$$(\hat{\psi}, \phi) = (\psi, \hat{\phi}), \quad \forall \phi \in S.$$

Thanks Fabio for the classes.